InvestiGator: explainable financial alerts for the retail investor

Anomaly detection and precedent retrieval over free data sources

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A dissertation submitted in partial fulfillment of the requirements for the degree of Master of Science, Specialisation Area of

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Porto, September 2026

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ISEP, Porto, September 2026

Dedicatory

To my family.

Abstract

When a stock price moves abruptly, retail investors need to determine whether the movement is unusual, whether its origin lies with the company or the market, and whether comparable precedents exist. Free applications display the percentage change, and the few that go further do not expose the evidence behind them; professional terminals do answer them, at a cost incompatible with this investor profile.

This dissertation presents InvestiGator, a financial alerting system built exclusively on free data sources, which answers the three questions and exposes the evidence supporting each answer. The system integrates four techniques: anomaly detection through a rolling z-score, decomposition of returns into market, sector and company components with beta shrinkage, precedent retrieval by semantic similarity with measurement of the observed outcome, and a supervised triage model trained on a purpose-built dataset.

Each component was evaluated against transparent baselines. Volatility-normalised detection proves consistent across companies; semantic retrieval outperforms the base rate in every sector assessed; the triage model does not outperform a volatility-only baseline, a negative result that is reported and analysed.

By design, the system does not produce price forecasts.

Keywords: Explainable Artificial Intelligence, Financial Alerts, Anomaly Detection, Semantic Retrieval, Retail Investor

Resumo

Quando o preço de uma ação se move de forma abrupta, o investidor particular necessita de determinar se o movimento é invulgar, se a sua origem é a empresa ou o mercado, e se existem precedentes. As aplicações gratuitas apresentam a variação percentual, e as poucas que vão além disso não expõem a evidência que as sustenta; os terminais profissionais respondem, a um custo incompatível com este investidor.

Esta dissertação apresenta o InvestiGator, um sistema de alertas financeiros assente exclusivamente em fontes de dados gratuitas, que responde às três questões e expõe a evidência de cada resposta. O sistema integra quatro técnicas: deteção de anomalias por z-score deslizante, decomposição do retorno em mercado, setor e empresa com encolhimento do beta, recuperação de precedentes por semelhança semântica com medição do desfecho observado, e um modelo supervisionado de triagem treinado sobre um conjunto de dados próprio.

Cada componente foi avaliado contra linhas de base transparentes. A deteção normalizada pela volatilidade revela-se consistente entre empresas; a recuperação semântica supera a taxa-base nos cinco setores avaliados; o modelo de triagem não supera uma linha de base baseada apenas em volatilidade, resultado negativo reportado e analisado.

Por desenho, o sistema não produz previsões de preço.

Palavras-chave: Inteligência Artificial Explicável, Alertas Financeiros, Deteção de Anomalias, Recuperação Semântica, Investidor Particular

Acknowledgement

To Prof. Luís Gomes, supervisor, and to Rafael Silva, co-supervisor, for the conversations that set the direction of this work, for the examples that showed me what already exists in the field, and for the guidance on writing the document. Some of the decisions described here were taken after one of those conversations.

To my family, for the support over these months.

To my colleagues at Sistrade, and to Sistrade, for the flexibility that made it possible to reconcile this master's degree with work, and for the conversations about the application while it was being built.

To those who follow the channel and rate the alerts they receive. That feedback is the only information in this work that did not come from an automatic measurement.

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List of Symbols

P_t	Closing price of session t	Eq. 3.1
r_t	Logarithmic return of day t, $\ln(P_t/P_{t-1})$	Eq. 3.1
μ, σ	Mean and standard deviation of the returns in the window preceding the day	Eq. 3.1
z_t	Rarity of the return of day t against the preceding window	Eq. 3.1
r_m, $\tilde{r}_s$	Daily return of the market index and of the sector index	Eq. 3.2
β_m, β_s	Historical sensitivity of the stock to the market and to the sector	Eq. 3.2
ε	Part of the return explained neither by the market nor by the sector	Eq. 3.2
$\hat{\beta}$	Beta estimate obtained by regression over the window	Eq. 3.3
β_{ref}	Reference beta towards which the estimate is shrunk	Eq. 3.3
$\text{SE}(\hat{\beta})$	Standard error of the beta estimate	Eq. 3.3
σ^2_{ref}	Reference variance that sets the strength of the shrinkage	Eq. 3.3
w	Weight of the Vasicek shrinkage, between zero and one	Eq. 3.3
cos	Cosine similarity between two sentence vectors	Section 3.5
h	Half-life of the recency decay applied to precedents	Section 3.5
k	Number of precedents retrieved and presented	Section 3.5
p_{bruta}	Triage score before calibration	Eq. 3.4
$p_{\text{calibrada}}$	Triage score after Platt calibration	Eq. 3.4
a, b	Learned parameters of the Platt calibration	Eq. 3.4
P, C	Precision and recall	Section 5.1
F_1	Harmonic mean of precision and recall, $2PC/(P+C)$	Section 5.1

List of Acronyms

AI Artificial Intelligence.

ARCH Autoregressive Conditional Heteroscedasticity.

BERT Bidirectional Encoder Representations from Transformers.

FNSPID Financial News and Stock Price Integration Dataset.

GARCH Generalized Autoregressive Conditional Heteroscedasticity.

HTTP HyperText Transfer Protocol.

LIME Local Interpretable Model-agnostic Explanations.

ONNX Open Neural Network Exchange.

PR-AUC Precision-Recall Area Under the Curve.

PSI Population Stability Index.

ROC-AUC Receiver Operating Characteristic Area Under the Curve.

RQ Research Question.

SBERT Sentence-BERT.

SHAP SHapley Additive exPlanations.

Chapter 1

Introduction

This chapter presents the context of the work described in this dissertation, the problem that motivates it, the research questions and the objectives defined, the scientific contributions and the organisation of the document.

1.1 Context

Buying shares has become a low-friction operation, available through mobile applications and without significant brokerage costs. Most adults in the United States now hold shares, directly or through funds and retirement plans (Gallup 2025), in a market whose capitalisation exceeds 62 trillion US dollars (Securities Industry and Financial Markets Association (SIFMA) 2025), as shown in Figure 1.1.

Ease of access has not extended, however, to the interpretation of information. When the price of an asset moves, the retail investor has limited instruments for determining what that movement means.

The behaviour of this segment is the object of current research, and two recent results delimit the problem addressed in this dissertation. The first is that retail investor attention is today a measured quantity, whose indicators are discussed and compared in the literature (Cahill, Zhangxin Liu, and Smales 2025): what determines what the investor observes is studied as a variable, and not as chance. The second is that retail participation in the market, and the conditions under which it trades, remain the subject of active review (Ernst and Spatt 2026). Chapter 2 develops the behavioural mechanisms that this literature documents, and which explain why the absence of context is more costly to this audience than to a professional.

What matters to retain from this body of work is what delimits the problem. The signals that capture the investor's attention arrive without context, and the decision that follows is taken under conditions the literature documents as unfavourable. Three concrete questions remain unanswered at the moment the investor observes a price movement:

1. **is the movement unusual?** A four per cent change is a rare event in a low volatility company and an ordinary day in a volatile one;
2. **does the movement originate in the company or in the market?** A fall that follows the market and an isolated fall are distinct situations;
3. **are there precedents, and what was their outcome?** The existence of comparable days in the past, and of what followed them, allows the observed movement to be situated.

Current free applications display the percentage change and answer none of the three. There are recent products that claim to go further, and Section 2.2 names them: what their own pages describe is a summary in natural language, and not individual past cases with the outcome measured. Professional financial information terminals do answer all three, at subscription costs incompatible with the investor profile in question. The gap does not follow from the unavailability of data, which exist in quantity and are free, but from the absence of verifiable context built on them.

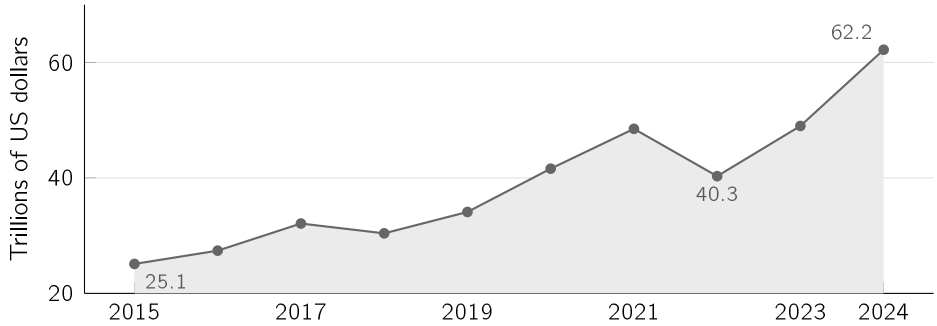

Figure 1.1: Capitalisation of the United States stock market, 2015–2024, in trillions of US dollars. Compiled from the data of the SIFMA Capital Markets Fact Book (Securities Industry and Financial Markets Association (SIFMA) 2025).

1.2 Problem description

This dissertation presents InvestiGator, a financial alerting system built exclusively on free data sources, which monitors a set of companies, detects events that depart from the usual behaviour of each one, and issues alerts accompanied by the explanation and the evidence that support them.

The system has two triggers, shown in Figure 1.2: a price movement that is unusual for the company in question, or the arrival of a relevant news item about it. In both cases, the message delivered identifies the event, the criterion by which it was considered unusual, the provenance of the data used, and the semantically closest past cases, with the observed behaviour of the price after each of them.

One design decision runs through the system and conditions all the others: **the system does not produce price forecasts**. It does not indicate future direction, does not issue buy or sell recommendations, and does not present price targets. This restriction has two justifications. The first is verifiability: a statement about a past event can be confirmed against the record at the moment it is read, whereas a forecast can only be assessed after the investor's decision has already been taken. The second is theoretical: the efficient market hypothesis, in its semi-strong form, makes the systematic anticipation of movements from publicly available information improbable. Chapter 2 develops it and identifies the corresponding literature. This dissertation does not test that hypothesis, nor does it rely on it to establish any conclusion.

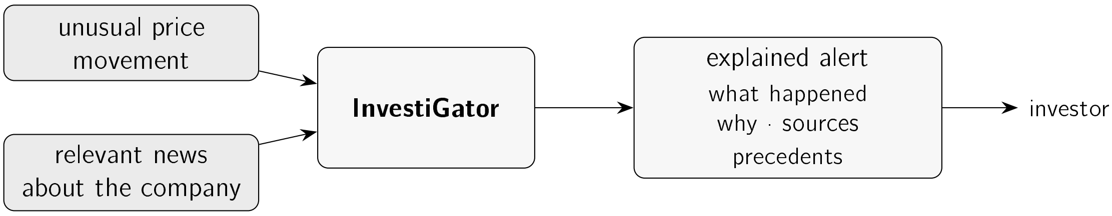

Figure 1.2: The two triggers of the system and the output they produce. The investor receives an alert accompanied by the evidence that allows it to be verified, and not merely the notification that the price has moved.

1.3 Objectives and research questions

The general objective of this dissertation is to design, build and evaluate an explainable financial alerting system for retail investors, using exclusively free data sources.

Three research questions follow from this objective, each associated with a measurement and a success criterion defined before the experiments were carried out:

- **Research Question (RQ)1, detection.** Can a simple and transparent statistical method detect unusual movements consistently across companies with distinct volatilities, and do so better than a fixed threshold rule?
- **RQ2, precedents.** Can the system retrieve past news from another company in the same sector better than trivial alternatives, and measure the subsequent behaviour of the price without resorting to information posterior to the moment of the query?
- **RQ3, triage.** Can a trained model decide which news items justify an alert better than a baseline built only on volatility?

The investor's second question, concerning the separation between the market component and the company-specific component, is addressed by a technique described in Chapter 3 but does not give rise to a research question. The reason is methodological: the split of a movement has no observable reference value against which it can be compared, since that quantity exists only relative to a model. Section 5.5 establishes what, in spite of that, remains measurable in the technique.

These questions take the form of four objectives:

1. to build the complete system, from data acquisition to the delivery of the alert;
2. to evaluate each component against transparent baselines;
3. to ensure that the explanations presented are faithful to what the system computed;
4. to report the results obtained, including those that do not confirm the initial expectations.

1.4 Scientific contributions

This being a dissertation in Artificial Intelligence (AI) Engineering, the contribution does not lie in the formulation of a new algorithm, but in the selection, integration and critical evaluation of established techniques until they constitute a working, explainable and reproducible system. Four contributions are identified:

1. **A complete system in operation**, with both triggers, built on free sources and actually deployed, and not a laboratory demonstration.
2. **A reproducible method for measuring the behaviour of the price following semantically close news items**, combining retrieval by sentence similarity with event study windows, without using information posterior to the moment of the decision. Chapter 2 identifies the two lines of work from which they come.
3. **A purpose-built dataset and a model trained on it**, with 79,753 examples constructed from the public Financial News and Stock Price Integration Dataset (FNSPID) corpus (Dong, Fan, and Peng 2024) and from price series, with the model, the calibration and the metrics kept as versioned artefacts.
4. **An evaluation of each component against transparent baselines**, including the result that contradicts the initial hypothesis, reported as such.

1.5 Structure of the document

Chapter 2 presents the state of the art, characterising the target audience, the tools currently available and the technical areas from which the components used are drawn. Chapter 3 describes the data and the four techniques that constitute the system. Chapter 4 describes the architecture of InvestiGator and the path of a news item from collection to delivery. Chapter 5 presents the experimental design and the results obtained for each research question. Chapter 6 synthesises the answers, identifies the limitations of the work and sets out the lines of future development.

Each mechanism described is illustrated on a real case, with the values actually computed by the system, and not on examples constructed for the purpose.

Chapter 2

State of the art

This chapter presents the scientific and technological framing of the work. It first characterises the audience the system addresses and the limitations that define the problem; it then examines the solutions currently available; and it finally surveys the technical areas from which the components used are drawn, situating the contribution of this dissertation in relation to each of them.

2.1 The retail investor and the attention problem

This section characterises the documented behaviour of the audience the system addresses, since it is the limitations of that behaviour, and not the size of the market, that delimit the problem treated here.

Chapter 1 established the scale of the universe in question. The characterisation that is missing is behavioural, and it begins with an observation that runs counter to the current portrayal of the retail investor as an agent who only reacts in panic. During the fall of March 2020, retail investors on a commission-free platform increased their aggregate positions rather than reducing exposure (Welch 2022). This is an audience that decides regularly, and that therefore benefits from better information at the moment of the decision.

The behavioural finance literature establishes that those decisions depart systematically from the rational ideal (Barberis and Thaler 2003). Two regularities bear directly on the design of an alerting system. The first is loss aversion (Kahneman and Tversky 1979), already stated in the previous chapter, whose consequence for this work is that the cost of the absence of context is asymmetric: it is greater precisely when the information is unfavourable.

The second is the attention effect, and it is the one that grounds the problem. Barber and Odean (2008) establish that retail investors are net buyers of the stocks that capture their attention, identified by three observable indicators, namely presence in the news, abnormal trading volume and an extreme return on a single day. The mechanism the authors propose is that of the search problem: faced with a very large number of alternatives, the investor does not assess them all and the choice falls on the subset that has been made visible.

The authors delimit the reach of the claim, which concerns an aggregate imbalance between purchases and sales and not a universal behaviour, and report the conclusion that justifies this work: the buying pattern so formed does not generate superior returns. The empirical basis is records from United States brokerages between 1991 and 1999, prior to trading by mobile phone and to the absence of commissions, so what is assumed transferable is the mechanism and not the measured values. The centrality of attention in this process is such that it became measurable through the volume of internet searches (Da, Engelberg, and Gao 2011).

The three indicators identified by the literature are all computed by this system, but the status of each differs. News and the extreme movement are the two triggers described in Chapter 1 and can, on their own, give rise to an alert. Volume is computed and presented as context, and does not interrupt the user. The reason for this asymmetry is empirical and is reported in Section 5.7: combining volume with the remaining signals improved triage in only one of three budgets considered, a result that does not support introducing a third route of interruption. The idea of fusing signals of distinct origins into a single indicator comes from a real-time information panel with that architecture (*World Monitor* 2026), suggested by the co-supervisor of this work, and enters the dissertation in the condition in which it was evaluated, that is, as a measured and not adopted alternative.

On the side of financial institutions, the adoption of artificial intelligence has ceased to be experimental, as documented by the survey of firms in the sector conducted by the Cambridge Centre for Alternative Finance (2026). That survey does not measure the population of products aimed at the retail investor, and therefore does not support claims about it; the characterisation of that population rests on what the providers themselves declare, and is the object of the following section.

2.2 Financial information tools for retail

This section examines the solutions currently available, assessing each one against a fixed criterion, namely the three questions stated in Chapter 1. Table 2.1 presents the result of that assessment.

Table 2.1: The categories of existing tool assessed against the three questions of Chapter 1. The classification *partial* indicates that the tool supplies raw material for the answer and leaves the interpretation to the user.

Tool	Is it unusual?	Company or market?	Has it happened?	Cost
Brokerage application	no	no	no	included
News aggregator	no	no	no	free
Sentiment application	partial	no	no	free or low
Automated portfolio manager	no	no	no	percentage of the portfolio
Professional terminal	yes	yes	yes	not published
This work	**yes**	**yes**	**yes**	**free**

The table makes the essential point. The row that answers all three questions exists and is the most expensive of all. The problem is therefore not scientific in nature, but one of access: the answers are known and are conditional on a subscription incompatible with the investor profile in question.

Three categories deserve individual examination, since each fails for a distinct reason and those reasons inform decisions taken in the following chapters.

The brokerage application is the most immediate instrument of access to the price, and it is the one whose public documentation describes the smallest set of indicators. It displays the day's percentage change and a price chart. It does not indicate whether that change is large for that particular stock, does not separate the company component from the market component, and has no memory of comparable events.

The price alerts of these applications share the limitation in a more pronounced form. They fire when a fixed threshold is crossed, a criterion that ignores the volatility proper to each stock and therefore produces frequent alerts in agitated companies and rare ones in calm companies. They further communicate that a level was crossed, and never why. This observation determines the methodological option described in Section 3.3, which replaces the absolute threshold with a measure relative to the recent norm of each company, and whose difference is quantified in Section 5.2.

Sentiment applications classify news as positive or negative and present the result of that classification. They supply raw material for the first question and go no further. Section 5.3 additionally presents a measurement of our own according to which the topic of a news item and the direction of the movement that follows it coincide less often than intuition suggests, which limits the reach of a sentiment score as an indicator of impact.

Automated portfolio managers answer above all a different question, that of the allocation of capital across a set of assets. The literature recognises genuine, though not uniform, benefits: D'Acunto, Prabhala, and Rossi (2019) measure an improvement in diversification and a reduction in volatility among investors who held fewer than five stocks before joining, whereas for those who already held more than ten the effect on diversification is close to nil. The containment of behavioural biases is the most consistently reported benefit (Cardillo and Chiappini 2024; D'Acunto, Prabhala, and Rossi 2019). The difference with respect to this work is one of object and not of transparency: an automated manager recommends a portfolio and executes it, whereas this system makes no recommendations and delivers evidence about a concrete event. Moreover, providing investment advice is a regulated activity, a boundary that the present work avoids by design, as discussed in Section 3.8.3.

There are, finally, recent products that claim to answer exactly the question treated in this dissertation. Robinhood Cortex, announced in March 2025, states on the provider's own page that it is intended to answer the question of why a stock is going up or down on a given day (Robinhood Markets 2025). Google Finance's key moments, presented in June 2026, propose to explain why a stock moved (Google 2026). This is the same question, posed by organisations with incomparably greater resources and user bases.

The difference lies in what the respective pages declare to be delivered. Both describe a summary in natural language, and neither describes the presentation of individual past cases, with dates, measured similarities and observed behaviour of the price after each of them, which is what this work delivers. The distinction claimed is not one of fluency, but of verifiability, and it concerns what the pages declare: the products were not tested, so nothing is asserted about what they actually deliver.

The most reasonable objection to the work consists in asking why a general-purpose artificial intelligence assistant would not solve the same problem. No comparison with any such assistant was carried out, so what follows is a design argument and not a result. An assistant without external access to market data lacks three elements. The first is the recent volatility of the stock, which allows one to determine whether a change is large. The second is the separation between the market component and the company-specific component. The third is a base of past cases over which to perform a retrieval.

On this last point, a language model recalls cases seen during training, which does not constitute retrieval and is not verifiable by the reader. There are large models specialised in financial text, whose capability is documented (Li et al. 2023). The output of any of them remains a statement that is read with ease and not checked.

The explainability literature gives reasons to distrust it. Lipton (2018) argues that interpretability does not designate a single property and that a system should declare which of them it offers. Rudin (2019) argues that an explanation produced over an opaque model is an approximation that may mislead. Neither deals with text generated by language models, so the transposition is the author's and what follows from it is a design reason and not a result of the literature. This is why language never produces facts in this system: the numbers come from the components that computed them, and the text is composed from those, as described in Section 4.5.

2.3 Anomaly detection in financial time series

This section situates the first decision of the system in the field of anomaly detection and states the families of method considered.

Anomaly detection is an established area, with a consolidated taxonomy that separates methods by the way they define normality (Chandola, Banerjee, and Kumar 2009; Pang et al. 2021). Three families are relevant for financial time series.

The statistical family defines normality from the local distribution of the data themselves, estimated in a rolling window, and flags as anomalous the observations that depart from it beyond a threshold. The explanation of a detection coincides with the computation that produced it, which makes it fully communicable to the user. In exchange, it assumes that the distribution of returns is locally stable.

The unsupervised learned family defines normality from the geometric structure of the data. Isolation Forest (F. T. Liu, Ting, and Zhou 2008) identifies as anomalous the points that a random partition isolates with few cuts; the Local Outlier Factor (Breunig et al. 2000) identifies them by their reduced density relative to that of their neighbours. Both dispense with labels and capture multivariate patterns that a rule over a single series does not capture, at the cost of the decision ceasing to be statable in one sentence.

The deep family learns a model of normality through a network (Pang et al. 2021). It offers greater representational capacity, requires larger volumes of data, and departs from the explainability requirement that structures this work.

A practical constraint conditions this whole choice. In neighbouring financial domains, such as fraud detection, unsupervised methods occupy a central position precisely because labelled anomalies are scarce and the patterns change over time (Ahmed, Mahmood, and Islam 2016). The same condition holds here: there is no dataset in which it has been recorded, day by day, which movements would justify an alert. This absence determines the way in which Chapter 5 can evaluate the component, and is the reason why the evaluation rests on a criterion derived from the data themselves rather than on human judgement.

The assumption of local stability on which the statistical family rests is confronted and not hidden. Conditional volatility models, introduced by Engle (1982) and generalised by Bollerslev (1986), formalise volatility clustering, that is, the tendency of calm and agitated periods to occur in runs. Both papers formalise the phenomenon over inflation series, so its transposition to stock returns is not assumed: Section 5.2 measures it on the data of this work, comparing the equally weighted window with a variant that assigns greater weight to recent observations. A rolling window follows the phenomenon in part, since the norm against which each day is compared is re-estimated daily from the near past. The choice does not eliminate the problem, and the measurement referred to quantifies what it costs to keep it.

The comparison between the statistical family and the learned family does not remain at the level of argument. Chapter 5 runs both on the same data and reports the result obtained.

2.4 Text representation and semantic retrieval

This section surveys the ways of representing text that make it possible to compare news by meaning, and states the criterion for evaluating retrieval that follows from them.

Comparing news by meaning depends entirely on the representation adopted, and that representation went through three generations, sketched in Figure 2.1 and detailed in Table 2.2.

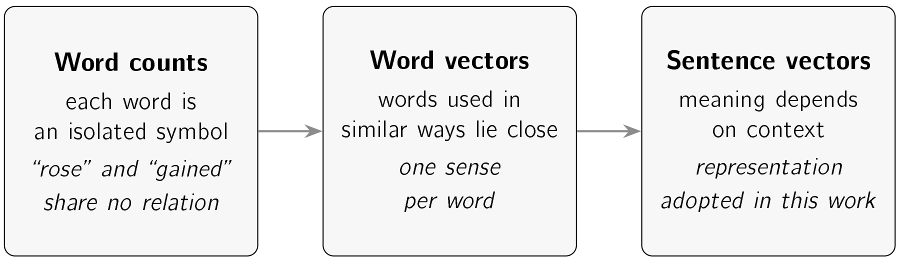

Figure 2.1: The three generations of text representation. Each one resolves a limitation identified in the previous one, and the third is the one that allows two sentences to be compared directly by the position of their vectors.

The first generation represents a document by the frequency of the terms it contains. The vector space model (Salton, Wong, and C. S. Yang 1975) is transparent and computationally cheap, and it is blind to synonyms: the terms "rose" and "gained" are, for this representation, symbols without relation. Within the same generation there is a second line, coming from a distinct probabilistic framework, that of ranking functions of which BM25 (Robertson and Zaragoza 2009) is the current representative; both remain strong baselines in information retrieval and share that blindness. In financial text there is also a specialised variant, the domain lexicons, which correct the fact that generic sentiment dictionaries misread words that are common in financial language (Loughran and McDonald 2011). They are entirely transparent, and are limited to the vocabulary that was compiled.

The second generation represents each word as a vector in a space where proximity reflects similar use, learned from local context (Mikolov et al. 2013) or from global co-occurrence (Pennington, Socher, and Manning 2014). This representation brings synonyms together, resolving the previous limitation, and introduces a new one: each word receives a single vector, regardless of the sentence in which it occurs.

The third generation removes that limitation. The Transformer (Vaswani et al. 2017) made it feasible to process the complete sentence with long-range dependencies, and Bidirectional Encoder Representations from Transformers (BERT) (Devlin et al. 2019) produced contextual representations, in which the vector of a word depends on those surrounding it. An additional step is needed for the problem treated here: comparing two sentences with a model of that nature would require processing them jointly, once per pair, which is not practicable over a base with tens of thousands of cases. Sentence-BERT (SBERT) (Reimers and Gurevych 2019) resolves this by producing one vector per sentence, computed only once, so that the comparison becomes an arithmetic operation between vectors that already exist.

The contribution of SBERT that matters to this work does not lie in the architecture but in the training objective: the vectors are learned so that the distance between two of them reflects similarity of meaning. The authors report that an encoder of the BERT family used directly to compare sentences produces weak representations for that purpose, at times falling behind the simple average of word vectors (Reimers and Gurevych 2019). It follows that an encoder tuned to another task does not inherit this property by virtue of having been trained in the appropriate domain.

Table 2.2: The approaches considered for representing financial news, with the representation each one produces and the implications of adopting it.

Approach	**Representation**	**Strengths and limitations**
Term counts (Robertson and Zaragoza 2009; Salton, Wong, and C. S. Yang 1975)	Weighted word frequency	Transparent and fast; blind to synonyms
Financial lexicons (Loughran and McDonald 2011)	Word lists with assigned polarity	Fully transparent; limited to the compiled vocabulary
Word vectors (Mikolov et al. 2013; Pennington, Socher, and Manning 2014)	One fixed vector per word	Brings synonyms together; a single sense per word
Sentence vectors (Devlin et al. 2019; Reimers and Gurevych 2019)	One vector per sentence, sensitive to context	Compares sentences directly; larger model, less legible internally
Domain models (A. H. Huang, Wang, and Y. Yang 2023; Zhuang Liu et al. 2020)	As above, tuned to financial text	Domain vocabulary; superiority on this task is not automatic
Large models (Li et al. 2023)	Text generation over broad knowledge	High capability; expensive, opaque and outside the free-of-charge constraint

For financial text there are models trained specifically for the domain (A. H. Huang, Wang, and Y. Yang 2023; Zhuang Liu et al. 2020). The natural assumption is that they are superior on this task for that reason. The assumption is not automatically true, and Section 5.3 measures it rather than assuming it.

The work belongs to the wider literature of natural language processing applied to finance. The review of Kearney and S. Liu (2014) covers the state of that area up to 2014, dominated by domain lexicons and by supervised classifiers over word counts; the contextual sentence representations used in this work are later and do not appear in that review. There is also a substantial line of research that uses text to anticipate price movements (Ding et al. 2015; Xing, Cambria, and Welsch 2018). The present work stands outside that line by design choice: text is used here to locate comparable earlier cases, whose outcome has already occurred and has already been measured, and not to generate a forecast.

Representing each news item by a vector turns the identification of comparable cases into an information retrieval problem. The proximity measure adopted is cosine similarity, whose mechanics are described in Section 3.5. The operation is exact, since the whole case base is scanned and the result ordered, so it does not depend on any approximation. At substantially larger scales, exhaustive search can be replaced by an approximate index (Johnson, Douze, and Jégou 2021) without changing the representation, which is a path for growth and not a present necessity.

Since retrieval returns an ordered list, its evaluation resorts to order-sensitive measures (Manning, Raghavan, and Schütze 2008). The one adopted in this work is precision at the first k results, that is, the fraction of those k results that are relevant. The definition of relevance, which is where an evaluation of this nature is decided, is established in Section 3.5 together with the baselines against which the result is compared.

2.4.1 Retrieval with generation, and why the system stops before it

The architecture in current use for answering a question from a body of documents is retrieval-augmented generation, in which a retriever selects passages and a language model writes the answer from them (Lewis et al. 2020). The literature on the family is extensive and has been systematised (Y. Huang and J. X. Huang 2026).

The system described here executes the first half of that architecture and stops before the second, and the reason is the one that runs through the work. Retrieval returns verifiable objects: a headline, a date, a measured similarity and an observed price movement. Generation converts those objects into a statement in prose, whose faithfulness to what was retrieved is itself a property to be evaluated, and which the recipient of this system has no means to confront. The option does not follow from a technical limitation, and the distinction should be drawn clearly: adding generation is possible and cheap; what is not cheap is the evaluation that it does not displace the meaning of the evidence. No comparison with a complete architecture was carried out in this work, so what is asserted here is a design reason and not a result. Section 4.7 describes the grounded generation layer that was in fact built and evaluated, and the reason why it is not exposed.

2.4.2 Financial news corpora and sector labels

Evaluating retrieval over financial news requires a body of text aligned with prices, and that alignment is what distinguishes a usable corpus from a collection of headlines. The dataset used in this work is FNSPID (Dong, Fan, and Peng 2024), which brings together news and the corresponding price series and whose construction is documented.

Relevance between two news items is approximated, in this work, by belonging to the same sector. The option replaces a non-existent human annotation with an objective criterion, and carries the corresponding limitation. The validity of a grouping is measurable, by the separation between groups (Rousseeuw 1987) or by the agreement between alternative groupings corrected for chance (Vinh, Epps, and Bailey 2010). Neither of those measures was applied to the sector map adopted. What the label guarantees is objectivity and reproducibility, and not that the sector is the most informative partition of the corpus.

2.5 Event studies and return decomposition

This section presents the instrument with which the effect of an event on the price is measured, as well as the statistical correction on which the separation between the company component and the market component depends.

The instrument adopted is the event study, whose formulation goes back to Fama et al. (1969). The behaviour of the method over daily returns was examined by Brown and Warner (1985), who compare alternative ways of estimating the expected return and conclude that the current procedures, based on estimating the market model by least squares, remain well specified in short windows around the event. The reach of this conclusion is delimited by the authors themselves: the simplest variant, which subtracts only the historical mean of the stock, loses statistical power and may become misspecified when the event dates coincide. It is this result that motivates the order of magnitude of the one, three and five day windows adopted in this work. The transposition is not direct, and the difference is declared in Section 3.5, according to which the value presented to the user is the raw return and not the abnormal return that those authors study.

The existence of a statistical relation between published text and market movements is a long standing observation (Tetlock 2007). The appropriate verb is that of evidence of a statistical relation, and not that of a demonstration of causality.

One methodological point conditions the second question of the work, that of the separation between the company and the market. That separation requires estimating the sensitivity of each stock to the market, and the estimate is noisy, especially in volatile companies or those with a short history. Blume (1971) establishes that the coefficients so estimated tend to regress towards the mean between successive periods, that is, that an extreme estimate in one period tends to be less so in the next; from this followed the practice, current but subsequent to the paper itself, of shrinking the estimate with a fixed weight. That practice applies the same correction to precise and imprecise estimates. The alternative adopted in this work weights the shrinkage by the imprecision of the estimate itself (Vasicek 1973), so that a reliable estimate is practically preserved and a fragile one is brought closer to the reference value. The formulation is presented in Section 3.4.

The theoretical framing that justifies the refusal to forecast also belongs to this area. The efficient market hypothesis, in its semi-strong form, holds that prices already reflect publicly available information (Fama 1970), from which it follows that future movements should not be systematically predictable from that information. It is one of the reasons why the system measures what has already happened rather than projecting what is to come; the second, of a practical nature, is that only the past is verifiable by the reader.

2.6 Case-based reasoning

This section frames the third technique of the system within an established line of artificial intelligence and delimits the part of that line the work implements.

Case-based reasoning (Aamodt and Plaza 1994) approaches a new problem through the retrieval of similar earlier problems and the use of what is known about them, dispensing with exclusive recourse to general domain knowledge. The source organises the method into a cycle of four processes, namely retrieve, reuse, revise and retain, and considers all four necessary to the complete resolution of a problem.

The technique described in Section 3.5 implements the first two and stops deliberately before the rest: it retrieves similar cases and presents what was measured on them, without adapting that outcome to the present case. Adaptation would produce an expectation about what is going to happen, which is precisely the forecast the work refuses. The partial implementation of an established cycle is, in this case, a consequence of the founding constraint and not a shortcoming.

The option finds additional support on the side of the reader. Research on explanation in the social sciences establishes that people look for contrastive and selective reasons, and not for an exhaustive enumeration of everything that contributed to an outcome (Miller 2019). A small set of concrete cases, with dates and outcomes, corresponds better to that expectation than a coefficient. It is for that reason that each alert presents a few precedents and the exact quantities that support the decision, rather than the totality of the values computed.

There is a second requirement, on the same side, that conditions not the content of the explanation but the frequency with which it is delivered. A recipient exposed to excessive warnings stops processing them, a phenomenon documented in domains where the cost of ignoring them is high (Ancker et al. 2017). The consequence is that the decision policy is not incidental but part of the design. An alert that is not read is worth the same as an alert that was not sent, and that is why Chapter 4 treats the suppression criteria with the care it devotes to the methods.

2.7 Explainability and trust in automated systems

This section situates the explainability requirement that runs through the work and presents the cautions that the literature associates with the presentation of explanations.

The demand that a system show how it reached a conclusion is an area in its own right, with consolidated reviews (Adadi and Berrada 2018; Barredo Arrieta et al. 2020). The area divides into two routes, shown in Figure 2.2: transparent models, interpretable in themselves, and *post-hoc* methods, which explain an opaque model already trained and which can be organised by the type of object they explain (Guidotti et al. 2018).

The two most widely used *post-hoc* methods operate in distinct ways. LIME fits a simple model in the neighbourhood of an individual prediction, locally approximating the behaviour of the opaque model (Ribeiro, Singh, and Guestrin 2016). SHAP attributes a prediction to the variables that produced it using Shapley values (Lundberg and S.-I. Lee 2017). Both are generic and both are useful.

Both share, however, a property that is decisive in this context. Faced with decisions of serious consequence, Rudin (2019) argues for abandoning opaque models in favour of models interpretable by construction, on the grounds that every *post-hoc* explanation is itself an approximation, and an approximation can lead to mistaken readings. The user receives in that case two objects whose relation they are not in a position to assess, namely a decision and an explanation that may not correspond to it.

This work follows the conservative route. Instead of explaining an opaque model *post hoc*, it resorts to components in which the explanation and the computation coincide. A departure

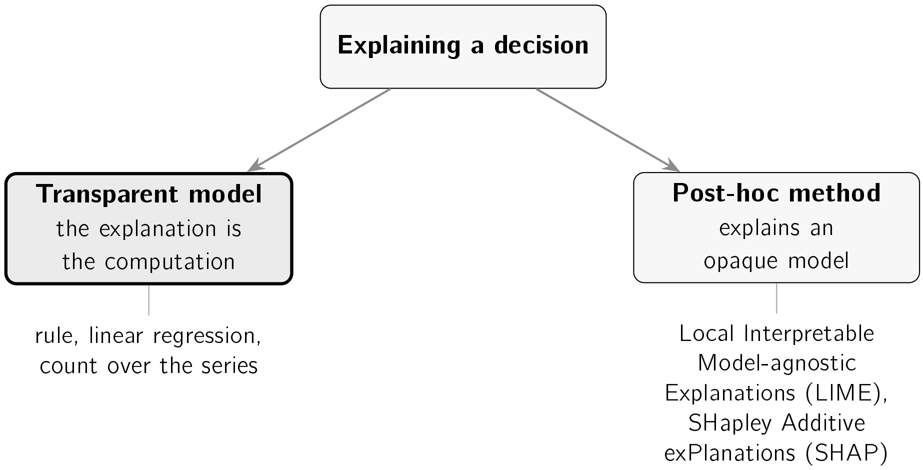

Figure 2.2: The two routes distinguished by the literature (Barredo Arrieta et al. 2020; Guidotti et al. 2018): building models that are transparent by design, or explaining an opaque model after training. This work follows the first.

from a recent norm, an additive split whose sum reproduces the observed movement, and a set of earlier cases with the outcomes measured are objects presentable in full. The explanation is not a second object produced over the decision: it is the decision, stated at length.

Two cautions from the literature close this section, and both condition the organisation of Chapter 5.

The first is that interpretability does not designate a single property (Lipton 2018), so a system should declare the type of interpretability it offers rather than claiming the term without qualification. This work offers transparency of mechanism, that is, the user can follow the computation that led to the decision, and does not offer a causal explanation of the event in the world.

The second concerns the effect of the explanation itself on the person who receives it. Trust in automated systems fails in two symmetric directions, with the user ignoring a correct warning or accepting an incorrect one, so the objective is not to maximise trust but to make it appropriate to the real capability of the system (J. D. Lee and See 2004). Bansal et al. (2021) add that the presence of an explanation increases the probability of accepting the system's answer whether it is right or wrong, that is, that the observed gain does not come from the person becoming better at distinguishing the two cases. The design consequences that follow are described in Section 3.8.3.

Furthermore, claims of interpretability demand rigorous evaluation and are not established by declaration (Doshi-Velez and Kim 2018). It is for that reason that Chapter 5 measures what is measurable by automatic procedure and explicitly identifies the part that remains unevaluated for lack of a controlled study with users.

2.8 Machine learning in a production environment

This section brings together the literature relating to the learned component of the system, organised according to the three questions its deployment raises.

The first question is that of determining against what stronger reference a simple model should be compared. Ensembles of trees trained by gradient boosting (Friedman 2001) capture interactions and non-linear effects that a logistic regression does not capture, and therefore play that role in Chapter 5. It is not claimed that they constitute the best existing method, since that would be a claim about an entire field.

The second question concerns the honesty of the value presented. A score is only interpretable as a probability if it is calibrated, and the raw scores of many classifiers are not; *post-hoc* calibration over a held-out validation block corrects that distortion (Niculescu-Mizil and Caruana 2005). The same source does not, however, favour the option taken in this work: it identifies logistic regression as already well calibrated to begin with and concludes that, above roughly a thousand calibration cases, isotonic regression matches or exceeds the sigmoid. The deployed model is a logistic regression with a validation block far above that threshold, so both statements apply to it. The option is justified by measurement and not by authority: the comparison of the two variants over the same validation set is reported in Table 5.2.

An alternative route was considered and not adopted. Conformal prediction returns a set of labels with a distribution-free guarantee, valid in finite samples (Angelopoulos and Bates 2023; Vovk, Gammerman, and Shafer 2005). Two clarifications determine that it is not applicable under the conditions of this work. The coverage of the current variant is marginal: it holds on average over cases, and not case by case. And the guarantee requires exchangeability between the calibration block and the test block, an assumption that the chronological split with embargo of Section 3.6 refuses by construction. Section 6.5 returns to the method as a path to explore.

The third question is that of the persistence of performance over time. A deployed model faces change in the distribution of the data it observes, and the concept drift literature establishes the taxonomy of that change and the strategies for detecting and adapting to it (Gama et al. 2014). This work measures drift on its own data rather than assuming it absent.

There is, finally, a cost that is captured by no quality metric. D. Sculley et al. (2015) establish that machine learning components accumulate hidden maintenance, in particular through entanglement between parts and through unstable, undeclared data dependencies. Chapter 6 returns to this point with an example measured in the present system, in which part of the learning cycle stopped without any error being recorded.

2.9 Synthesis and positioning of the work

This section locates the work with respect to the review presented and states the scope decisions that follow from it.

The review allows the space in which the work is inscribed to be delimited precisely, and that space is not a technical void. Every component used exists, is published and is evaluated, as Table 2.3 makes explicit.

The gap is therefore not one of method, but of integration under constraint. The products that answer the three questions are professional terminals, unavailable free of charge or at a price accessible to the retail investor. Their price is not published in a citable form, and it is unavailability, and not a concrete value, that supports the argument. The free products

Table 2.3: Every component of the system exists in the literature in isolation. What does not exist is their combination under the zero-cost constraint, with the evidence delivered to the user at the moment the statement is presented.

Question	What the literature offers	What the target audience lacks
Is it unusual?	Anomaly detection, statistical and learned (Chandola, Banerjee, and Kumar 2009; F. T. Liu, Ting, and Zhou 2008)	A measure relative to the company's own norm, communicated in accessible language
Is it the company or the market?	Event study and sensitivity estimation (Brown and Warner 1985; Vasicek 1973)	Availability outside a professional terminal
Has it happened before?	Sentence representation and retrieval (Manning, Raghavan, and Schütze 2008; Reimers and Gurevych 2019)	A case base with measured outcomes, and not recalled ones
Why believe it?	Explainability and the trust literature (J. D. Lee and See 2004; Rudin 2019)	The evidence delivered alongside the statement and verifiable at the moment of reading

that claim to answer the third deliver a statement without the evidence that supports it. And the academic components that would solve each part were evaluated in isolation, over reference corpora, and not as a system in continuous operation against real sources.

This observation determines the distinction that runs through the work and that Chapter 4 takes up when justifying the absence of a language model in the composition of the messages. A generated sentence is a statement: it presents itself to the reader in a fluent register and gives no means of confronting it with what supports it. A sentence composed from computed objects is a statement accompanied by the evidence, in which every value presented points back to the measurement that produced it and the reader can confirm each element at the moment of reading. This is the property the system seeks to preserve, and it is what fixes the nature of the contribution. It is not the formulation of a new method. It is the selection, integration and critical evaluation of existing methods in a working system, under declared constraints, and with the results that do not favour the options taken included in the evaluation.

Four scope decisions follow from this positioning, and each one deliberately excludes a class of functionality.

Answer the three questions, and only those. No functionality is incorporated that does not serve one of them, which keeps the system at a size compatible with explaining it in full.

Produce no price forecasts. Neither direction, nor target, nor recommendation. Besides the verifiability reason stated in Chapter 1, the theoretical framing of Section 2.5 applies (Fama 1970).

Manage no portfolios and give no advice. The system does not know the user's positions and does not tell them what action to take, which avoids collecting personal data and keeps the work outside the boundary of regulated financial advice.

Use free sources exclusively. The constraint makes the system accessible to the audience it addresses and turns the resulting limitations into measured and reported quantities.

The last decision has a real cost that should be stated without mitigation. News arrives late and the coverage of events is incomplete. Chapter 4 quantifies both limitations and Chapter 6 assesses their consequences.

Chapter 3

Methods and materials

Chapter 2 established that the three questions of the retail investor have an answer in the literature but not in any free tool. This chapter describes the options taken to answer them. It first presents the research methodology and the datasets. It then describes the four methods selected, one for each decision the system has to take. It indicates the tools employed. And it finally states the technological and social challenges of the domain.

3.1 Research methodology

This dissertation is framed within design science research (Hevner et al. 2004), in which the object of study is an artefact built to solve a class of problem and the knowledge produced resides both in the artefact and in the process of its evaluation. It is the appropriate framing when the contribution is a system, and it differs from a purely empirical study in that the object evaluated does not exist before it is built.

The design science literature establishes that the evaluation of the artefact must be rigorous and constitute a central activity of the process, and not a final confirmation step (Hevner et al. 2004; Peffers et al. 2007). This requirement is met through three commitments made before the experiments were carried out.

The first consists in comparing each component with a simpler alternative, selected and declared before the measurement is carried out, so that the result can be negative. The second consists in fixing the success criterion of each comparison together with the metric, and not after observing the values obtained. The third consists in evaluating the artefact twice: over held-out historical data and, subsequently, in operation, over the decisions the system takes autonomously in production. The third evaluation is the least common of the three and is the one that produces the result described in Section 5.6.

The requirement is not met in full, and the gap is declared. Utility, in the sense in which the design science literature employs the term, is utility for whoever uses the artefact, and that property is determined with users. What this dissertation measured is the faithfulness of the explanations produced, a property that belongs to the system; utility, which depends on the relation between the system and whoever reads it, remains unmeasured.

The system takes four distinct decisions, shown in Figure 3.1. The first three correspond to the three investor questions stated in Chapter 1; the fourth is not a question of the investor, but of the system itself, and exists because answering the first three does not imply interrupting the user.

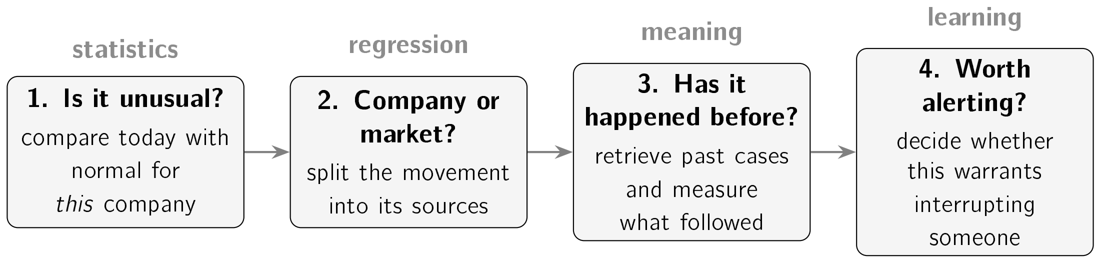

Figure 3.1: The four decisions of the system and the family of techniques associated with each. The first three correspond to the questions stated in Chapter 1. Each column corresponds to a section of this chapter, between Section 3.3 and Section 3.6.

3.2 Datasets

This section describes the datasets used in the training and evaluation of the system, as well as the records produced by the system in operation.

The historical layer rests on FNSPID (Dong, Fan, and Peng 2024), a public corpus of financial news that associates a date and a company with each headline. The corpus contains no prices, no market reaction and no indication of relevance: that information is built in this work, by aligning each headline with the corresponding price series. From the alignment comes the training set of the triage model, with 79,753 examples.

The evaluation of the anomaly detector uses a distinct dataset, made up of three years of daily quotes. The choice of a separate set is deliberate: the evaluation requires variety of volatility across companies, whereas the list monitored in production is a product decision. Evaluating the detector on exactly the set it was built to serve would produce a measurement with no informative value.

All prices used are closes adjusted for splits and dividend distributions. The adjustment is not a detail in the period considered, which contains a four-for-one split of Apple and two of Tesla: on unadjusted series, the detector would flag falls of thirty to eighty per cent on days with no event at all.

Table 3.1 brings together the datasets used, and Table 3.2 the company sets, whose counts differ for reasons described in the respective column.

3.3 Detection of unusual movements

The first decision of the system consists in determining whether the movement observed on a day is unusual for the company in question. A fixed threshold rule, of the kind that flags changes above three per cent, is inadequate for this purpose, because the same percentage corresponds to events of very different rarity depending on the volatility of the company.

The method adopted belongs to the statistical family of the anomaly detection taxonomy (Chandola, Banerjee, and Kumar 2009), which defines normality from the recent distribution of the data themselves. The movement of the day is compared with the recent behaviour of the same stock and expressed in standard deviations:

Table 3.1: Datasets used in the work, with the time span and the size read from the files. The last three rows correspond to the system in operation.

Dataset	Span	Size
Historical		
Triage training set	2018-01-02 to 2023-12-18	79,753 examples, 14 companies
Prices for the detection evaluation	2023-06-01 to 2026-06-01	15 companies, ≈ 750 days each
Operation		
Deployed precedent base	one year of own collection	38,214 cases with measured impact
Alerts delivered	2026-07-09 to 2026-08-13	367 messages
Triage decision log	2026-07-22 to 2026-08-20	36,925 decisions

Table 3.2: Company sets used. The counts differ because each set serves a distinct purpose. The sets are not nested: the list monitored in production contains two companies absent from every historical set.

Set	Companies	Purpose
Extended sector map	17	Universe covered by the movement decomposition
Canonical historical map	15	Relevance label in the retrieval evaluation
Prices for detection	15	Variety of volatility in the detector evaluation
Triage dataset	14	Result of the alignment between news and prices
covered by the training block	13	Companies actually present in training
covered by the test block	9	Companies present in the evaluation
List monitored in production	12	Product decision

$$z_t = \frac{r_t - \mu}{\sigma} \tag{3.1}$$

where r_t is the logarithmic return of the day, $r_t = \ln(P_t/P_{t-1})$, and μ and σ are the mean and the standard deviation of the returns of the twenty preceding days. The use of logarithmic returns follows the current convention in financial series, since it makes returns additive over time and symmetric between rises and falls. The values presented therefore differ from those displayed by a stock application: a logarithmic return of +19.82% corresponds to a price change of +21.92%.

The window slides each day, and the day being assessed is excluded from the window that defines its own norm, as shown in Figure 3.2. Without this exclusion, the day would contribute to the mean and to the standard deviation against which it is compared, attenuating its own anomaly. The same discipline prevents the use of any information posterior to the day being assessed.

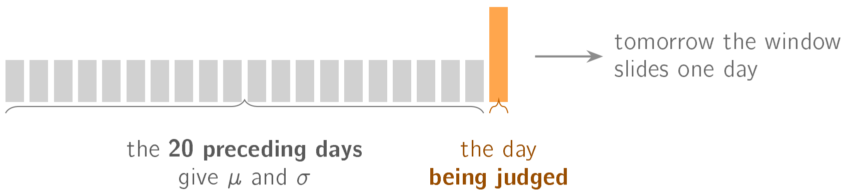

Figure 3.2: The window that establishes the usual behaviour of the stock. The day being assessed is **left out** of the computation of the norm against which it is compared, a separation that prevents the use of information unavailable at the moment of the decision.

When the standard deviation of the window is zero, Equation 3.1 is undefined. The system distinguishes two cases: holding the value constant does not constitute an anomaly, whereas a departure from that flat norm is flagged without a z value being assigned to it, stating that the preceding variance was zero. This distinction prevents the guard against division by zero from silencing precisely the first movement after a constant period.

Applying the method to the largest movement of the period studied, recorded by Tesla on 24 October 2024, produces a daily return of +19.82% against a mean of −0.92% and a standard deviation of 2.72% in the twenty preceding days, which corresponds to $z = +7.61$. The value does not follow from the absolute size of the movement, but from the fact that it exceeds by seven and a half times the usual oscillation of the stock in those weeks.

3.4 Decomposition of the observed movement

The previous technique quantifies the size of the movement, but does not identify its origin. The second decision of the system consists in splitting the observed return into its market, sector and company-specific components:

$$r = \underbrace{\beta_m r_m}_{\text{market}} + \underbrace{\beta_s \tilde{r}_s}_{\text{sector}} + \underbrace{\varepsilon}_{\text{company}} \tag{3.2}$$

The variables r_m and $\tilde{r}_s$ denote, respectively, the daily return of the market index and that of the sector, whereas β_m and β_s quantify the historical sensitivity of the stock to each of those sources. The company-specific component is defined as the residual, so that the sum of the three parts always reproduces the observed movement. The market is represented by the exchange-traded fund that tracks the broad United States index, and the sector by the corresponding sector fund, chosen for being the most liquid of their class and for having a long history in free sources.

Three aspects determine the validity of the split.

The first is the orthogonalisation of the sector factor. A sector fund and the market index move in strongly correlated ways, and introducing them jointly in the regression produces unstable coefficients through collinearity. The factor used is therefore not the sector return, but the residual of a first regression of the sector on the market.

The second is the temporal restriction of the estimation. The window ends on the day before the day being explained, by the same discipline applied to detection.

The third is the treatment of noise. A sensitivity estimated over twenty observations is imprecise, and extreme estimates in agitated windows produce meaningless splits. The finance literature documents that estimated betas regress towards the mean between successive periods (Blume 1971), from which followed the practice of shrinking the estimate with a fixed weight. That practice applies the same correction to precise and imprecise estimates. The alternative adopted weights the shrinkage by the imprecision of the estimate itself (Vasicek 1973):

$$\beta = w\hat{\beta} + (1 - w)\,\beta_{\mathrm{ref}}, \qquad w = \frac{\sigma^2_{\mathrm{ref}}}{\sigma^2_{\mathrm{ref}} + \mathrm{SE}^2(\hat{\beta})} \tag{3.3}$$

When the fit is precise, the standard error is small, w approaches unity and the estimate is practically preserved; when the window is noisy, the standard error grows and the estimate falls back on the reference value. The adjustment is gradual and not dichotomous. The original formulation estimates β_{ref} and σ^2_{ref} from the cross-sectional distribution of betas; in this work they are fixed, and each has its own justification. For the market, $\beta_{\mathrm{ref}} = 1.0$ is adopted, which corresponds to the assumption that the stock follows the index in the absence of reliable local information, and is the value towards which the cross-sectional mean of the betas tends by construction of the index itself. For the sector factor, already orthogonalised with respect to the market, $\beta_{\mathrm{ref}} = 0.0$ is adopted, since the residual of an orthogonalisation has zero mean and the absence of sector exposure is therefore the neutral assumption. The common standard deviation of 0.5 determines how quickly the transition between the sample estimate and the reference value takes place: it is fixed so that a window with a standard error of that same order receives approximately equal weight from the two origins, which places the point of balance within the range of standard errors actually observed in twenty-day windows. Estimating these values from the sample was not done, and the reason is declared: twelve to seventeen companies do not constitute a representative cross-section, so a prior estimated over them would have an empirical appearance without being one.

Three safeguards complete the estimation. Below ten usable observations the regression is not attempted, unit sensitivity to the market and zero to the sector being assumed, a situation that is declared to the user. An estimate whose absolute value exceeds ten is treated as a numerical failure of the window and not as an economic sensitivity, falling back in the same way; the guard is meant for degenerate windows and does not replace the shrinkage, which is the mechanism that handles noise. Finally, the constant term is recomputed over the already shrunk coefficients, since the value returned by the regression is centred on the raw coefficients and reusing it would decentre the model actually reported.

Figure 3.3 presents the split of a real movement, in which the market and sector components act in the direction opposite to the observed movement.

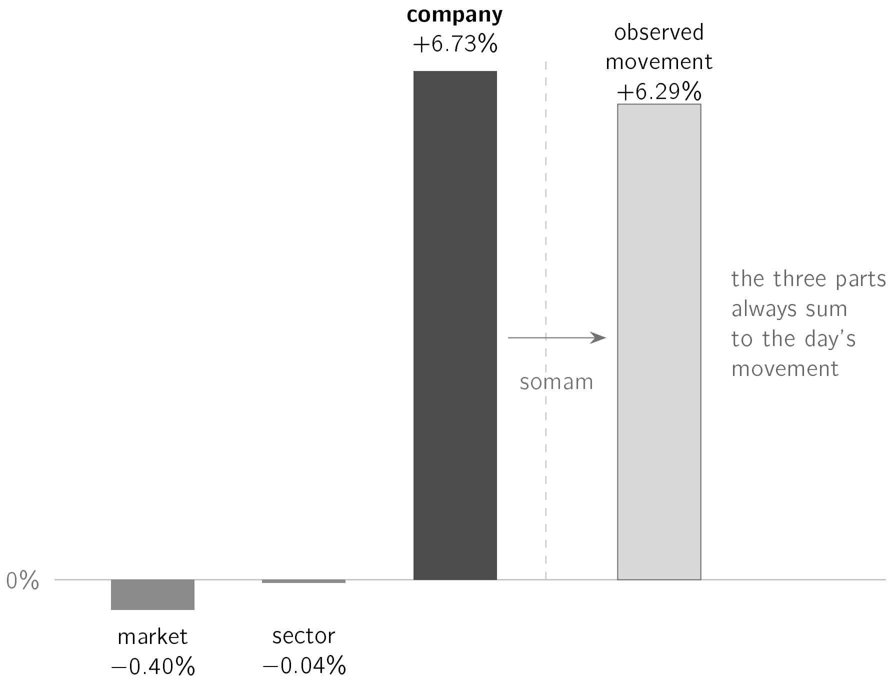

Figure 3.3: Split of one day of AMD, that of 15 August 2026, in logarithmic returns. The market and sector components acted downwards, so the rise is attributed entirely to the company. The equality between the sum of the parts and the observed movement holds by construction and, for that reason, establishes nothing about the quality of the fit, a point taken up in Section 5.5.

In this case, recorded by AMD on 15 August 2026, the logarithmic return of +6.2944%, corresponding to a simple price change of +6.50%, splits into −0.3992% of market, −0.0362% of sector and +6.7298% specific to the company, with $\beta_m = 2.0143$ and $\beta_s = 1.5888$ estimated over the twenty preceding days. The information conveyed exceeds the percentage alone: the stock rose despite the context acting in the opposite direction, and the market beta indicates a stock that historically amplifies the movements of the index.

The exact sum of the three parts does not constitute validation of the method. The equality holds by construction, since the specific component is defined as the residual, and a meaningless split would sum just as well. The informative measure of the quality of the fit is the coefficient of determination in the estimation window, whose value is reported in Chapter 5.

3.5 Precedent retrieval

The third decision consists in identifying past news items comparable with the news item under analysis and measuring the subsequent behaviour of the price. Search by word matching is inadequate for two symmetric reasons: news items with the same meaning may share no vocabulary, and news items with common vocabulary may deal with distinct subjects.

The representation adopted is a sentence embedding model (Reimers and Gurevych 2019), which converts each headline into a 384-dimensional vector whose position in the space reflects the meaning of the sentence. The relevant contribution of this model is not the architecture, but the training objective: the vectors are learned so that the distance between two of them corresponds to similarity of meaning. An encoder of the same family trained for another task does not inherit this property, an observation that Chapter 5 confirms experimentally.

The proximity between two sentences is measured by cosine similarity, that is, by the angle between the vectors and not by the distance between their endpoints, which makes the measure independent of the length of the text. The vectors are returned with unit norm, so the cosine reduces to the inner product, a property that allows the complete case base to be scanned without approximation. On two real pairs from the case base, two headlines about falls in technology and semiconductor companies obtain $\cos = +0.956$, and a headline about Peloton compared with another about the impact of the pandemic on the automotive industry obtains $\cos = -0.086$. The similarity threshold used in production is 0.45.

The ordering of the candidates is not, however, the cosine alone. Cases close in topic but distant in time describe a market regime that is no longer the prevailing one, so the ordering applies an exponential decay over the age of the case, in the form $\cos \times 2^{-\mathrm{age}/h}$, with a half-life h of one hundred and twenty days. A case one year old thus enters with an approximate weight of 0.12 and prevails only in the absence of more recent material on the same topic.

The decay orders and is not displayed: the value shown next to each precedent is always the cosine, since presenting a composite score under the name of similarity would produce a number the user cannot verify. The two decisions compose in series, so the 0.45 threshold applies to a set already selected by the ordering, and not to the neighbours with the highest cosine. Figure 3.4 represents the property this measure exploits: headlines with the same meaning occupy nearby positions even when they share no vocabulary, whereas headlines that share vocabulary and deal with distinct subjects end up far apart.

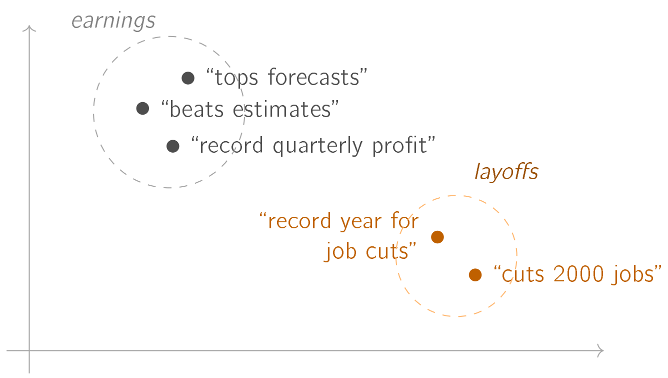

Figure 3.4: Each sentence corresponds to a point in the space. Sentences of close meaning occupy neighbouring positions even when they share no vocabulary, whereas sentences with common vocabulary ("record") and distinct subject end up far apart. The representation is two-dimensional to allow it to be read; the model uses 384 dimensions.

The measurement of the outcome adapts the event study (Brown and Warner 1985): with the day of the news fixed, the price change is measured at one, three and five days. Two detailed decisions condition the result. News published outside trading days is aligned with the first following trading day. The measurement begins at the close of the day of the news, and not at the open, a conservative option that avoids attributing to the news the part of the movement prior to its publication.

The value presented to the user is the raw return and not the abnormal return the source normalises. The option is deliberate and has a declared cost. The raw return is verifiable by the user in their own application, but it incorporates the movement of the market over the period. In periods of generalised decline, the precedents appear to have worse outcomes than those attributable to the news. The market is discounted where it is indispensable, in the construction of the label described in the following section.

3.6 News triage

The fourth decision determines which news items justify interrupting the user. Unlike the previous ones, it has no obvious analytical formulation, and constitutes the natural point for learning from the data.

The label formalises the question as the occurrence of an abnormally large movement after the news. The return of the stock over the following three days is measured, the movement of the market over the same period is subtracted, and the example is classified as positive when the absolute value of the residual exceeds two per cent. The use of the absolute value is deliberate: the model learns about materiality and never about direction, which preserves the founding constraint of the work.

Subtracting the movement of the market assumes unit sensitivity, which is an inconsistency with the technique described in Section 3.4, which refuses precisely that assumption. In a high-beta stock, the residual retains part of the movement of the market, and the label may classify as material a day on which only the market moved. All the families of model evaluated in Chapter 5 are trained and evaluated against the same label, so the comparison between them is internally consistent under this definition of target. That does not guarantee, however, that the ordering survives an alternative definition, since the error introduced may be correlated with the company and with volatility.

The model receives nine inputs: the volatility of the twenty preceding days, the momentum of the five preceding days, the return of the day itself, the length of the headline and five sector indicators. An additional variant incorporates the 384-dimensional vector of the headline. Figure 3.5 groups the inputs by the level of information each one describes, a distinction that Chapter 5 shows to be decisive.

Seven inputs describe the company and vary slowly or are constant; one describes the day and is identical for every news item of that day; and only one, the length of the headline, distinguishes two headlines of the same company on the same day.

The split of the data is chronological and not random, as shown in Figure 3.6. A random split would allow training on news posterior to that used in the evaluation, inverting the temporal order the system faces in operation. Between consecutive blocks there is a five-day embargo, necessary because the label of a news item depends on the following three days: without the embargo, the label of the news items at the end of a block would be determined by days belonging to the following block. The embargo discards 820 of the 79,753 rows, that is, 1.03%, leaving 28,574 examples for training, 17,710 for validation and 32,649 for testing.

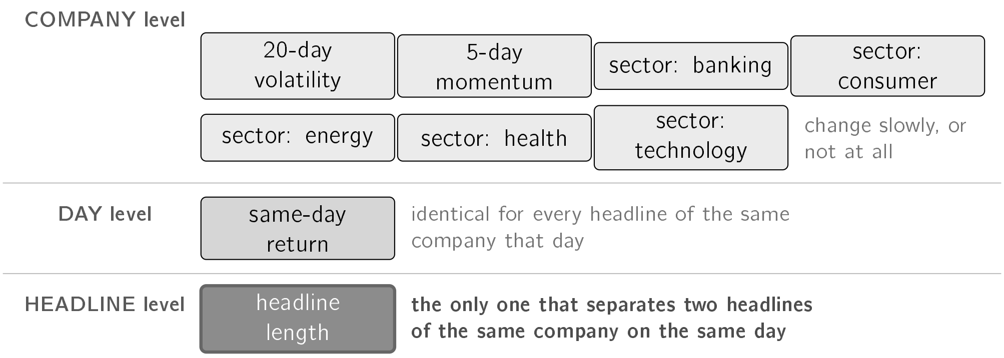

Figure 3.5: The nine inputs of the deployed model, grouped according to what each one allows to be distinguished. Seven characterise the **company** and vary slowly or stay constant; one characterises the **day** and takes the same value for every news item of that day; **only one** varies between consecutive headlines. From this composition follows the expectation that the model does not separate two news items of the same company on the same day, an expectation the ablation of Section 5.4.5 confirms.

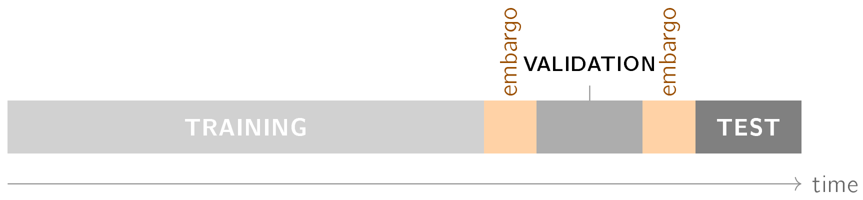

Figure 3.6: The split of the data follows **chronological order** and not a draw. Between consecutive blocks a five-day *embargo* is inserted, since the label of each news item depends on the following three days: without it, the end of the training block would be determined by days belonging to the following block.

The test block contains more examples than the training block. The cut is made on trading days, according to a proportion of seventy, fifteen and fifteen per cent, and the asymmetry results from the density of news, which grows markedly over the period covered by the corpus. Splitting by days, and not by number of examples, is necessary so that news items of the same day are not distributed across distinct blocks.

The score produced by the model is not, in itself, a probability. Since the value is presented to the user, it is converted through *post-hoc* calibration, fitting a two-parameter sigmoid over the held-out validation block (Niculescu-Mizil and Caruana 2005):

$$p_{\text{calibrated}} = \frac{1}{1 + e^{-(a\, p_{\text{raw}} + b)}} \tag{3.4}$$

The learned values are $a = 3.700$ and $b = -2.313$. The negative sign of b indicates that the model, without calibration, produces values systematically above the observed frequency. Part of this behaviour follows from the reweighting of the classes during training, which shifts the scores upwards by construction.

The calibration is fitted on a block whose prevalence of positive cases differs from that of the test block, a consequence of the chronological split. The effect is measurable and is reported in Chapter 5. The transformation is monotonic and preserves the ordering between news items, which is what the decision policy uses; it is the absolute value presented that is affected.

3.7 Tools and infrastructure

The system is written in Python 3.12. The dependencies that produce the results of this dissertation are pinned to exact versions, and not to ranges, so that the environment can be recreated on another machine: `numpy` 2.1.3, `pandas` 2.2.3, `scikit-learn` 1.9.0, `matplotlib` 3.11.0, `sentence-transformers` 5.6.0 and `torch` 2.12.1 in the CPU variant. No result requires a graphics processing unit, a consequence of the options described in this chapter.

All data sources are free. Prices come from a chain of alternative providers consulted in a fixed order, since no free source is sufficiently reliable on its own; the system always records which one answered. News comes from three complementary providers, whose comparison is presented in Chapter 4. Delivery uses the Telegram bot interface.

3.8 Technological and social challenges

This section states the risks that a system with these characteristics raises and the design decisions that respond to each of them.

3.8.1 Privacy and data protection

An investment portfolio reveals income, risk aversion and life plans, information that the General Data Protection Regulation treats as deserving reinforced protection when it allows a person to be profiled.

The design decision that responds to this risk is minimisation: the system does not know the users' positions and does not ask for them. There is no portfolio, there are no positions and no amounts, and the public channel does not identify recipients. An alert about a company is identical for every reader. The cost of this option is the impossibility of personalisation, since the system alerts on a fixed list of companies and not on the positions of whoever reads it.

There is one exception worth naming. Users who choose to receive personalised alerts through the Telegram bot are associated with two elements: the conversation identifier assigned by the platform and the list of companies selected. No name, contact or amounts are stored. The processing rests on the express request of the user, who begins it by subscribing and ends it through a command that deletes the record.

3.8.2 Security

An autonomous system that consumes external services holds credentials. The keys reside in environment variables and in the platform's secret store, never in the repository, and the text sent to the logs passes through a function that masks parameters with a credential name. This second safeguard responds to the fact that the error messages of communication libraries reproduce the address of the request, and that some interfaces carry the key in that address.

The sentence encoder is fetched on demand and checked against a checksum fixed in the code. Without that verification, a corrupted or substituted file would silently modify the similarity criterion used by the system.

3.8.3 Ethical and social questions

The founding constraint of the work is also its main ethical safeguard. A system that announced the future behaviour of the price would issue a forecast not verifiable at the moment of reading; a system that indicated instruments to buy would enter a regulated activity. This boundary is one of design, drawn out of prudence and not from an analysis of the applicable law: there was no legal opinion, and operating the system as a service would require obtaining one first.

The guarantee applies to what the system writes and not to what it reproduces. Third-party headlines are quoted in full, since they constitute the fact that triggers the alert and allow verification at the source, and some are themselves speculative. From this follows a limitation without a solution within the scope assumed: the relevance filter determines whether a headline is about the company, and not whether it is factual. The distinction between news and opinion would require a classifier the system does not have.

Repeated interruption is the second risk. The effect is documented in clinical decision support systems, where Ancker et al. (2017) measure that the probability of accepting an alert decreases as the volume of alerts received increases, and that the effect is accentuated by repetition of the same alert. The domain is distinct and what transfers is the mechanism: whoever is interrupted frequently stops distinguishing the interruptions, from which point a correct warning and an incorrect warning have the same effect. The system responds with a daily alert budget, a rising threshold for repeated alerts of the same company, and suppression of news that reproduces a story already communicated.

The third risk follows from the explanation itself. Trust in automated systems fails in both directions, with the user ignoring a correct warning or accepting an incorrect one (J. D. Lee and See 2004), and the presence of an explanation increases the acceptance of wrong answers (Bansal et al. 2021). The system therefore presents past cases individually, with date and outcome, rather than summarising them into an average liable to be read as a forecast. That this option produces appropriate trust is a hypothesis and not a result, since the controlled study needed to verify it was not carried out.

3.8.4 Use of artificial intelligence tools

This subsection declares, as the Statement of Integrity refers, how generative artificial intelligence tools were used in this work and for what purpose.

The conception is the author's: the problem, the research questions, the constraints that define the system — free sources only, explainability first, and the refusal to forecast prices —, the design of each experiment, the success criterion fixed before running it, and every decision about what to build, keep or discard. No conclusion in this document was delegated.

The execution was assisted, and to a substantial extent. Those tools wrote and restructured a considerable part of the system's code and of its tests, implemented the evaluation procedures, drafted and edited prose in this document, and conducted reviews that found defects in the software and in the text. Where a measurement contradicted a claim of the author, the claim was withdrawn rather than defended, and Appendix A records those that were withdrawn.

The content of this document was reviewed by the author. It, the errors that remain in it, and its defence are his responsibility.

Chapter 4

Implementation: InvestiGator

This chapter describes the realisation of the system: its architecture, the way each method of Chapter 3 is put into effect, the policy that decides what is communicated to the user, the construction of the message delivered, and the life cycle of the learned component.

Chapter 3 presented four methods in isolation. Composing them into a working system introduces problems that none of them presents on its own: the unavailability of the sources, the volume of candidates per cycle, the coherence between the values computed and the text presented, and the silent degradation of the components over time. The description that follows is organised around those problems.

The system developed is called InvestiGator and is in continuous operation, delivering alerts to a messaging channel and making the record of its decisions available in a web application. The values presented in this chapter were read from the log of the system in operation, and the dates on which each measurement was taken are indicated alongside each one.

4.1 System architecture

The architecture of InvestiGator was defined according to the following requirements, concerning cost, modularity, verifiability and resilience:

- to operate exclusively on free data sources, with no subscriptions and no per-request costs;
- to isolate each method in a component with a single responsibility, replaceable without changing the others;
- to guarantee that any value presented to the user originates in the component that computed it, and not in a second independent derivation;
- to tolerate the unavailability of any external source without interrupting the service;
- to record every decision taken, including the decisions not to communicate.

The system is organised into five components, shown in Figure 4.1: detection, retrieval, triage, explanation and delivery. The decomposition of the movement is a sixth component, activated only on the path started by a price movement, since the split applies to the return of the day and there is no object to split when the event that triggers the assessment is a news item.

One construction rule runs through the whole implementation: the text delivered to the user is composed from the objects the components produced, and not written independently.

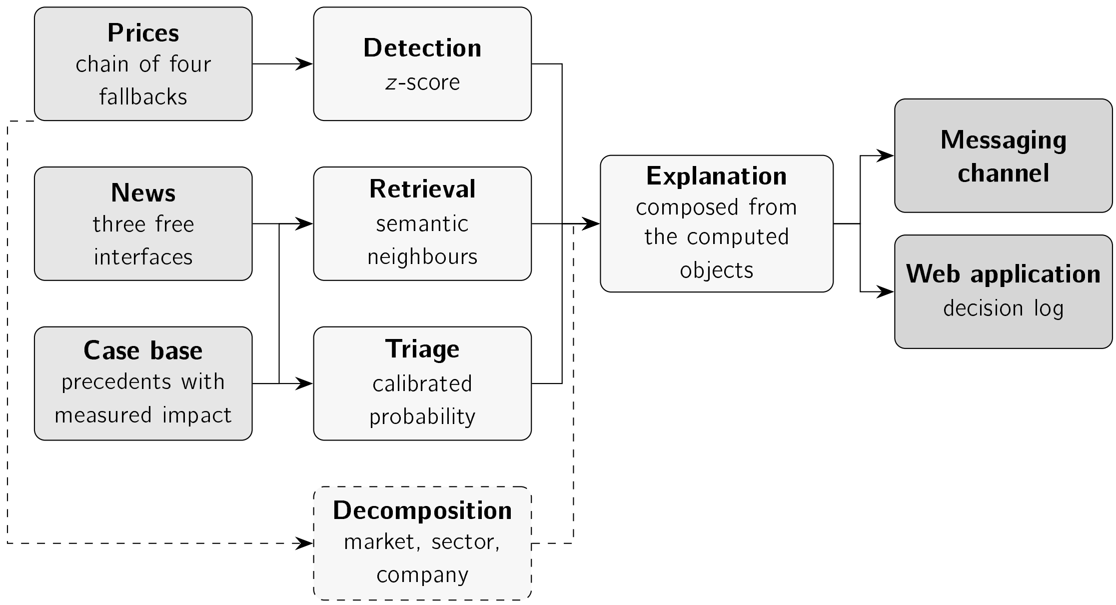

Figure 4.1: Components of the system and their connections. On the left, the data sources; in the centre, the methods of Chapter 3 and the component that composes the text; on the right, the two delivery channels. The decomposition is dashed because it is the only component that does not take part in every assessment.

When the detector computes a z value equal to $+7.61$, it is that value the message presents. The alternative, which would consist in keeping one place where the text is written and another where the quantity is computed, admits divergence between the two without that divergence being observable. The constraint eliminates that possibility by construction.

4.1.1 Communication between components

The components communicate by direct passing of objects within a single process, and not by network protocol. The decision follows from the usage profile: the system serves a small list of companies, with a sixty-second assessment cycle, and the volume does not justify the coordination cost of a distributed architecture. The separation between components is ensured by explicit interfaces, which preserves replaceability without introducing communication latency.

Two exceptions operate across process boundaries. Delivery uses the bot interface of the messaging channel, by HyperText Transfer Protocol (HTTP) request. The web application that exposes the decision log runs as a separate process and reads the same data repository, without writing to it. This second separation guarantees that consulting the log does not interfere with the assessment cycle.

4.2 Data acquisition

The zero-cost constraint determines the acquisition decisions. A retail investor who considers brokerage costs high does not subscribe to a financial information terminal, and the usefulness of the system for the recipient defined in Chapter 1 depends on the system operating without that cost. Free sources present, however, three systematic limitations: they block access for excess of requests, they answer with an error without warning, and they publish late. The decisions described in this section respond to those limitations.

4.2.1 Quotes

Prices are obtained through a chain of four alternatives, consulted in a fixed order, the first that answers with valid data being adopted. The identifier of the source that served each value is recorded together with the value, since the provenance of a quantity is part of its explanation.

The order of the chain is neither arbitrary nor definitive. One of the sources was demoted after observing the content it returned, and an additional candidate was rejected for answering with an anti-robot check, which makes it unusable in automated operation. Both decisions resulted from measurement and not from declared preference.

The redundancy of the chain has a consequence that goes beyond availability. Without it, a day on which the single source blocks access is indistinguishable from a day on which the market did not open, and the system has no way of separating the two situations.

4.2.2 News

News is obtained per company, from three free interfaces. The material received includes a high volume of unusable content: automatic lists of the day's largest movers, texts that mention the company only in passing, and press releases from law firms. A relevance filter, applied on entry, requires the headline to name the company explicitly and rejects the known patterns of automatically generated text.

The use of three sources rather than one was assessed by measurement, over twelve companies and three days. The criterion adopted is not the number of headlines returned by each source, but the number of headlines that survives the relevance filter, since material rejected on entry never comes to exist for the system. Table 4.1 presents the result.

Table 4.1: Material delivered by each free source after the relevance filter is applied. The *exclusive* column indicates the headlines that one source brings and none of the others brings. That value is an upper bound, since the comparison between headlines is made by textual matching after normalisation, and two sources that describe the same event in different words produce two headlines counted as distinct.

Source	Relevant	Precision	Freshness	Coverage	Exclusive
Finnhub	432	**35%**	15.8 h	**12/12**	401
Alpha Vantage	141	24%	**9.3 h**	**12/12**	119
Polygon	429	27%	52.6 h	8/12	**418**

The three sources have complementary profiles, and it is that complementarity, and not the aggregate volume, that justifies using them together. Finnhub has the highest labelling precision and covers all the companies. Alpha Vantage has the lowest median age of the news, at 9.3 hours against 15.8, which makes it the relevant source for reducing the delay of the alerts. Polygon contributes the largest number of exclusive headlines, with a median age of 52.6 hours, which makes it suitable for feeding the case base and unsuitable for alerting in useful time. That is the use assigned to it.

The three sources together produce 970 distinct relevant headlines, against the 432 obtained with Finnhub alone. The gain concentrates on the companies with the lowest individual coverage, which are precisely the cases in which the system previously had no material to comment on.

Aggregation does not, however, guarantee a reduction in latency in general. The three sources publish with delays of their own, and the gain in a concrete case depends on which of them observed the event first. The median freshness improves, and that improvement is measured; the claim that alerts now arrive earlier would require a new end-to-end latency measurement over an extended period, which was not carried out.

4.2.3 Historical base and deployed base

Precedent retrieval requires a body of past cases with a known outcome. The system uses two distinct sets, with distinct purposes, and their separation matters for reading the results of Chapter 5.

The first is FNSPID (Dong, Fan, and Peng 2024), a public set of financial news aligned with the corresponding prices. From the alignment described in Section 3.2 result 79,753 examples between 2018 and 2023, each with the subsequent price movement. It is on this set that the methods are evaluated.

The second is the case base the deployed system consults, and it results from merging two origins that should not be confused. The first is a reconstruction of one year of history, obtained by running the collection and maturation components over the past, with 38,214 cases; the separation from the log of the running system is deliberate, since a replayed case is not an observed case. The second is the live base, fed by the scan, with 11,445 cases. Retrieval scans both together. The vectors are stored as a single-precision matrix mapped on disk. The format is not indifferent: the same representations in the language's native structures exceeded the memory available in the execution container, and the change of format was the condition that made their use possible.

4.3 The path of an event

The system admits two distinct paths, depending on whether the event that triggers the assessment is a news item or a price movement. The two converge in the explanation component and use the same decision policy, described in Section 4.4.

4.3.1 Path started by a news item

The path started by a news item comprises nine stages, shown in Figure 4.2. Five of those stages are decision points at which the candidate can be discarded, in which case the system does not communicate.

The operation of the path is illustrated by an alert delivered on 12 July 2026, concerning Meta, with the headline *"Mark Zuckerberg Said Meta's AI Bets 'Haven't Come to Fruition Yet' as Shares Fell 5%"*. The headline names the company and matches none of the automatic text patterns. The date falls within the two-day interval required. The three closest precedents are Microsoft of 2023-11-07, with similarity 0.46, Nvidia of 2023-08-02, with 0.51, and Nvidia of 2023-09-21, with 0.46; the best of them is above the 0.45 threshold.

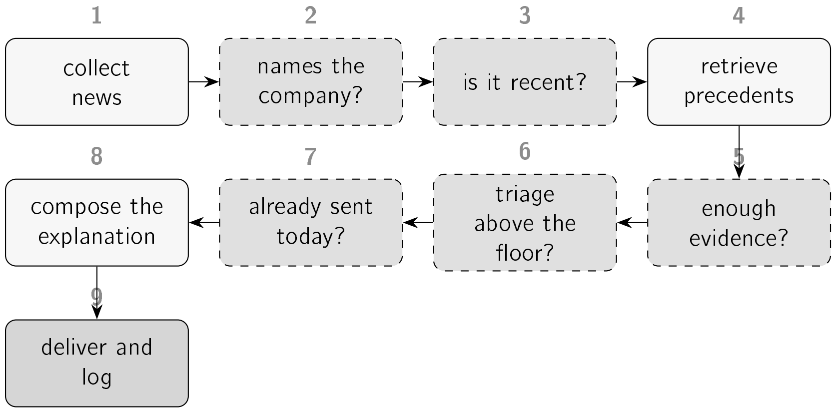

Figure 4.2: The nine stages of the path. The dashed boxes correspond to decision points at which the candidate can be discarded. Section 4.4 quantifies the proportion of candidates eliminated at each one.

The triage model assigns a probability of 54%. And the alert is the first concerning the company on that day. The message is then composed and logged.

The precedents retrieved in this case belong to companies distinct from the one the alert concerns, and are presented nonetheless. The option is deliberate and follows from the formulation of the second research question: comparability is established by the topic of the news and not by the identity of the company. News items concerning investment in artificial intelligence at large technology companies are cases comparable with one another.

The outcomes of the three precedents diverge, with movements of +2.70%, −3.87% and +5.05%. The system presents the three individually, and not their average, precisely for that reason. The average hides the most relevant piece of information in the set, which is the absence of a common outcome across similar cases.

4.3.2 Path started by a price movement

The second path is substantially shorter. The z-score is computed by two routes. After the close of each session, the system assesses the return of the closed day of each company on the watched list against the norm of the twenty preceding days, as described in Section 3.3. During the session, it assesses the return in progress, obtained from the current quote against the previous close, against the same daily norm, formed in that case by the twenty most recent complete days, since the day in progress does not yet belong to the series. The second route brings the detection forward without changing the definition of normality, and is what allows a movement to be communicated while the session is under way; it is also the one that does not depend on the historical series source, whose bar for the day may arrive late. On either route, crossing the threshold is a sufficient condition for assessment.

It is on this path, and only on this one, that the decomposition of the movement intervenes. A line is added to the alert with the split computed according to Section 3.4, in the form *"NFLX* −2.11% *today* = −0.08% *market* · −0.32% *sector* · −1.71% *company itself"*, followed by a sentence that names in plain language the dominant part. When estimating the sensitivity to the market is not possible, the line is presented all the same, declaring that unit sensitivity was assumed and that the split is indicative.

Two capabilities were added to the path after the system entered operation, both from observing its behaviour in production. The first consists in assigning severity levels: a z value of 1.6 and a value of 3.2 came to be described by distinct terms, instead of producing the same wording. The second consists in comparing with the remaining companies of the same sector in the same session, since an alert that reports a fall without indicating that the whole sector fell conveys incomplete information.

The second correction resulted from reading the first thirty alerts delivered. In nine of them the text was internally contradictory: one line indicated that the movement appeared to be common to the sector, and the split presented two lines below attributed the largest part to the company. The two lines were correct in isolation, and the contradiction resulted from the comparison with peers considering only the direction of the movement of the neighbouring companies, and not its magnitude. The comparison came to consider both.

4.4 Decision policy

Communicating every candidate assessed is equivalent to communicating nothing. A channel that delivers fifteen messages a day stops being read, and the system loses the usefulness that justifies its existence. The decision points of Figure 4.2 respond to this constraint.

4.4.1 Observed selectivity

The joint effect of the decision points is considerable. Over the news dated between 1 and 3 September 2026, the system assessed 743 distinct headlines from twelve companies and delivered 15 alerts, which corresponds to a ratio of roughly one in fifty. The window is short, and the decision log that supports it keeps three days, which fixes its limit. Figure 4.3 presents the distribution by company.

The budget is, in the configuration in operation, the constraint that actually binds. Over the production decision log, and on each of the six days measured, between 77.7% and 99.8% of the assessments end there, against a maximum of 4.6% at the similarity threshold and of 14.0% at the same-day repetition check. The reading is not that the remaining gates are useless: it is that the reduction in volume that matters has already occurred before them, in the relevance filter and in the selection of one headline per company per cycle, and that from there on the governed quantity is the daily total. It is also the direct consequence of the policy change described in Section 5.6: the model ceased to exercise a veto and came to rank within the budget, so the budget being the active constraint is the policy behaving as it was designed, and not a departure from it.

Two clarifications condition the reading of these numbers, and the second is the more important. The first is that the unit is the distinct headline: the system re-assesses the same headlines at every cycle, so the log contains 1,321 rows for the 743 headlines, and counting assessments would inflate the funnel. The second is that the total of fifteen alerts does not measure the raw material available, but the policy: the daily budget of five was used in full on all three days, so fifteen is the ceiling the policy imposes and not an observed quantity. What the distribution by company measures is the ordering within that ceiling, and that is not imposed: no company has a quota of its own.

The behaviour on a single day makes it possible to locate where candidates are eliminated. Figure 4.4 distributes the 5,060 assessments recorded on 15 August 2026 by the decision

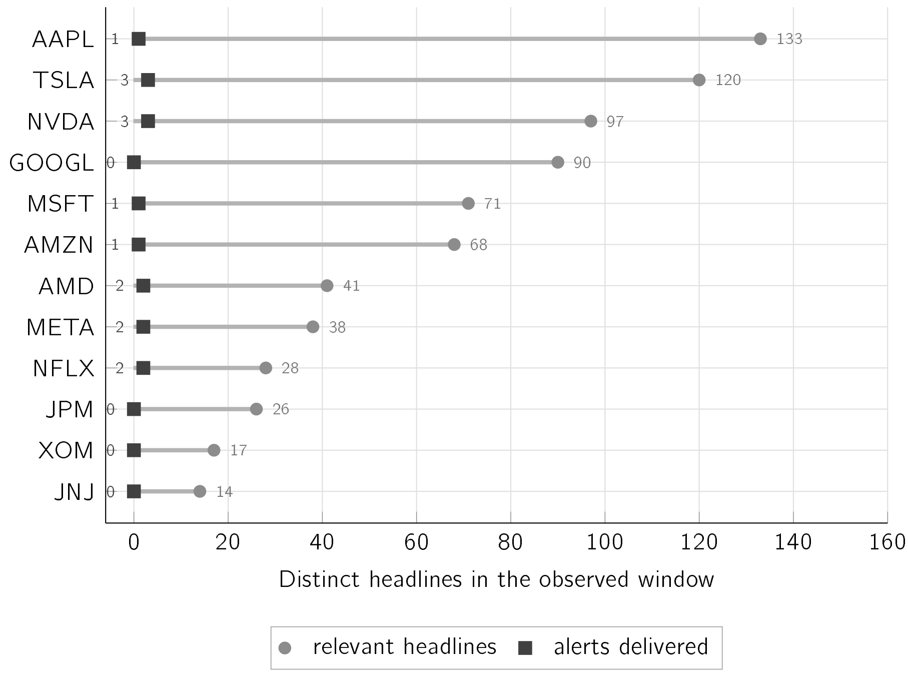

Figure 4.3: Distinct headlines assessed and alerts actually delivered, by company, between 1 and 3 September 2026. The total is 743 headlines and 15 alerts, and the two quantities are counted over the same window. The total of alerts is the ceiling the daily budget imposes, and not a measurement; what the figure shows is the distribution of that ceiling, which the policy does not fix. Eight of the twelve companies receive at least one alert, and the company with the most headlines assessed is not the most alerted. Section 5.6 examines the effect of this policy over a much larger window.

point at which each one ended. The log read corresponds to part of the day and not to the complete day, so the values should be read as proportions.

Reading these values exposes an instrumentation defect. The 333 assessments that crossed every decision point do not correspond to 333 distinct candidates approved for delivery. The system re-assesses the same headlines at every sixty-second cycle, and a headline already delivered that day crossed every decision point again at every cycle. Delivery was prevented at the end, by a duplication check that, unlike all the others, was not recorded as a stage.

The consequence matters for the purpose of the log. The view built to make the system's decisions inspectable reported 333 approvals on a day on which five messages were delivered. The duplication check became a stage with an identifier of its own, like the rest, and there is an automatic test that fails if it is omitted again. The values of Figure 4.4 predate that correction.

4.4.2 Thresholds and their provenance

Each decision point uses a constant, and the origin of those constants is heterogeneous. Table 4.2 distinguishes the values derived from an analysis from the values chosen, a distinction that matters because the two cases admit defences of a different nature.

The similarity threshold requires separate treatment, since it is the least grounded of the seven. In a scan measured in July 2026, when the watched list had ten companies and

Table 4.2: Constant associated with each decision point and the origin of its value. The first point uses no constant, since it corresponds to a textual matching rule. The daily budget acts after the decision points shown in Figure 4.2 and constitutes, in the configuration in operation, the mechanism that actually limits the volume.

Decision point	**Value**	**Provenance**
Unusual movement	$\lvert z \rvert \geq 1.5$	Chosen, and distinct from the value of 3.0 under which the detector is evaluated in Chapter 5. The evaluation fixes a demanding value in order to characterise the method; the deployment adopts a more permissive value as a product decision, offset by the severity levels that distinguish 1.5, 2.0 and 3.0 in the wording presented. The two values answer different questions.
Names the company	—	Textual rule, defined after inspecting the material that was crossing the initial filter.
Is it recent	2 days	Chosen, so as to cover a weekend.
Enough evidence	0.45	Chosen. It is the point that eliminates the largest number of candidates.
Triage above the floor	50%	Derived from a cost analysis, which produces 0.49 under the condition of symmetric cost between an uncommunicated movement and a false alarm; the deployed value is the rounding of that result. In the configuration in operation it exercises no veto: it ranks the candidates of the same cycle.
Daily budget	5 per day	Chosen. The five slots are assigned in order of arrival and not to the five highest-scoring candidates of the day, since the decision is taken cycle by cycle, without knowledge of later candidates. It is additional to a limit of two messages per company per day, which remains in force and is checked after this one. The two answer distinct problems: the per-company limit prevents saturation by a single name, and the global budget prevents twelve companies at two messages each from producing twenty-four interruptions in a day.
Stepped floor	second alert: 0.64	The configuration keeps the derived values 0.49 and 0.64, and the first is explicitly ignored when the global budget is active: the first daily alert of each company has no triage floor. The mechanism thus acts only as a limiter of repetition within the same company.

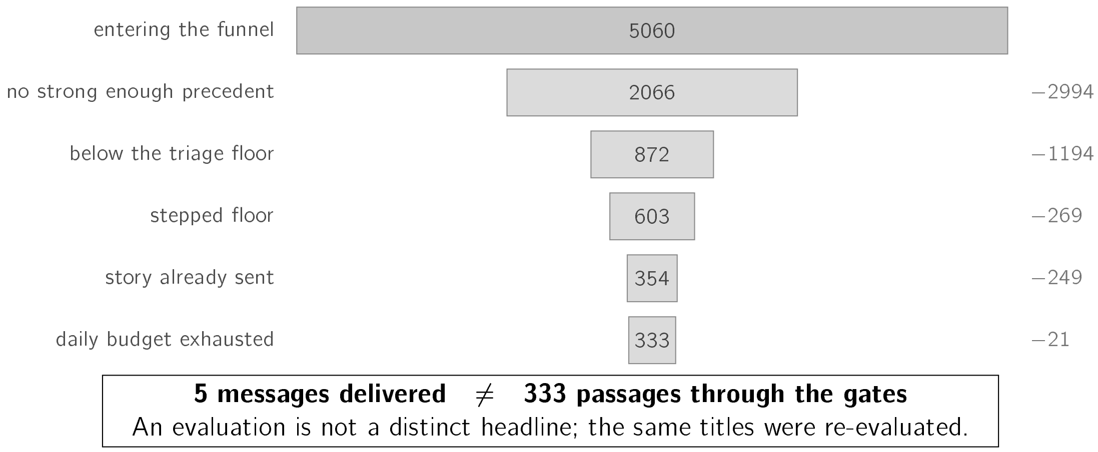

Figure 4.4: The 5,060 assessments recorded during part of 15 August 2026. Each band represents what remains after the gate indicated, and the value on the right what that gate eliminated; the eliminations sum to the 5,060 assessments and leave 333 passages. The five messages delivered are a separate count, not a category of assessments. The daily budget was used in full. The date is also that of the day on which the triage floor ceased to exercise a veto and came to rank, so the band of 1,194 assessments eliminated at that point belongs to the configuration prior to the one described in Table 4.2; on the following days that point stops eliminating candidates.

the daily budget did not yet exist, seven candidates were eliminated at this point, four of them by differences of 0.04 or less, and it was then the point that eliminated the most. It ceased to be: under the policy in operation, the order of magnitude of what each point eliminates is that of Figure 4.5. The hypothesis that higher similarity values correspond to more informative precedents was tested, and the hypothesis is not confirmed: the agreement in direction between the past case and the case under assessment is at the level of chance on both sides of the threshold. The threshold controls the volume of alerts and ensures thematic coherence, and does not select cases with greater predictive value.

4.5 Construction of the explanation

The message delivered is not a single sentence, but a structure with four parts, always present and always in the same order:

1. the observed fact, expressed in values, with the source indicated;
2. the past cases retrieved, presented individually, with date, similarity and subsequent movement;
3. the caveat that the cases presented are observed history and not a forecast;
4. the estimated probability of a market reaction, with no indication of direction.

The order responds to two constraints. The fact precedes the statistic because an alert that opens with a z value loses the reader on the first line, and the recipient defined in Chapter 1 does not have the training that would make that value immediately interpretable. The estimate occupies the last position because it is the only quantity in the message that

DECISION POINT	VALUE	ORIGIN	OF WHAT IT ELIMINATES
Names the company	`rule`	☐ chosen	before the log
No news at all	--	☐ chosen	0 to 3.9%
Is it recent	2 days	☐ chosen	0 to 3.5%
Enough evidence	0.45	☐ chosen	0 to 4.6%
Triage above the floor	50%	■ derived	no longer vetoes
Same story	--	☐ chosen	0 to 1.8%
Already sent today	--	☐ chosen	0 to 14.0%
Stepped floor	0.64	■ derived	never acted
Daily budget	5 per day	☐ chosen	77.7 to 99.8%

The point that eliminates almost everything is one of the chosen ones, and it is so on every day measured. Of the two whose value was derived from a cost analysis, the triage floor ceased to veto on 15 August 2026 and the stepped floor remains armed but never acted, because the budget runs out before a company reaches a second alert. The relevance filter acts before a stage log exists, so the fraction it eliminates is not observable here.

Figure 4.5: Each decision point, the origin of the value it uses and the fraction of assessments that ends there, measured over the production decision log across six days, between 19 and 21 August and between 3 and 5 September 2026. The column presents the range across those days and not a single value, since the log keeps only three days at a time and the fractions vary with the day. The unit is the assessment and not the news item: the system re-scores the same headlines at every sixty-second cycle, so these values measure where the system's time is spent and not how many distinct stories each point eliminated. The distinction between a derived value and a chosen value is the same as in Table 4.2. The reading the figure allows, and that the table alone does not give, is that the reduction in volume is not made by the points whose value was derived: it is made by the budget, whose value was chosen, and so it is on each of the six days. The values come from the decision log published with the system.

refers to the future, which ensures that a reading interrupted before the end retains facts and history.

On the path started by a price movement, the first part also carries the statistical characterisation in plain language, since in that case the observed fact is the movement itself. On the path started by a news item, that position is occupied by the headline and the link to the article.

The number of precedents presented is likewise a design decision. The system displays three of the cases retrieved and not all of them, and the reason is not available space. Research on explanation establishes that people look for selective and contrastive reasons, and not exhaustive enumerations of the factors that contributed to an outcome (Miller 2019). A list of thirty cases would be more complete and less usable. The same principle determines that the message displays the quantities that support the decision and not the full computation that produced them.

Figure 4.6 reproduces an alert exactly as it was delivered, copied from the channel log without editing. The headline it quotes is reproduced verbatim, since translating or paraphrasing quoted material would break the verifiability that the system exists to preserve.

```
[1] News alert for AMZN (Amazon) (2026-08-13)
  "Prediction:  Amazon Will Join Apple in the $4 Trillion Club Before 2030"
[2] 3 similar past headlines.  Their 5-day move ranged -2.73% to -2.73%
  (average -2.73%):
  -2.73% in 5d · AAPL 2026-08-05 (8d ago) · "If You Invested $100 In Apple
    Stock 10 Years Ago, You Would Have This Much Today" (sim 0.56)
  -2.73% in 5d · AAPL 2026-08-05 (8d ago) · "Apple’s Selloff Is Overdone"
    (sim 0.51)
  -2.73% in 5d · AAPL 2026-08-05 (8d ago) · "Apple’s Buyback Is Retiring
    Less Stock Just As The Memory Bill Grows" (sim 0.47)
[3] 3 of 3 shown cases moved down - topic-similar past cases, not a
  prediction for this news (an observed pattern, not a forecast).
  Observed past outcomes after similar news, not a price prediction and
  not advice.
[4] Materiality estimate (learned triage):  57% chance of an unusually
  large move over the next few days, in EITHER direction - raised by
  recent volatility (20d) and sector; lowered by today’s own move and
  recent momentum (5d).  This estimates whether the market reacts, never
  which way:  it is not a price forecast and not advice.
```

Figure 4.6: Alert delivered on 13 August 2026, reproduced from the log without editing. The numbers in square brackets identify the four parts of the structure. Every value comes from the component that computed it: the similarity, from the retrieval engine; the five-day movement, from the impact measurement; and the probability of 57%, from the calibrated model. Part [1] contains a headline that is itself a forecast, with a target and a horizon; it is third-party content reproduced literally, and Section 3.8.3 discusses the reach of the no-forecast guarantee in that context. Part [2] displays the similarity as a raw value, faithful to what the engine computed and not interpretable by whoever reads it.

The probability presented on the last line is not the raw score of the model. It results from its conversion into a probability, by fitting a sigmoid over a held-out validation block (Niculescu-Mizil and Caruana 2005). The distinction is operationally relevant: a raw score orders the candidates without admitting interpretation in isolation, whereas a calibrated probability establishes that, among the headlines to which the system assigns 57%, approximately that fraction does show an unusual movement. It is the difference between a quantity usable only internally and a quantity that can be presented.

The guarantee has declared limits. Section 5.4 reports the mean error of this probability and records that the calibration is fitted on a block whose prevalence does not coincide with that of the block to which it is applied, which shifts the scale by about five percentage points in the optimistic direction. The ordering between candidates is not affected, since the transformation preserves the order; the absolute value announced is.

4.5.1 Disagreement between the claim and the evidence

The alert of Figure 4.6 presents three precedents with an identical impact of −2.73%, and an identical date. The coincidence is not accidental: the impact is measured per pair formed by company and day, so three headlines of the same company on the same day share the same value by construction. The wording *"3 of 3 shown cases moved down"* presents as agreement between three cases what is one day observed three times.

The extent of the problem was measured over the 247 news alerts with precedents delivered between 11 July and 13 August 2026. In 36.8% of them the cases displayed rest on a number of distinct days lower than the number of cases presented, and in 11.3% the three cases correspond to the same day. Of the 120 alerts that claim unanimity of direction, 23.3% rest on a single day.

The effect is circumscribed. No metric reported in Chapter 5 is affected, since precision at k counts headlines and direction agreement is measured pair by pair over the historical corpus. What is affected is the strength the alert claims before the user, and Chapter 6 records the fact as a limitation.

4.5.2 Web application

The same information is made available in a web application, which allows what was delivered and, above all, what was not, to be consulted. It is in that application that the decision log described in Section 4.4 resides. The design of the interface is not a contribution of this dissertation and is therefore not described; presenting it is justified by its being the only place in which the system's absence of communication is observable. Figures 4.7, 4.8 and 4.9 present, in that order, the state of the day, the evidence for one company, and the record of a decision not to communicate.

The company of Figure 4.8 is an illustrative case of the investor's second question, through the disagreement between the percentage change and the split. The stock recorded a fall of 0.30%, and the company-specific part was +0.53% — positive. The fall came from the market, at −0.61%, and from the sector, at −0.22%. The percentage change alone indicates that the company fell and leads to the search for an internal cause that does not exist; the split shows that the company rose against a falling market. The verdict classifies the day as ordinary: 204 of the last 250 trading days recorded an equal or larger change.

Figure 4.9 shows the same requirement on the side of silence. On that day, the seven companies that generated no alert all stopped at the same gate: the daily budget of five alerts, exhausted at 14:02 UTC. The important reading is not the number, but the form: the company crossed eight gates, each stated by the criterion it satisfied, and stopped at

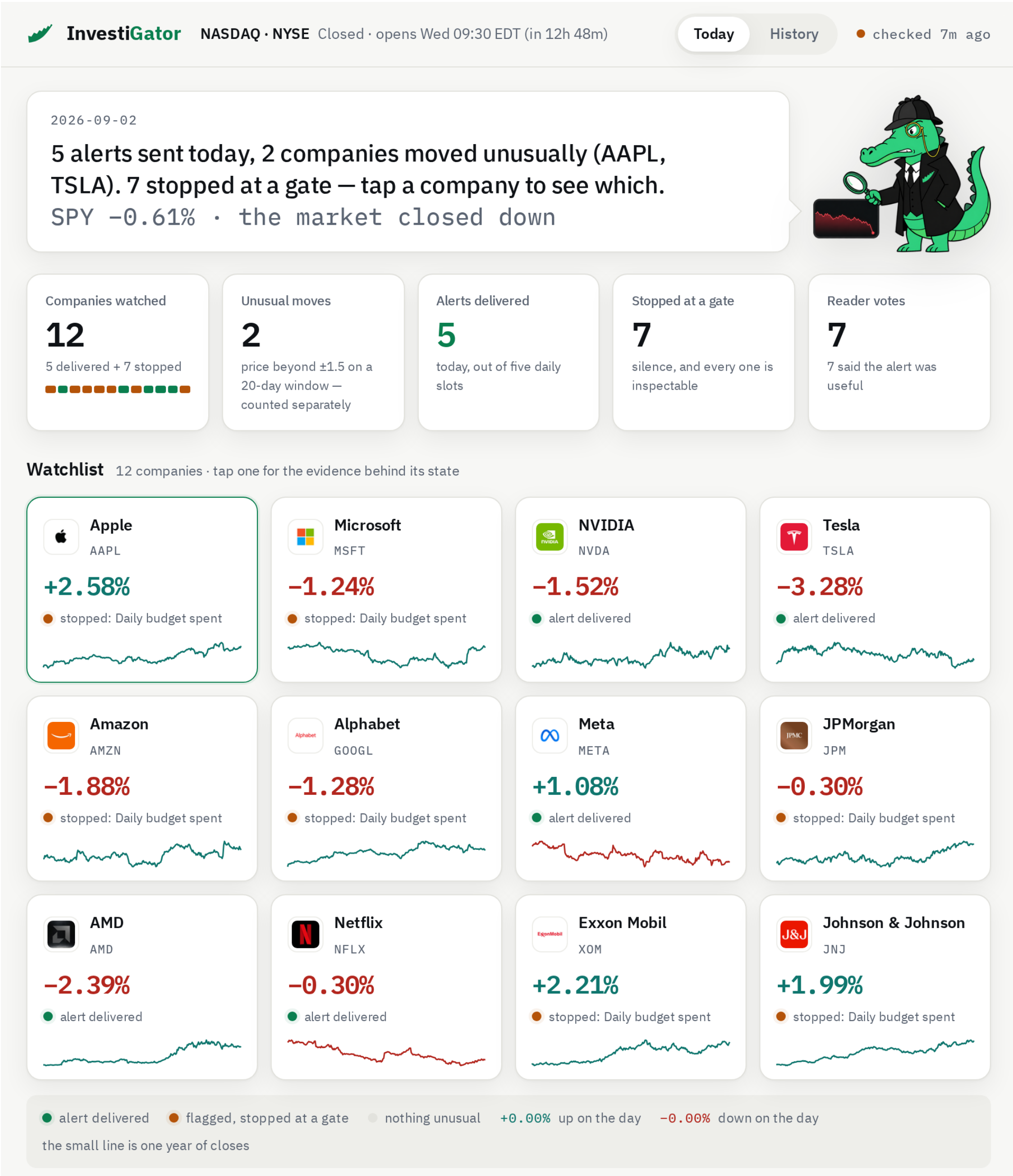

Figure 4.7: Web application on 2 September 2026, with the United States session closed. The five indicators split the twelve watched companies into five alerts delivered and seven decisions not to communicate; the count of out-of-the-ordinary movements measures something else, the price, and is therefore presented as a separate count. Each company declares its state on the surface, and every colour used has a legend, including that of the mascot, which follows the daily change of the market index used in the decomposition. The captures are generated from a snapshot of the application itself, and not collected by hand; the capture width is 960 pixels, since that is the width at which the interface text stays legible after being reduced to the text box of this document.

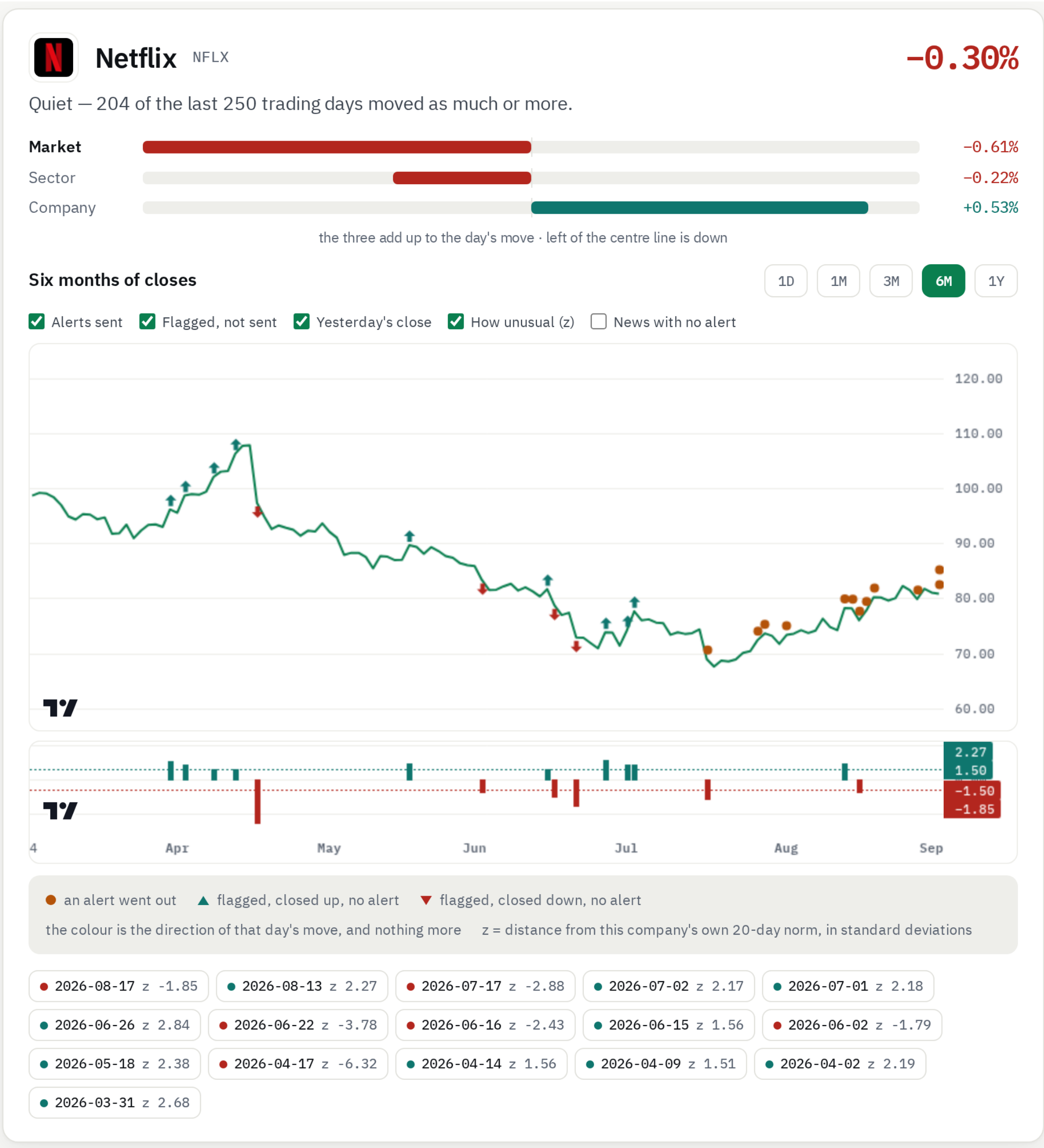

Figure 4.8: View of one company: the verdict in words before any number, the split of the movement into the three parts, and the price series. The marks separate the two things this work distinguishes: the dot marks a day on which an alert was delivered, and the arrow a day the detector flagged that no message accompanied. The lower band shows the z value of those days against the ± 1.5 threshold, on an axis of its own since it shares no unit with the price. The direction of the arrow is that of the close and nothing more; the date and the z value of each day are on the cards below, and not on the line, so that they do not overlap. The interface opens on the current day; the figure shows the six-month range, since that is the one in which the distinction between flagging and communicating becomes observable.

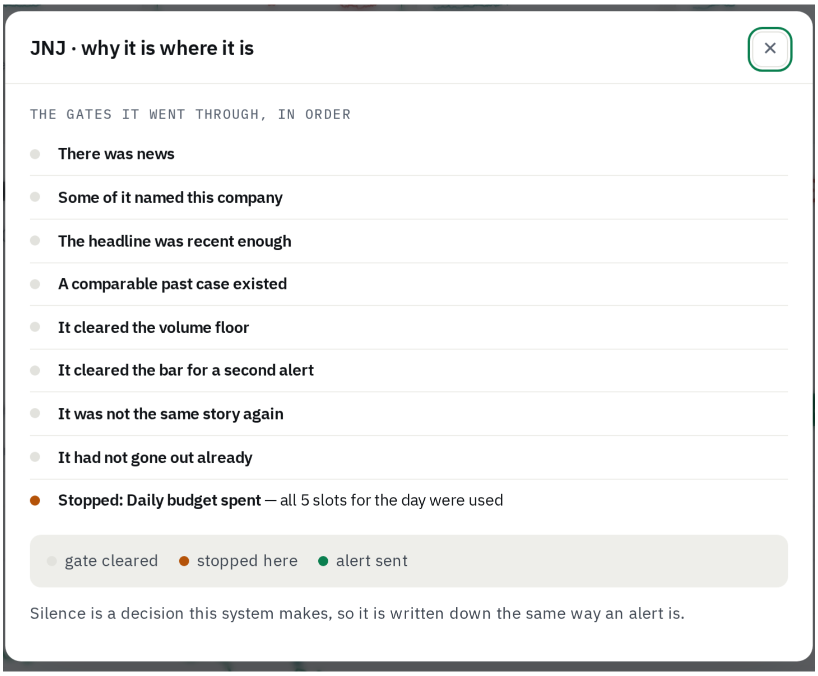

Figure 4.9: The gates a company crossed, in the order of the policy of Section 4.4. The gates crossed are stated by what the company satisfied; only the one at which it stopped is stated by the rejection, with the concrete reason. Silence is here a recorded decision, and not an absence of information.

a design constraint declared from the outset. None of the gates crossed is presented as a failure, which avoids the inverse reading that the previous version of the interface allowed.

4.6 Deployment and operation

The system runs as a permanent process, with a sixty-second scan cycle, on hosting credits included in an academic plan. The first version used a free best-effort scheduler, whose effective period was approximately one and a half hours. The change to a permanent process was motivated by reducing the delay of the alerts.

Latency was measured over the 278 news alerts delivered between 29 July and 30 August 2026 that record a publication time and a send time. The median of the complete path is 353 minutes, with a ninetieth percentile of 964 minutes. The breakdown locates that time without ambiguity: the median between publication and detection is 353 minutes, and the median between detection and the arrival of the message is 5 seconds, with a ninetieth percentile of 16 seconds. The system delivers in seconds what the source gives it.

The separation by configuration does not confirm the expectation that motivated the change. Over the 28 alerts delivered by the scheduler the median is 196 minutes; over the 250 delivered by the permanent process it is 402 minutes. The shorter cycle did not reduce the observed latency. The comparison is not interpretable as an effect of the cycle, since the two configurations did not operate in parallel, the news sources and the calendar period changed between them, and the samples are of very unequal size. What the measurement establishes is that shortening the cycle would only pay off if the time were in the cadence with which the system consults the source, and it is not.

Three causes contribute to the delay in discovery, and this history does not separate them. The first is the source listing the news hours after the time it itself declares. The second is that the most recent item in the feed is not the most recent *relevant* one: in a sample collected on 7 August 2026, the feed of one of the companies carried 250 headlines with the most recent published at 11:39, but the most recent among the 30 relevant ones was from 08:14. The alert goes out on the correct headline, and it is that headline that is old. The third is that the daily budget is already spent, which does not appear in this measurement because an alert that is not delivered records no send time. Only the first would be mitigated by a paid news service, and its contribution is not measured.

All these values are a lower bound on the latency felt by the user, since the publication time is the one declared by the source and not the instant of the event.

4.6.1 Growth of the case base

Every relevant headline observed by the scan is kept as a candidate precedent. Eight days later, when the impact on the price becomes observable, the candidate is measured by the same alignment rule of Section 3.5 and joins the case base. The waiting interval is not inefficiency: it is the condition that prevents a case from knowing its own outcome. Figure 4.10 represents the cycle.

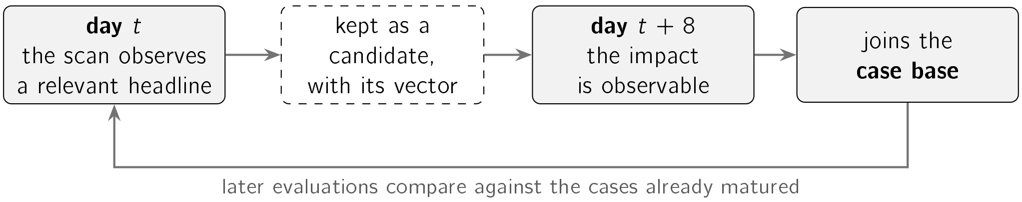

Figure 4.10: Path of a headline between its observation and its availability as a precedent. The eight-day interval follows from the impact of a case being observable only after the price movement. The cycle keeps up to date the evidence the system presents, and not the parameters of the model: no component is retrained.

The mechanism updates the evidence displayed and not the parameters of the model. In the concept drift taxonomy of Gama et al. (2014), the system has detection of the change and not adaptation to it.

It was found, with the system already in operation, that this cycle had been interrupted for nineteen days without any error being recorded. The two processes run on separate machines whose file system is reinitialised at every start, and the intermediate maturation files were not being published to the data repository. It corresponds to the class of cost that D. Sculley et al. (2015) call an undeclared data dependency: each component fulfils its contract, and the assumption that links them is expressed nowhere. Chapter 6 returns to the case as a limitation.

4.7 Life cycle of the learned component

The system trains a single model, the nine-input logistic regression described in Section 3.6. Measured against a one-variable baseline, that model performs worse, a result reported in Section 5.4.

The engineering contribution does not lie in the size of the model, but in the infrastructure that makes that result, and the negative conclusion it supports, defensible. That infrastructure has seven components, each of which exists because its absence would produce a favourable and unverifiable result. Figure 4.11 represents the two halves of the cycle.

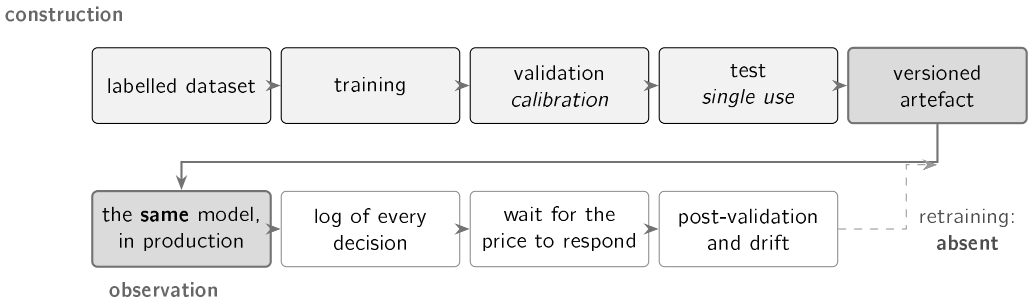

Figure 4.11: The two halves of the cycle. In the upper half, construction: the data are labelled, the split is chronological, the calibration is fitted on the validation block and the test block is used only once. In the lower half, observation: the model in production is the model evaluated, every decision is logged, and days later it is confronted with the movement actually observed. The dashed link would close the cycle and does not exist: drift is measured and is not corrected.

Construction of the dataset. FNSPID supplies dated headlines, and the prices come from a distinct source. The alignment of the two series, the assignment of a Saturday news item to the corresponding trading day, and the derivation of the label from the future movement constitute the work that produces the 79,753 examples of Section 3.2.

Declaration of the label as a decision. The existence of an abnormal movement in the following three days is not a fact present in the data: it is a definition, with a chosen threshold and horizon. Section 5.4 measures nine alternative definitions instead of assuming the one adopted.

Verification of the temporal separation. To the chronological blocks and the embargo is added an executable test: if the future of an example is perturbed, its inputs must stay unchanged and its label must change. The absence of leakage thus goes from a claim to a property verified at every run.

Calibration with measured bias. The output of the classifier is a value in the unit interval, which serves both to rank candidates and to exclude them. The Platt fit runs on a block that never took part in training, and the prevalence discrepancy between that block and the test block is declared in Chapter 5 instead of being left implicit.

Versioning of the artefacts. The trained model, the calibrator and a descriptor of the run, containing the size of each block, the prevalence, the seed and the metrics, reside in the repository. A metric unaccompanied by the artefact that produced it is not reproducible.

Identity between the model evaluated and the model deployed. The sentence encoder used in the evaluation depends on a library whose size exceeds the production container. Instead of substituting another model, which would make the dissertation describe a system distinct from the one in operation, the same model was exported to a lightweight execution format and the fidelity of the conversion was measured.

Observation of the deployed system. Every triage decision is logged. Once the necessary days have passed, a second procedure labels those decisions by the same rule used in training and measures the contribution of the model. It was by this route that the absence of contribution reported in Section 5.4 was established. The drift of the inputs is measured by the same mechanism.

4.7.1 Components developed and components consumed

The only model consumed from outside is a pre-trained sentence encoder, used without any tuning. The work claims no contribution over that component beyond its selection by measurement, carried out against four alternatives, among them a larger model, two more recent encoders and a model specific to the financial domain, which had the worst performance of the set. The dependency is essential, since there is no semantic retrieval without sentence representation, and it is at the same time replaceable: the interface that isolates it has an alternative implementation that degrades to lexical overlap when the model is not available.

Everything else was developed within this work. The labelled dataset and the alignment procedure that protects it from temporal leakage. The detector, with the estimation window that excludes the day being explained. The split of the movement with shrinkage of the coefficients. The precedent base with measured impact. The triage classifier and its calibration. The decision policy and the constants of Table 4.2. The layer that composes the text from the computed objects. And the mechanism that observes the system after deployment.

The system does not integrate a language model in the production of the text delivered to the user. The absence is deliberate and follows from the argument developed in Section 2.7: a generated sentence is a statement, whereas the system described delivers statements accompanied by the evidence that supports them.

A grounded generation layer is, even so, implemented and evaluated, and is not exposed. In it, every sentence must cite a fact from the evidence bundle that accompanies it, and a check rejects any sentence whose number does not belong to the fact it cites. Over a corpus of twenty-three attempts to circumvent that check, all were rejected, and over eight faithful texts none was; in the sections actually composed not a single delivered violation was recorded. The reason for not exposing it is not the performance of that check, but the nature of the guarantee: in the text delivered to the user the composition starts from a closed vocabulary, whereas the check of the generated layer enumerates what it forbids, and an enumeration of the forbidden is weaker than an enumeration of the permitted. No comparison with a complete retrieval-augmented generation architecture was carried out, so nothing is asserted about what it would add.

4.7.2 The missing link: retraining architecture

The cycle does not include retraining. The parameters correspond to the period from 2018 to 2023 and were not readjusted, even though drift is measured and the population stability index places the main input in the band of significant displacement. The operation would be technically possible, since new labels derive from observed prices; the impediment is that retraining with the production log would change the values reported in this document, and assessing that change requires observation time which, at the date this chapter was written, did not exist.

The instrumentation that would make it possible was built and deployed on 4 September 2026, and the log came to store, per decision, the value of each input, the date of the last price bar used and the identity of the model. Two restrictions limit what can be drawn from it, and both were fixed before any candidate existed. The first is that only decisions whose price bar is prior to or equal to the date of the news are usable, since a later bar would describe a market that has already observed the outcome the label measures. The second is that the unit of analysis is the pair formed by company and day, and not the headline, which substantially reduces the number of independent observations available in a short window. Nothing is asserted in this document about the result of that comparison.

The absence of execution does not, however, dispense with defining the architecture. Figure 4.12 presents the continuous training cycle designed for this system, with the mechanisms that would make it safe. Three decisions characterise it.

The first is the **trigger**, conditional and not temporal. Retraining is set off when the population stability index of any input exceeds 0.25, the conventional threshold of significant displacement, or after three months without review. The second is the **rolling window**: training covers the twenty-four months preceding the trigger, and validation and test the following periods, with the five-day embargo of Section 3.6. The split stays chronological, because the problem does not cease to be temporal just because the model is new.

The third, and the one that distinguishes a continuous training cycle from dangerous automation, is the **promotion gate**. A newly trained model does not replace the model in force by virtue of being more recent: it replaces it only if it beats it over the same held-out test block, and if the confidence interval of the difference excludes zero. A model that does not pass that gate is logged and discarded, and the model in force remains with the drift declared. It is the difference between adapting to change and chasing noise.

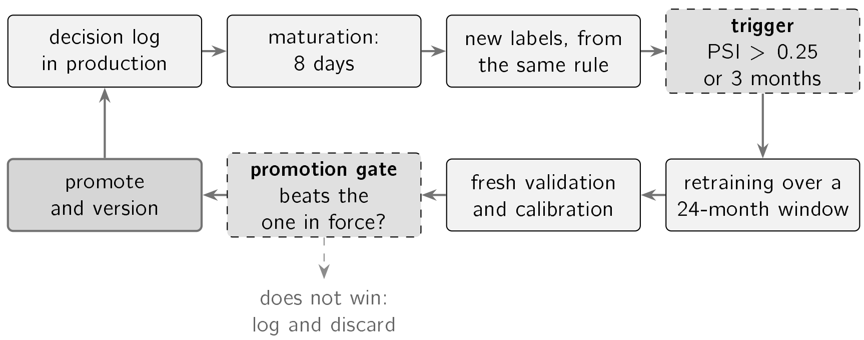

Figure 4.12: Continuous training cycle defined for this system and not executed within this work. The trigger is conditional and not temporal; the window is rolling and the split stays chronological with embargo; and the promotion gate prevents a new model from replacing the model in force without beating it over a held-out block. The dashed link is the rejection path, and its existence is what distinguishes the cycle from automation that chases noise.

The cycle likewise includes no form of human ground truth. The three labels used in this work, corresponding to abnormal movement, membership of the same sector and percentile of movement, are verifiable approximations that were never confronted with the judgement of a human assessor. It is the limitation on which the others most depend, and Chapter 6 returns to it.

Chapter 5

Case studies

The two previous chapters described the methods chosen and the system that puts them to work. This one determines what each of them is worth. The experimental design and the metrics used are described first. Four case studies follow: three correspond to the research questions, and the fourth to the behaviour of the system in operation, where the choices of the previous chapters meet data that were not available when they were made.

5.1 Experimental design and metrics

This section fixes the form common to all the case studies and defines the measures employed, with particular attention to the value each one takes in the absence of information.

Each case study follows four steps, shown in Figure 5.1: the question, the measurement procedure and the alternative against which the comparison is made, the result obtained, and what that result does not allow to be concluded. The fourth step is what distinguishes an evaluation from a demonstration. The values measured over frozen historical sets are produced by automatic procedures with a fixed seed and reproduce exactly. Two classes of value do not have that property, and are identified where they appear. The values read from the system in operation are dated snapshots, which grow with the channel. Those that depend on quotes readjusted by the source may vary in the third decimal place.

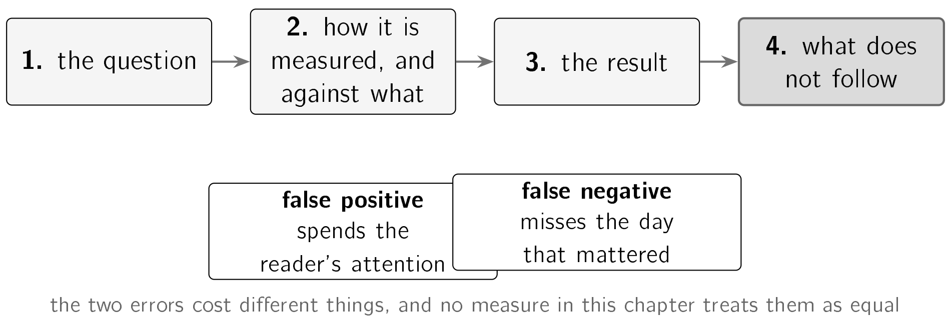

Figure 5.1: The form followed by all the case studies. The fourth step delimits the reach of each result. Below, the two types of error that the measures of this section balance, and the distinct cost of each one in the application domain.

All the measures rest on the four combinations between what the system claims and what is observed. Precision is the fraction of correct calls among the occasions on which the system spoke, $TP/(TP+FP)$; recall is the fraction of the relevant cases that were flagged, TP/(TP+FN). The two vary in opposite directions, and a rule that flags almost everything reaches high recall with low precision.

F_1 is the harmonic mean of the two, $2PC/(P+C)$, and stays close to the worse of the two values, which prevents an extreme from being rewarded. For the z-score, with precision 0.381 and recall 0.800, it gives 0.516; for the fixed threshold rule, with precision 0.122 and recall 0.992, it gives 0.218.

A precisão@k aplica-se à recuperação, que devolve uma lista ordenada de que o utilizador lê apenas o início. Mede a fração dos k primeiros resultados que são relevantes (Manning, Raghavan, and Schütze 2008), com $k = 5$, valor que corresponde ao que cabe num alerta.

A Precision-Recall Area Under the Curve (PR-AUC) aplica-se à triagem, que devolve uma probabilidade e não uma decisão. Cada limiar possível produz um par de precisão e cobertura, e o conjunto desses pares desenha a curva de precisão e cobertura; a área sob essa curva resume o modelo em todos os limiares simultaneamente, em vez de o julgar num limiar escolhido manualmente. É preferível à percentagem de acertos porque, com classes desequilibradas, responder invariavelmente que a notícia não é material acerta 62.2% das vezes sem qualquer aprendizagem.

O índice de Brier observa o valor da probabilidade e não a ordenação, sendo a média do quadrado do erro entre a probabilidade anunciada e o resultado observado, com valores menores a corresponder a melhor desempenho. O quadrado penaliza de forma mais do que proporcional os enganos confiantes, que é a propriedade adequada num sistema que exibe percentagens a um destinatário sem meios para as questionar.

Uma medida lida sem o valor que lhe corresponde na ausência de informação conduz à conclusão errada, e este capítulo documenta três ocasiões em que isso sucedeu. A Tabela 5.1 fixa esse valor para cada medida.

Table 5.1: Each measure, the question it answers, the value a method without information obtains, and the case study in which it is used. The central column is the one that prevents the optimistic reading of an isolated percentage.

Measure	**Answers**	**Value without information**	**Used in**
Firing-rate spread	Does the detector behave the same way in distinct companies?	zero is the optimum, and is attainable without information	RQ1
F_1	Is it right when it speaks, and does it speak when it should?	depends on the data	RQ1
Precision@5	Of the five presented, how many are relevant?	0.240, measured by random choice	RQ2
PR-AUC	Does it rank well, whatever the threshold?	0.378, the prevalence of the test block	RQ3
Brier score	Does the percentage announced match the observed frequency?	0.622, obtained by always announcing one	RQ3

The first row of Table 5.1 is the only measure in the chapter that uses no labels. The firing-rate spread is the difference between the rate at which the detector flags in the company where it speaks most and the rate in the company where it speaks least. It rests on a single principle: a universal detector should behave in a similar way in distinct companies. Its least obvious property, and one that Section 5.2 develops, is that the optimum is attainable without any information.

5.2 Detection consistent across companies

The first case study answers RQ1: a rolling z-score detects unusual movements more consistently across companies than a fixed threshold rule. The answer is affirmative, with a margin of more than twenty times on the principal measure.

5.2.1 Procedure

The evaluation uses three years of daily quotes from fifteen large-capitalisation companies, between June 2023 and June 2026, which corresponds to approximately seven hundred and fifty days per company. The set is distinct from the list of twelve companies monitored in production, for the reason stated in Section 3.2: the evaluation requires variety of volatility, and the deployed list is a product decision.

The principal argument uses no labels. A detector meant to flag rare days should fire at a similar rate in distinct companies; if it fires frequently in one and rarely in another, what it is measuring is the volatility of the company and not the rarity of the day. As a supporting argument an approximate label is used, which classifies as extreme a day in the ninety-ninth percentile of the company's own returns. The label is imperfect, being itself relative to volatility, and therefore occupies the secondary position.

5.2.2 Result

Figure 5.2 summarises the two measures. It is the first that supports the conclusion: what distinguishes the methods is not the level at which they fire, but the fact that the fixed threshold follows the volatility of each stock while the z-score stays almost constant across them.

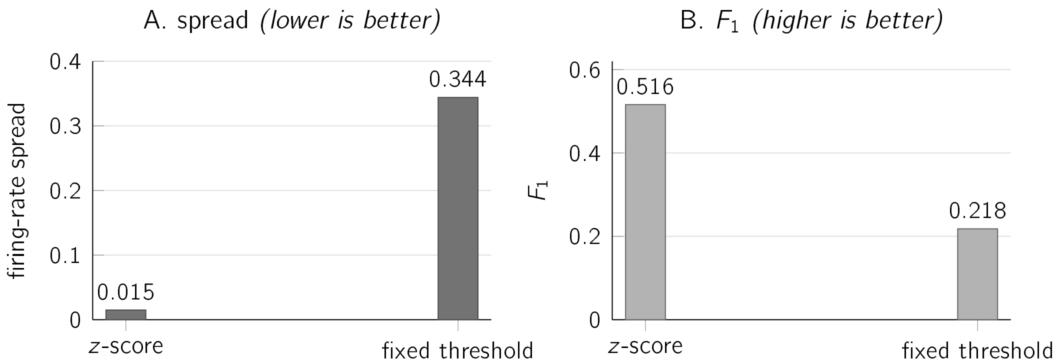

Figure 5.2: The two measures, side by side. In A, the difference between the maximum and the minimum rate across the fifteen companies; the fixed threshold flags 0.9% of the days in the calmest company and 35.3% in the most volatile, while the z-score stays between 1.6% and 3.1% in all of them. In B, the performance against the approximate label.

The firing-rate spread is 0.015 for the z-score and 0.344 for the fixed threshold, a difference of more than twenty times. Against the approximate label, the z-score obtains $F_1 = 0.516$ and the fixed threshold 0.218.

The comparison involves an asymmetry of treatment. The window of the z-score is varied later in this section, whereas the three per cent threshold of the fixed rule is the conventional value and was not optimised. What the comparison establishes with confidence is not that no fixed threshold would obtain a better result, but that a fixed threshold measures a quantity different from the one intended, because the same percentage corresponds to distinct rarities in distinct companies. This argument does not depend on the value chosen.

5.2.3 What the principal measure does not exclude

The firing-rate spread has zero as its optimum, and that optimum is attainable by a method with no information at all. A detector that flagged in each company a fixed fraction of days chosen at random would show a practically null spread, beating the z-score precisely on the measure this section elects as principal. The claim follows from the construction of the method and requires no execution.

The consequence is not that the measure is invalid, but that it must be read together with the second, and that is why there are two. The spread excludes the fixed threshold, which fires as a function of the volatility of the company and not of the rarity of the day. F_1 excludes random firing, which sits at the level of chance against any label. Neither measure is sufficient on its own, and only the rolling rule satisfies both.

An asymmetry in the standard of evidence applied throughout the chapter should also be recorded. The values of this section and of the next are points, without confidence intervals, whereas those of Section 5.4, where the result that contradicts the initial hypothesis resides, come with resampling by groups. The partial justification lies in the order of magnitude of the differences observed here, which no plausible resampling would reverse. The justification is not complete, since days of the same company are also not independent observations in this case, for the same reason of volatility clustering invoked later on.

5.2.4 Comparison with learned detectors

The most demanding comparison sets the statistical rule against machine learning methods designed to detect anomalies. Two were evaluated, over the same information and under the same discipline of not using information posterior to the day assessed: *Isolation Forest* (F. T. Liu, Ting, and Zhou 2008) and the *Local Outlier Factor* (Breunig et al. 2000). Both ran in a single configuration, without a hyperparameter search, which limits the reach of their defeat for the same reason invoked later for the tree model. What is established is that these detectors, parameterised in this way and over these inputs, do not beat the rule; not that no parameterisation would. Figure 5.3 presents the result.

The transparent rule beats both methods, with almost twice the F_1 and with substantially better consistency across companies. This is the first of three cases, throughout the chapter, in which a more sophisticated technique is compared under equivalent conditions with a simpler one and does not obtain a better result. The choice of the simple method thus becomes justified by measurement and not by preference.

The explanation does not lie in the quality of the methods. Both determine what is unusual from the geometry of the data, by opposite routes. *Isolation Forest* identifies the rare point

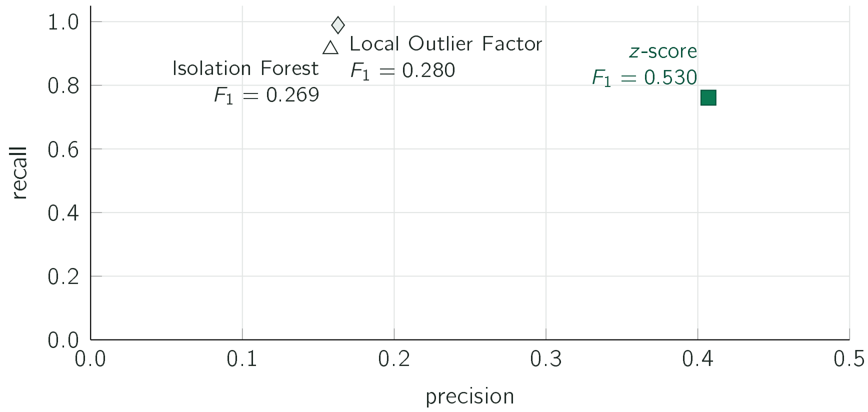

Figure 5.3: The three detectors over the same data and under the same protocol, restricted to the region all three can score. The two learned methods occupy the region of near-total recall and low precision; the rule trades some recall for precision. The F_1 beside each point summarises that trade-off. The firing-rate spread follows the same pattern, at 0.017 for the rule against 0.140 and 0.183 for the learned detectors.

by how easily successive random cuts separate it from the rest (F. T. Liu, Ting, and Zhou 2008). The *Local Outlier Factor* compares the density in the neighbourhood of each point with that of the points surrounding it (Breunig et al. 2000).

These are non-parametric methods, outside the statistical family that Chandola, Banerjee, and Kumar (2009) characterises by the modelling of the distribution of normal data, and they were designed for sets with many interacting variables. The problem treated here has another form: one series of returns per company, whose relevant structure is temporal. Volatility clustering, formalised by Autoregressive Conditional Heteroscedasticity (ARCH) and Generalized Autoregressive Conditional Heteroscedasticity (GARCH) models (Bollerslev 1986; Engle 1982), moves over time the threshold of what constitutes an unusual day; a rolling window follows that movement by construction, whereas a detector fitted to the complete set has to infer it.

A note on the compatibility between the values: the z-score appears with $F_1 = 0.530$ in this comparison and with 0.516 in the previous one. The two values measure the same detector under distinct protocols. The second covers all the days assessed, while the first is restricted to the region all three detectors can score, the only one in which the comparison between them is fair. The spread follows the same rule, being 0.017 under this protocol and 0.015 under the previous one.

5.2.5 Two comparisons that do not confirm the deployed choices

Two additional experiments were designed to justify options taken, and neither of them confirmed them. Both are reported as they came out.

The first concerns the volatility estimator. The alternative to giving the same weight to the twenty days of the window consists in giving greater weight to the recent days, through an exponentially weighted average. Figure 5.4 presents the result.

With identical recall, the alternative obtains $F_1 = 0.664$ against 0.516 for the deployed estimator. The mechanism is understandable. Large movements occur in clusters. A

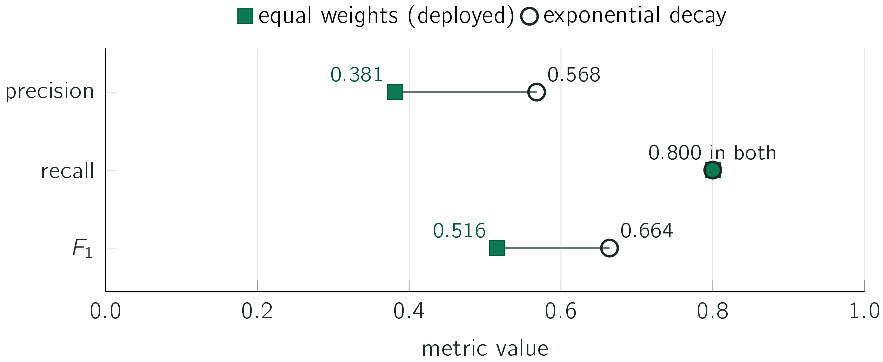

Figure 5.4: The two estimators, with the same recall; each segment links the deployed result to the alternative on the same measure. The exponentially weighted alternative removes almost half the false alarms, raising precision from 0.381 to 0.568, and is even more consistent across companies, with a spread of 0.012 against 0.015. The system keeps the estimator that obtained the inferior result, for the reason set out in the text.

twenty-day average with equal weights dilutes a shock over twenty observations and keeps flagging the following days for weeks; an average that favours the recent past absorbs it immediately.

The system keeps the estimator that obtained the inferior result. The reason is the one that runs through the work: the mean and the standard deviation of the last twenty days can be explained in two sentences to any reader, and exponentially decaying weights cannot. In a tool whose central argument is explainability, a gain in precision obtained at the cost of the explanation is not unequivocally a gain. It is recorded, however, as a choice and not as a result: the alternative is measured, the gain is quantified, and adopting it would be a small change should explainability cease to be the dominant constraint.

The second experiment concerns the size of the window, and Figure 5.5 presents it.

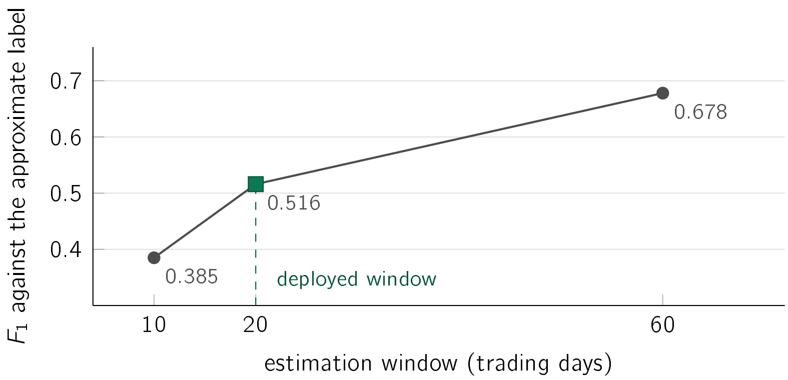

Figure 5.5: Performance against the approximate label for three window sizes. The value grows monotonically with the size of the window, and the deployed window is not the one that obtains the best result. The text sets out the two reasons why the longer window was not adopted, the second being a reservation about the validity of this measurement itself.

A sixty-day window obtains $F_1 = 0.678$ against the 0.516 of the twenty-day window. Two reasons justify keeping the shorter window. The first is responsiveness: a sixty-day window takes three months to react to a change of regime, and a company moving from calm to volatile would continue to be judged by the previous norm.

The second concerns the validity of this particular measurement. The label is built from the percentile of movement of the company itself, and is therefore relative to volatility; a longer window produces a less adaptive norm, which flags the extreme tail more often, which is precisely what this label rewards. Circularity exists, and it favours the long window by construction. This is why the principal argument of this section is the consistency of the firing rate, which uses no labels. The formulation that survives is that the twenty-day window was chosen for responsiveness, and that the only measurement available on this point does not support it.

5.2.6 Limits of the conclusion

The label is approximate and relative to volatility, which is why the argument that carries weight is the one about consistency, which does not depend on it. The fifteen companies are all large-capitalisation and were selected in 2026, with the evaluation running over the past: they survived by construction, and none of them was delisted, absorbed or dropped from the list. What is measured is the behaviour of the method in companies that persisted, and not in the population from which they were drawn. The comparison covers this set and this period only, and nothing in it allows the result in another market or another decade to be anticipated.

5.3 Precedent retrieval

The second case study answers RQ2: retrieval by meaning finds past news from another company in the same sector better than trivial alternatives. The answer is affirmative in the five sectors assessed, and the per-sector formulation is necessary, since the aggregate value is misleading in both directions.

5.3.1 Procedimento

Evaluating retrieval systems faces the absence of a list of correct answers, since no set of pairs of news items is annotated as related. A verifiable approximation is therefore used: a retrieved news item is considered relevant if it belongs to the same sector as the query news item. The approximation is imperfect, because two news items from the same sector may deal with distinct subjects, but it is objective and was fixed before the results were observed.

A further constraint is imposed that makes the task harder: the retrieved news item must belong to a different company. Without it, the system would perform well by returning news from the company itself, which almost always belongs to the same sector by construction. The corpus contains 3,714 news items distributed across five sectors, collected over twenty-seven days, and the measurement uses five hundred queries per repetition across five repetitions with distinct seeds. The span of the corpus delimits what the measurement establishes: twenty-seven days do not support claims about temporal generalisation. The protocol does not require the candidate to be prior to the query, and only 31.1% of the neighbours are. The check under the causal constraint was made in aggregate over the historical corpus, where the margin over chance holds, and not per sector.

5.3.2 Resultado

A Figura 5.6 apresenta o método utilizado, uma variante de maior dimensão e as alternativas triviais.

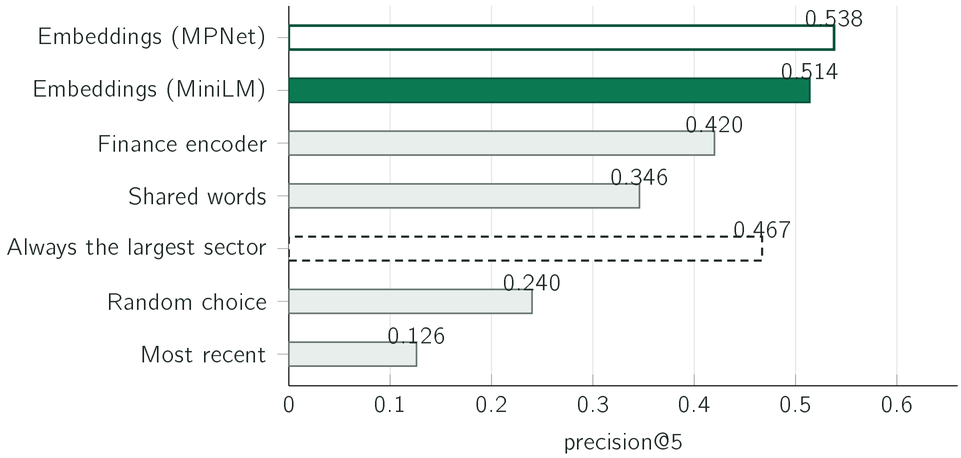

Figure 5.6: Precisão@5 do MiniLM implantado, em verde cheio, contra seis alternativas. O contorno verde identifica a variante MPNet de maior dimensão; as barras cinzentas reúnem duas linhas de base, uma linha lexical e um codificador específico do domínio financeiro. A alternativa tracejada de devolver sempre notícias do setor maior obtém 0.467 sem qualquer modelo, valor discutido no texto e que obriga a ler o agregado com reserva.

A escolha aleatória acerta em 24.0% dos casos, e é esse o valor de referência, e não zero, uma vez que os setores existem em número finito e têm dimensões distintas. A procura por palavras em comum eleva o resultado para 34.6%, e a comparação de significado para 51.4%.

A linha lexical corresponde à forma mais simples dessa família: contagem de palavras projetada por dispersão e comparada pelo cosseno, sem ponderação pela raridade dos termos, sem saturação da frequência e sem normalização pelo comprimento do texto. Pertence à tradição que vai do modelo de espaço vetorial (Salton, Wong, and C. S. Yang 1975) às funções de ordenação probabilísticas como o BM25 (Robertson and Zaragoza 2009), situando-se no degrau inicial dessa família. Os dezassete pontos percentuais que a representação de frases (Reimers and Gurevych 2019) acrescenta medem, por conseguinte, a distância a um ponto de partida elementar e não a distância a um sistema lexical afinado. A comparação com uma função de ordenação bem parametrizada constitui trabalho por realizar, e é a que tornaria esta conclusão mais forte.

O codificador específico do domínio financeiro obtém 0.420, o valor mais baixo de todos os modelos avaliados. O modelo avaliado é o ponto de controlo público `ProsusAI/finbert`, identificado desta forma por ser o artefacto exato sobre o qual a medição incide; pertence à família de codificadores financeiros de que Zhuang Liu et al. (2020) e A. H. Huang, Wang, and Y. Yang (2023) são as versões documentadas em publicações revistas por pares. Foi utilizado com a média dos seus vetores como representação da frase, que é a forma de o aplicar sem novo treino. Trata-se de um codificador ajustado para classificação de sentimento e não para posicionar frases num espaço onde a distância signifique semelhança, e é precisamente esse objetivo de treino que o Sentence-BERT acrescenta (Reimers and Gurevych 2019). A medição estabelece que este modelo de domínio, utilizado desta forma, obtém resultado inferior a um modelo genérico treinado para semelhança; não estabelece que conhecimento de domínio seja irrelevante para a tarefa.

5.3.3 A alternativa trivial que o valor agregado esconde

O corpus não está equilibrado: 1,736 das 3,714 notícias, quase metade, pertencem ao setor da tecnologia. Existe por isso uma estratégia que dispensa qualquer modelo e não observa sequer a consulta, consistindo em devolver invariavelmente cinco notícias de tecnologia. Acerta em todas as consultas de tecnologia e em nenhuma das restantes, pelo que a sua precisão iguala exatamente a fração de consultas que são de tecnologia, 0.467. Medida contra esta alternativa, e não contra o acaso uniforme, a margem do método reduz-se de +0.274 para +0.047, e a linha lexical de 0.346 passa a situar-se abaixo do valor de referência.

O valor agregado é, contudo, o elemento enganador, e não o método. A estratégia trivial obtém 1.000 nas consultas de tecnologia e 0.000 em todas as outras, devolvendo notícias de semicondutores a quem consulta sobre uma petrolífera. O que ela explora não é uma propriedade do problema, mas a composição deste corpus.

A comparação que resiste a esta objeção é a que se realiza dentro de cada setor, onde a estratégia trivial não tem por onde se apoiar. O painel A da Figura 5.7 apresenta-a.

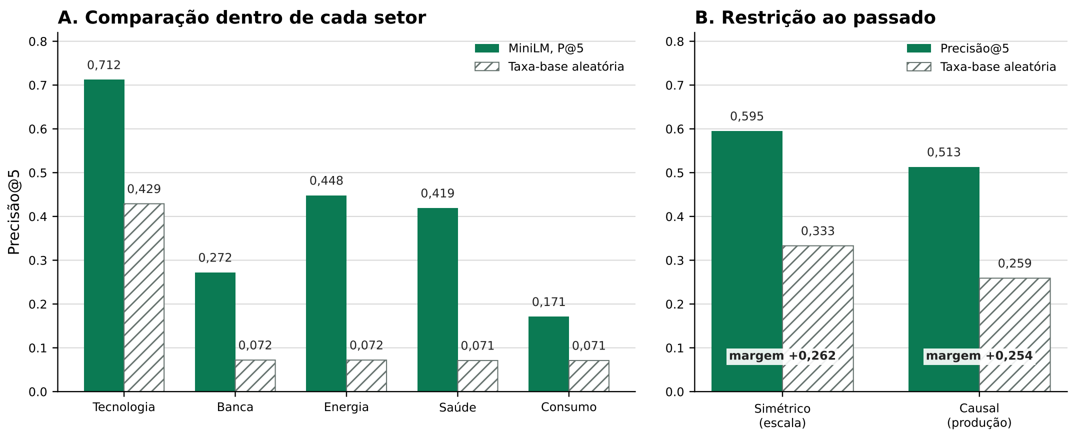

Figure 5.7: Em A, a precisão@5 e o valor de acaso correspondente dentro de cada setor: o método situa-se acima do respetivo valor de referência nos cinco. Em B, o protocolo simétrico do teste de escala e o protocolo causal que reproduz a restrição da produção, cada um ao lado do seu valor de referência. A precisão desce de 0.595 para 0.513 e o valor de acaso de 0.333 para 0.259, pelo que a margem varia apenas de +0.262 para +0.254.

O valor de acaso é de 0.429 na tecnologia e situa-se em torno de 0.072 em cada um dos quatro setores restantes. O método obtém 0.712 na tecnologia, 0.448 na energia, 0.419 na saúde, 0.272 na banca e 0.171 no consumo. Onde o valor agregado é dominado, o método acrescenta 0.283; onde o corpus é escasso, obtém várias vezes o valor de referência. O agregado subestima o método e sobrestima a alternativa trivial pela mesma razão, que é a de quase metade das consultas provir do único setor onde o valor de referência já era elevado. A afirmação que o trabalho sustenta é por isso a mais estreita: a recuperação semântica supera a taxa-base dentro de cada um dos cinco setores, e o valor agregado não é a forma adequada de o enunciar.

A variante de maior dimensão obtém 0.538 ± 0.011 contra 0.514 ± 0.015 da utilizada, sobre cinco sementes; a diferença é pequena face à dispersão e nenhum teste de significância foi realizado. Dois codificadores mais recentes obtêm 0.504 e 0.513, indistinguíveis do implantado. A adoção do modelo mais pequeno em produção decorreu do custo computacional e não de uma vantagem de qualidade, constituindo uma troca explícita cujo preço em precisão está quantificado. A alternativa aparentemente mais razoável, que consiste em apresentar as notícias mais recentes, obtém 0.126, o valor mais baixo de todos, o que é coerente com a ausência de qualquer razão para a notícia mais recente sobre outra empresa estar relacionada com a notícia em análise.

5.3.4 Comportamento à escala e sob a restrição da produção

A avaliação anterior incide sobre um corpus recente e de dimensão limitada. Para verificar que o resultado não depende desse corpus, a mesma medição foi repetida sobre o conjunto histórico completo, com aproximadamente oitenta mil notícias de seis anos. Neste protocolo simétrico, que ainda admite candidatos posteriores à consulta, a precisão@5 sobe para 0.595. A subida não deve, porém, ser lida na íntegra como ganho: o valor de acaso sobe igualmente, de 0.240 para 0.333, porque a composição por setores é distinta. Medida contra o valor de referência respetivo, a margem é de +0.274 no primeiro caso e de +0.262 no segundo. O resultado estabelece que o método não se degrada quando o corpus passa de recente e curto para seis anos, e não que melhore com mais dados.

O protocolo anterior concede uma liberdade que o sistema em produção não possui. A única restrição imposta é a de o candidato pertencer a outra empresa, o que permite que seja posterior à consulta, e esse caso é comum: no corpus recente, 38.7% dos vizinhos devolvidos têm data posterior à da consulta e outros 30.2% são do mesmo dia. O sistema implantado não dispõe dessa liberdade, uma vez que a base de precedentes só recebe um caso oito dias após a sua observação, conforme a Secção 4.6.1 descreve.

Não é utilização de informação futura no sentido corrente. O critério de acerto é a pertença ao mesmo setor, e o setor de uma notícia não varia com o tempo, pelo que a direção temporal não pode inflacionar a precisão. O que está em causa é a correspondência entre o valor reportado e a tarefa que a questão de investigação descreve, que fala em notícias passadas. A medição repete o protocolo acrescentando à máscara a condição de o candidato ser estritamente anterior à consulta, e reproduz o valor simétrico na mesma execução para que a comparação seja emparelhada. Nenhuma das 2,500 consultas ficou sem passado suficiente. O resultado consta do painel B da Figura 5.7: a precisão desce 0.082, o valor de referência desce quase outro tanto, e a margem varia 0.008. O método conserva praticamente toda a vantagem quando obrigado a operar como opera em produção.

Esta é a terceira ocasião no capítulo em que a leitura de uma precisão sem o valor de referência correspondente conduziria à conclusão errada. Sucedeu na passagem do corpus recente para o completo, sucede na precisão dentro do orçamento tratada na Secção 5.4, e sucede aqui. A precisão é uma quantidade sem significado próprio, que apenas o adquire ao lado da alternativa sem informação.

5.3.5 Limites da conclusão

A pertença ao mesmo setor não equivale a relação entre notícias, sendo uma aproximação construída por ser verificável e não por corresponder exatamente à propriedade pretendida.

A segunda observação tem consequências para o produto: a recuperação capta tema e não direção. Duas notícias podem tratar do mesmo assunto e o preço evoluir em sentidos opostos, que foi o que sucedeu no exemplo apresentado no Capítulo 4. A Figura 5.8 quantifica-o.

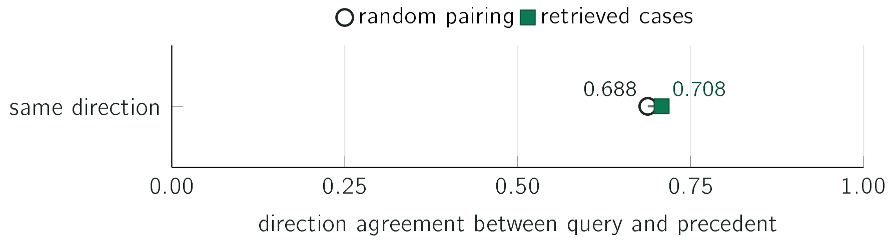

Figure 5.8: Fração de casos recuperados cujo preço evoluiu no mesmo sentido que o da notícia de consulta, contra o valor obtido por emparelhamento aleatório sob as mesmas restrições, numa escala completa entre zero e um. A diferença é de 0.020. O sistema apresenta por isso os casos individualmente, e o texto do alerta declara que a semelhança é de tema.

A concordância de direção entre a notícia de consulta e os casos recuperados situa-se em 0.708 contra um valor de acaso de 0.688. É por esta razão que o sistema apresenta os casos um a um e nunca os resume a uma média, que seria lida como previsão, e que o texto do alerta declara explicitamente tratar-se de casos semelhantes quanto ao tema.

5.4 Triagem de notícias

O terceiro caso de estudo responde à RQ3: um modelo treinado prioriza notícias melhor do que uma linha de base baseada apenas em volatilidade. A resposta é negativa, e é o resultado com maior valor informativo do trabalho.

5.4.1 Procedimento

O conjunto de dados contém 79,753 exemplos entre 2018 e 2023, cada um correspondente a uma notícia com o contexto de mercado do dia e o rótulo definido na Secção 3.6. A divisão é cronológica com embargo, e a prevalência de casos positivos no bloco de teste é de 0.378. A métrica principal é a PR-AUC, adequada quando as duas classes estão desequilibradas.

Para efeitos práticos, uma diferença de PR-AUC inferior a 0.02 é tratada como indistinguível, mesmo quando o intervalo emparelhado exclui zero. O limite é uma escolha do autor, declarada antes dos resultados, o que obriga à sua aplicação nos dois sentidos.

Dois cuidados de montagem asseguram que a comparação não favorece a conclusão obtida. O primeiro é a inclusão de um conjunto de árvores treinado por reforço do gradiente (Friedman 2001), o mais expressivo dos modelos comparados, por captar interações e efeitos não lineares que a regressão logística não capta por construção. Este modelo foi executado com os parâmetros por defeito da biblioteca, sem procura de hiperparâmetros, o que limita o alcance da sua derrota: estabelece que o texto não produz vantagem nesta montagem, e não que a não produziria num modelo não linear afinado. O segundo cuidado é a conversão de todas as pontuações em probabilidades por calibração *a posteriori* sobre um bloco de validação reservado (Niculescu-Mizil and Caruana 2005), sem a qual a comparação pelo índice de Brier não seria válida entre modelos.

Uma entrada não está disponível no mesmo referencial em produção. O retorno do próprio dia corresponde, no conjunto histórico, ao movimento completo do dia da notícia, ao passo que em produção a notícia é pontuada antes de esse dia encerrar. O resultado apresentado avalia por isso o modelo com a variável diária completa e não demonstra paridade com a pontuação intradiária. A direção da assimetria é conhecida. A entrada em causa pertence ao bloco de contexto, presente nos modelos de resultado inferior. A linha de base vencedora usa uma só entrada: a volatilidade dos vinte dias anteriores, fechada na véspera. O referencial mais generoso encontra-se do lado dos modelos que a comparação não favorece, e a sua correção só poderia tornar o resultado negativo mais forte.

5.4.2 Resultado

A Figura 5.9 apresenta os cinco modelos e a referência sem informação comparados. As curvas completas, de que estes valores são a área, constam dos artefactos identificados no Apêndice A. Nelas a curva da volatilidade situa-se acima das restantes em quase toda a extensão do eixo, o que estabelece que a diferença não é artefacto do valor agregado.

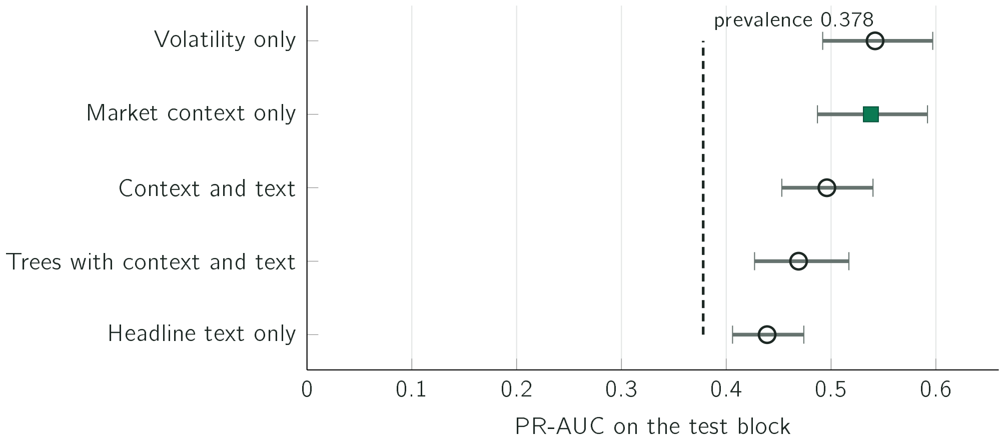

Figure 5.9: Área sob a curva de precisão e cobertura de cinco modelos, com intervalos marginais a noventa e cinco por cento obtidos por reamostragem dos 1,951 grupos. A linha tracejada assinala a prevalência do bloco de teste, que é o valor obtido por um modelo que atribua a mesma probabilidade a todos os exemplos. Os valores são pontos, e os intervalos são amplos e sobrepõem-se: [0.492; 0.597] para a volatilidade, [0.487; 0.592] para o contexto e [0.453; 0.540] para o contexto com texto. A comparação que a subsecção seguinte sustenta é a diferença emparelhada entre modelos sobre as mesmas reamostragens, e não a distância entre dois pontos.

O valor mais elevado pertence ao modelo mais simples de todos, o que utiliza apenas a volatilidade recente. Acrescentar-lhe o restante contexto não altera o resultado de forma distinguível, com 0.538 contra 0.542; acrescentar o texto do título deteriora-o de forma distinguível, com 0.496. O índice de Brier acompanha a mesma ordenação, com 0.218 para o modelo de volatilidade contra 0.229 para o modelo com texto e 0.622 para o modelo que anuncia sempre a unidade. A hipótese inicial, segundo a qual o texto da notícia traria informação ausente dos preços, não é confirmada sobre este corpus.

5.4.3 Três verificações à montagem experimental

O resultado negativo foi submetido a três verificações destinadas a determinar se decorre da montagem experimental.

A primeira respeita ao ajuste do modelo de texto. A comparação foi repetida concedendo ao texto as melhores condições disponíveis, com regularização afinada, redução do bloco de texto a trinta e duas dimensões e um modelo de linguagem especializado em finanças. O melhor resultado do texto sobe para 0.533 e permanece abaixo dos 0.542 da volatilidade.

A segunda respeita ao ruído de amostragem. O bloco de teste foi reamostrado mil vezes por grupos constituídos pela empresa e pelo dia, uma vez que notícias do mesmo dia sobre a mesma empresa partilham o rótulo e não constituem observações independentes. A diferença entre a volatilidade e a variante com texto é de +0.048, com intervalo entre +0.032 e +0.066, que exclui zero e supera o limite prático de 0.02. É a única comparação desta secção que o supera.

O alcance deste intervalo é, contudo, limitado: a reamostragem realiza-se dentro de um único bloco de teste, com duzentos e vinte e um dias de negociação, pelo que mede o ruído de amostragem nesse período e nada estabelece sobre a estabilidade entre períodos. A distinção não é teórica neste trabalho, dado que a Secção 5.6 reporta um deslocamento significativo na distribuição da entrada principal. Uma origem rolante com três ou quatro cortes sucessivos responderia à pergunta mais larga, e não foi realizada.

A terceira respeita à definição do rótulo, que utiliza um limiar de dois por cento e um horizonte de três dias, ambos escolhidos pelo autor. A comparação foi repetida para nove combinações de limiar e horizonte, e a Figura 5.10 apresenta o resultado.

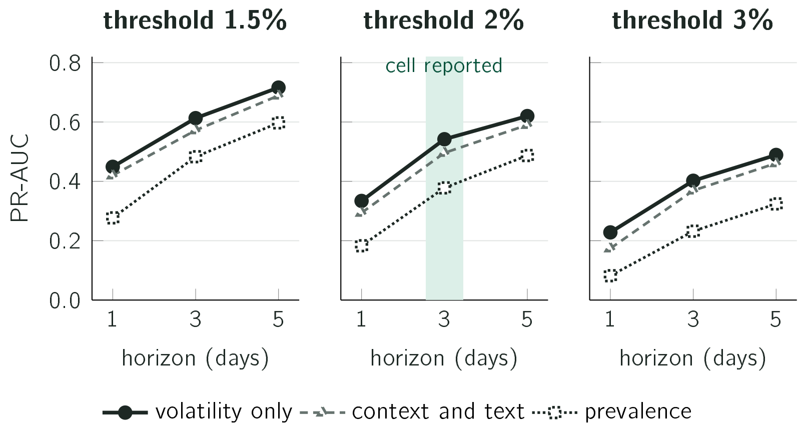

Figure 5.10: Desempenho das duas famílias sob os três horizontes, em três painéis correspondentes aos limiares, com a prevalência de cada célula como referência. A faixa verde identifica a definição utilizada no restante capítulo. A volatilidade iguala ou supera o modelo com texto nas nove combinações, com prevalências entre 0.082 e 0.597.

O resultado negativo resiste às três verificações. A volatilidade iguala ou supera o modelo com texto nas nove definições de rótulo, com prevalências entre 0.082 e 0.597, o que retira à escolha concreta de limiar e horizonte o estatuto de ponto de ataque.

Uma quarta verificação incide sobre a composição dos blocos e conduz a uma conclusão que atua nos dois sentidos. Os três blocos quase não partilham empresas. O bloco de treino decorre entre 2 de janeiro de 2018 e 3 de março de 2022 e contém treze empresas; o bloco de teste decorre entre 2 de fevereiro e 18 de dezembro de 2023 e contém nove. Cinco empresas do treino não figuram uma única vez no teste, e uma empresa do teste, responsável por 17.1% das suas linhas, nunca figura no treino. Para as empresas presentes em ambos os lados, a taxa de positivos desloca-se de forma acentuada, e a Apple passa de 0.448 no treino para 0.183 no teste. Estes valores leem-se diretamente das colunas de identificação e de bloco do conjunto congelado e não exigem qualquer execução adicional.

Não se trata de utilização de informação futura nem de erro de divisão: o corte é realizado por dia, e a proporção respeita a definida na Secção 3.6. É a composição do corpus que varia com o tempo, pela mesma razão que torna o bloco de teste maior do que o de treino, uma vez que as empresas sobre as quais se escreve em 2018 não são as mesmas de 2023.

A favor do resultado negativo pesa o seguinte. O sinal do modelo é essencialmente a identidade da empresa, conforme a Secção 5.4.5 estabelece, e essa identidade é estimada sobre um conjunto de empresas quase ausente do bloco onde é avaliada. A volatilidade dos vinte dias anteriores, fechada na véspera, é comparável entre empresas e não depende da identidade de nenhuma. Contra o resultado, a comparação não separa a hipótese de o modelo ser pior da hipótese de a identidade não transferir entre estes dois períodos, separação que exigiria uma divisão de composição constante. A mesma constatação explica, sem recurso ao acaso, a prevalência de 47.0% da validação contra 37.8% do teste, que correspondem a conjuntos de empresas distintos e não apenas a períodos distintos.

5.4.4 Utilidade sob a restrição de orçamento

O produto apresenta no máximo cinco alertas por dia, pelo que a pergunta com significado operacional é a de qual a fração de acertos entre as cinco notícias escolhidas em cada dia. A Figura 5.11 isola duas referências que parecem equivalentes, mas não são. O modelo e a alternativa baseada em volatilidade são comparados na Figura 5.18.

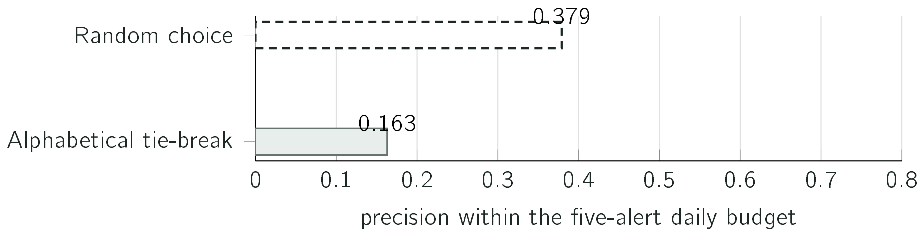

Figure 5.11: Duas formas de escolher as cinco notícias do dia, sobre o mesmo bloco de teste. O desempate alfabético não mede escolha às cegas; a escolha aleatória obtém 0.379 ± 0.017 sobre quarenta sementes. As duas políticas escolhem conhecendo o dia completo, pelo que os valores são limites superiores do que uma política em linha consegue obter.

A referência alfabética exige uma explicação: a designação original, «alertar sempre», induz em erro. O modelo que alerta sempre atribui a mesma pontuação a todas as notícias, e quando todas empatam a métrica seleciona pela ordem em que as linhas surgem no ficheiro, que está ordenado por data e por nome da empresa. Esse valor de 0.163 mede a escolha por ordem alfabética e não a escolha aleatória, e todas as linhas que seleciona pertencem à empresa que ocupa a primeira posição no alfabeto. Comparado com o valor de referência correto, o ganho do modelo é de 1.67 vezes e não das quase quatro vezes que a primeira leitura sugeriria. A lição é geral e aplica-se ao restante capítulo: verificar o que a linha de base mede é tão necessário quanto verificar o que o modelo faz.

A comparação com a volatilidade, na Figura 5.18, contém o resultado incómodo. Uma tabela de treze constantes, correspondentes à volatilidade média de cada empresa, calculada exclusivamente sobre o bloco de treino e sem leitura de um único título, obtém 0.662 e supera o modelo treinado. A comparação tem uma reserva que a torna menos limpa do que parece: 17.1% das linhas do bloco de teste pertencem a uma empresa ausente do treino e recebem a mediana global em vez de uma constante própria. É, ainda assim, o terceiro caso do capítulo em que o método mais simples obtém melhor resultado. O que a evidência sustenta é que ordenar por volatilidade compensa, e não que aprender compense.

5.4.5 Ablação das entradas do modelo

O resultado anterior sugere que o modelo implantado reconhece a empresa e não a notícia. A verificação dessa hipótese consiste em tentar reproduzi-lo com uma tabela de consulta que atribua a cada empresa um único número fixado no bloco de treino, correspondente à fração das suas notícias que se revelaram materiais. Essa tabela não observa o título, não observa o dia e devolve invariavelmente o mesmo valor para a mesma empresa. A Figura 5.12 compara-a com o modelo implantado e desmonta o modelo entrada a entrada, sob o protocolo congelado.

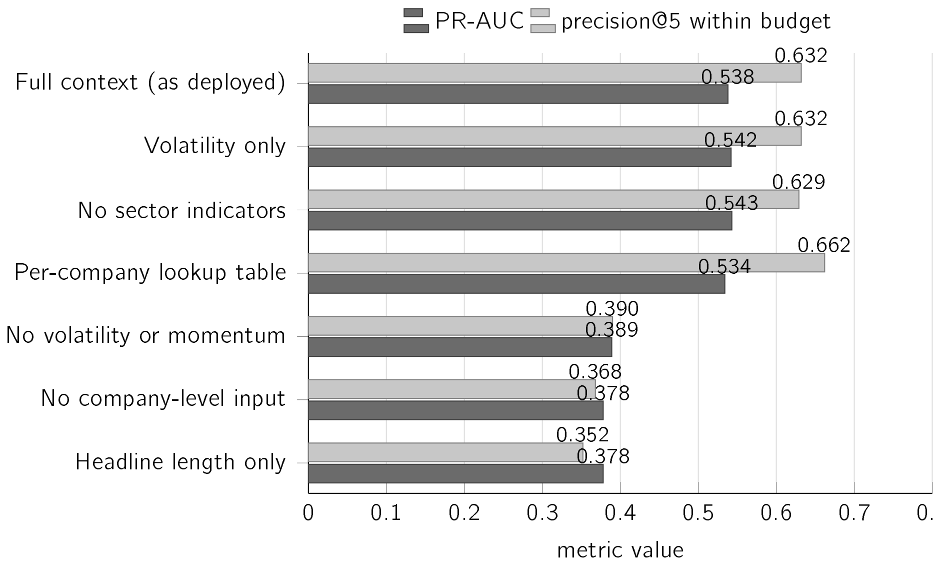

Figure 5.12: Cada linha corresponde a uma variante com um subconjunto das entradas ou a uma referência alternativa; as linhas não formam uma sequência de remoções cumulativas. A linha decisiva é a da tabela de consulta, que não observa a notícia e iguala o modelo implantado na primeira medida e o supera na segunda. As duas linhas inferiores, que não dispõem de qualquer entrada de nível de empresa, situam-se exatamente na prevalência do bloco de teste.

Três leituras resultam da figura, e nenhuma depende de interpretação.

A tabela de consulta iguala o modelo e supera-o numa das medidas. Na PR-AUC obtém 0.534 contra 0.538, uma diferença de 0.004; na precisão dentro do orçamento, que é a medida mais próxima da utilização real, obtém 0.662 contra 0.632. Esta segunda diferença não foi acompanhada de reamostragem por grupos, ao contrário das diferenças de PR-AUC desta secção, pelo que se reporta como observação e não como superioridade estabelecida. Um número por empresa, fixado meses antes e nunca atualizado, produz o mesmo efeito que o modelo treinado.

Sem as entradas de nível de empresa não subsiste sinal. Retirando-as, a PR-AUC desce para 0.378, que é precisamente a prevalência do bloco de teste e o valor definido na Secção 5.1 como correspondente à ausência de informação.

A única entrada que descreve a notícia nada acrescenta. O comprimento do título, isoladamente, produz igualmente 0.378, o que torna a única informação por título recebida pelo modelo implantado indistinguível da ausência de informação.

Estas leituras obrigam a separar duas afirmações. A conclusão científica da RQ3 mantém-se: a variante com texto dispõe de trezentos e oitenta e quatro números por título, possuindo por isso informação real sobre a notícia, e ainda assim obteve 0.496 contra os 0.542 da volatilidade, sendo essa a comparação que responde à questão. A decisão de produto, por seu turno, estava errada por razão distinta: o que foi implantado como porta de decisão foi a variante sem texto, escolhida por uma métrica que não formulava a pergunta de que o produto necessitava.

A manutenção da variante de nove entradas não decorre de esta apresentar o melhor valor entre as que cabiam no contentor de execução. Pelo critério fixado antes dos resultados, quatro valores são indistinguíveis entre si: os 0.542 da volatilidade, os 0.543 da variante sem indicadores de setor, os 0.538 da variante implantada e os 0.534 da tabela de consulta. As diferenças situam-se entre 0.004 e 0.009, muito abaixo do limite de 0.02. Aplicar esse limite apenas quando desfavorece uma conclusão seria não o ter fixado. A única diferença que o critério separa nesta comparação é a que distingue estas quatro variantes do bloco com texto, de +0.048 com intervalo entre +0.032 e +0.066. A escolha entre as quatro é, por isso, inteiramente de explicabilidade e não tem preço mensurável em PR-AUC: apenas a variante de nove entradas permite ao alerta apresentar as contribuições que elevam e reduzem a pontuação, conforme a Figura 4.6 exibe.

A lição de método permanece: a PR-AUC sobre o conjunto completo pode assumir valor elevado quando a variação entre empresas domina, mesmo na ausência de qualquer capacidade de distinguir notícias dentro da mesma empresa. A métrica respondia à pergunta sobre a ordenação do conjunto, enquanto o produto necessitava da pergunta sobre a distinção entre duas notícias da mesma empresa.

5.4.6 O texto como acréscimo e não como substituto

Comparar a volatilidade, o contexto e o texto isoladamente responde a qual das representações é melhor, e não a se o texto acrescenta informação. A resposta a esta segunda pergunta exige somá-lo a uma base que conheça a empresa sem conhecer a notícia.

A variante que combina contexto e texto não constitui essa base, por somar o texto a um bloco que contém informação da notícia através da entrada de retorno do próprio dia. A base adequada é a tabela de consulta por empresa, e a razão é metodológica e não de desempenho: ela contém tudo o que o modelo conhece sobre a empresa e nada sobre a notícia, pelo que somar-lhe o título isola a contribuição do texto. A tabela de consulta não é o melhor preditor conhecido, dado que na PR-AUC a volatilidade isolada se situa acima dela, com 0.542 contra 0.534.

O critério foi fixado antes da medição: um acréscimo apenas conta se o intervalo de confiança a noventa e cinco por cento da diferença, obtido por reamostragem por grupos constituídos pela empresa e pelo dia, excluir zero. A representação de texto não foi afinada para esta experiência, utilizando a mesma redução a trinta e duas dimensões da verificação anterior, de modo que o resultado não decorra da procura da configuração conveniente. A reamostragem utiliza mil repetições sobre 1,951 grupos. A Figura 5.13 apresenta os três intervalos.

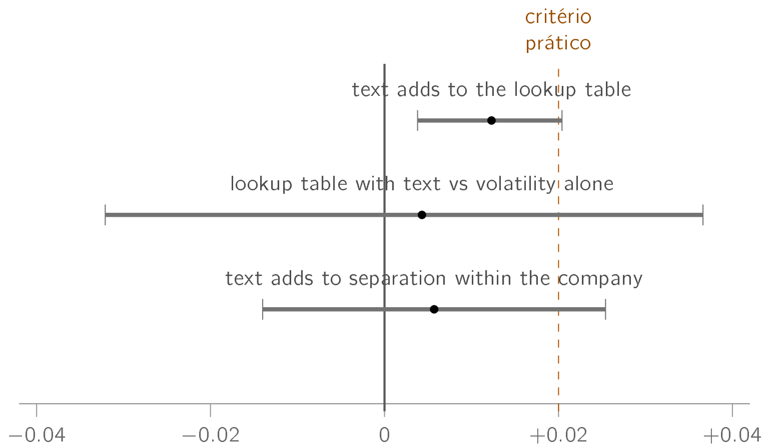

Figure 5.13: Diferenças de desempenho e respetivos intervalos de confiança a noventa e cinco por cento, obtidos por reamostragem por grupos. As duas primeiras linhas são diferenças de PR-AUC e a terceira é uma diferença de capacidade de ordenação dentro de cada empresa. Apenas a primeira exclui zero, e fá-lo abaixo do critério prático fixado antes da medição.

O título acrescenta a essa base +0.012 de PR-AUC, com intervalo entre +0.004 e +0.020, que exclui zero. É a primeira ocasião em que uma representação de texto apresenta um acréscimo detetável nesta tarefa, e o valor situa-se abaixo do critério prático de 0.02 fixado antes dos resultados.

Três reservas delimitam o alcance deste acréscimo, e todas são medições. A primeira é que a soma não supera a volatilidade isolada: o valor obtido é de 0.547 contra 0.542, e a diferença tem intervalo entre −0.032 e +0.037, que contém zero. A segunda é que o acréscimo não chega ao produto. A precisão dentro do orçamento de cinco alertas por dia é de 0.662 com e sem o texto, sem uma casa decimal de diferença. O acréscimo existe na ordenação do conjunto completo e desaparece quando apenas cinco notícias por dia são selecionadas.

A terceira é que o texto continua a não separar notícias da mesma empresa. A capacidade de ordenação dentro de cada empresa vale 0.500 por construção para a tabela de consulta e sobe para 0.512 quando se lhe soma o título. Para o modelo implantado é de 0.502, com intervalo entre 0.462 e 0.542, e sobe para 0.508 com o texto, num acréscimo de +0.006 cujo intervalo, entre −0.014 e +0.025, contém zero. Nenhum dos dois se distingue do acaso. A informação que o texto traz distingue empresas e períodos, e não notícias.

A resposta à RQ3 mantém-se negativa e torna-se mais precisa. Nenhum modelo com texto supera a volatilidade, e o texto não é inútil: acrescenta ao que o modelo já conhece sobre a empresa, de forma pequena e mensurável. O que impede esse acréscimo de alterar o sistema não é a ausência de informação, mas a natureza da informação, que distingue empresas quando o produto necessita de distinguir notícias. O trabalho seguinte requer um rótulo por notícia, e não apenas um modelo de maior dimensão.

Uma reserva final aplica-se ao próprio método. Este é mais um modelo avaliado sobre o mesmo bloco de teste, que já serviu várias comparações do capítulo, e quanto mais vezes um conjunto de teste é observado, mais provável se torna encontrar nele uma diferença pequena. As duas defesas utilizadas, que consistem em não afinar a configuração e em declarar o critério antes da medição, reduzem esse risco sem o eliminarem. A formulação que subsiste é a mais fraca das possíveis: não é possível rejeitar que o título acrescente informação, e o acréscimo, a existir, é inferior ao critério prático fixado antes da medição. Trata-se de um limite superior de um efeito e não de uma descoberta.

5.4.7 Limites da conclusão

O resultado não estabelece que o texto de notícias seja inútil. Foi avaliada uma única representação, constituída por títulos sem o corpo do artigo, sobre um único corpus. Com textos completos e maior volume de dados a resposta poderia ser distinta, o que constitui trabalho futuro.

Existe ainda uma hipótese alternativa que estas medições não excluem. O rótulo desconta o movimento do mercado assumindo sensibilidade unitária, pela incoerência declarada na Secção 3.6. Numa ação de sensibilidade elevada, o resíduo da subtração retém movimento do mercado, pelo que o erro do rótulo não é aleatório: é maior nas ações mais sensíveis ao mercado, que são tipicamente também as mais voláteis. Ora a linha de base vencedora é exatamente a volatilidade. Com o que está medido não é possível separar a hipótese de a volatilidade prever melhor a materialidade da hipótese de a volatilidade prever melhor o erro introduzido pelo rótulo. A experiência que resolveria a questão consiste em reconstruir o alvo com sensibilidades estimadas e encolhidas, como faz a Secção 3.4, e repetir as seis famílias. Não foi realizada. O nível e a ordenação das métricas ficam condicionados ao rótulo utilizado.

5.5 O que permanece mensurável na decomposição

A decomposição do movimento não dá origem a uma questão de investigação. Esta secção estabelece a razão dessa ausência e o que, apesar dela, permanece falsificável na técnica.

A diferença face às três técnicas anteriores é a existência de um alvo externo. Para a deteção existe um critério de movimento extremo, o que permite contar acertos e falhas; para a recuperação existe uma noção verificável de relevância, de que decorre a precisão@k; para a triagem existe um rótulo construído a partir de preços posteriores. Em todos estes casos a técnica pode revelar-se errada contra um alvo que lhe é exterior.

Na decomposição esse alvo não existe. Nenhuma fonte de dados publica qual foi a parcela verdadeira do mercado no movimento de uma ação num dado dia, uma vez que essa quantidade não é observável e apenas existe relativamente a um modelo. Fixadas as sensibilidades, a repartição é uma identidade contabilística e não uma previsão suscetível de sair certa ou errada. A construção de uma medida de exatidão neste ponto produziria um valor sem referente.

A técnica não fica por avaliar, mas por avaliar dessa forma, o que obriga a determinar o que nela permanece falsificável. São duas propriedades, e ambas se medem sobre as dezassete empresas do mapa de setores, que é o conjunto percorrido pelo procedimento de avaliação e não a lista monitorizada em produção. A Figura 5.14 apresenta as duas.

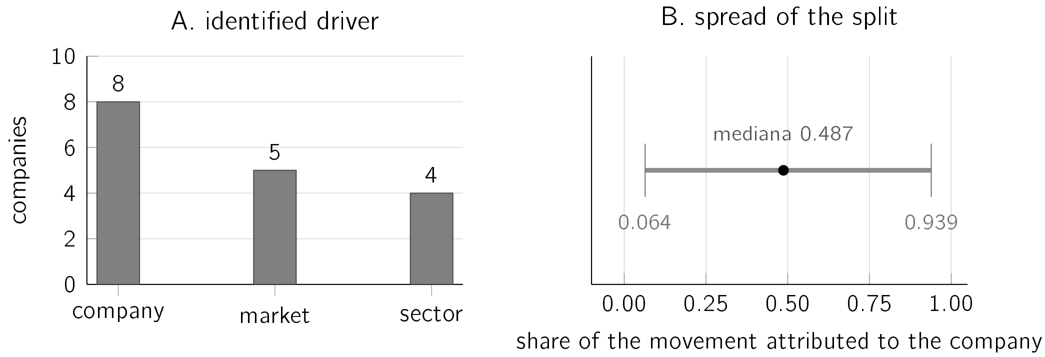

Figure 5.14: As dezassete empresas do mapa de setores, decompostas no mesmo dia. Em A, a componente identificada como maior parcela do mesmo sinal que o movimento observado. Em B, a dispersão da quota atribuída à empresa. A repartição produz respostas distintas para empresas distintas no mesmo dia, que é a condição mínima para a sua utilidade.

A primeira propriedade é a capacidade de discriminar. Uma decomposição que atribuísse a totalidade do movimento à empresa em todos os dias seria inútil ainda que aritmeticamente correta. No dia medido, o motor identificado foi a própria empresa em oito casos, o mercado em cinco e o setor em quatro, e a quota do movimento atribuída à empresa apresentou mediana de 0.487, com extremos em 0.064 e 0.939.

A segunda propriedade é a qualidade da estimação das sensibilidades, e é a que pode efetivamente estar errada. A medida apropriada é o coeficiente de determinação na janela de estimação, cuja mediana foi de 0.460. Não se trata do coeficiente do ajuste por mínimos quadrados, que, calculado dentro da amostra e com termo constante, nunca poderia ser negativo, mas do coeficiente do modelo efetivamente utilizado, com as sensibilidades já encolhidas e o termo constante recentrado. É esta a medida adequada, por corresponder ao modelo cuja repartição é apresentada ao utilizador, e é por essa razão que pode assumir valor negativo. Numa das dezassete empresas assumiu, o que significa que, naquela janela, o mercado e o setor não explicavam a variação daquela ação melhor do que a sua própria média, e que a repartição desse dia assenta num ajuste que não descreve os dados.

O resultado não estabelece que a repartição esteja correta pelo facto de as parcelas somarem. Elas somam sempre, com diferença nula até à precisão da máquina, porque a parcela da empresa é definida como resíduo, e uma repartição sem qualquer significado somaria igualmente. A soma exata constitui uma verificação de implementação e não uma validação de método. O resultado também não estabelece que o valor apresentado ao utilizador seja fiável em todos os dias, o que no caso de coeficiente negativo não sucede, e é por essa razão que a limitação consta do Capítulo 6.

5.6 O sistema em operação

O quarto caso de estudo incide sobre o comportamento do sistema depois de implantado, sobre as decisões que toma autonomamente. É a avaliação menos comum das três descritas na Secção 3.1, e a que produz o diagnóstico que alterou o sistema.

5.6.1 O que a porta de decisão estava a separar

A porta implantada era um limiar fixo sobre a probabilidade calibrada, que deixava passar as candidatas acima de 0.50. A pergunta com significado é a de saber se esse limiar separava títulos ou separava empresas, e a medição foi realizada sobre as 4,366 decisões efetivamente tomadas em produção. A Figura 5.15 apresenta o resultado.

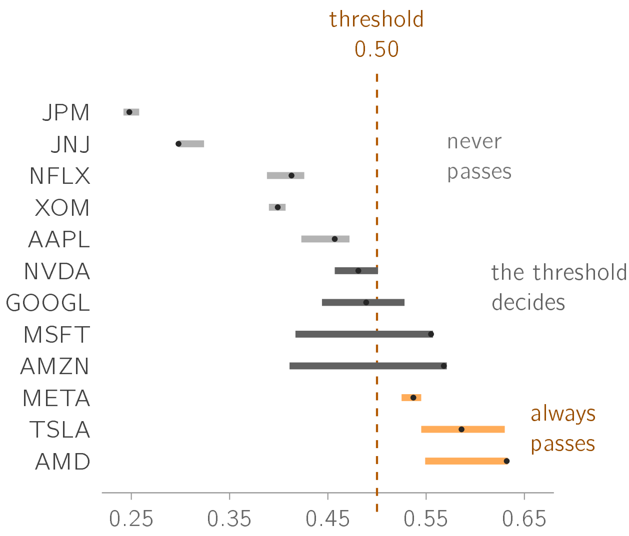

Figure 5.15: Intervalo de todas as pontuações atribuídas pelo modelo a cada empresa em produção, com a mediana assinalada. As barras claras nunca alcançam o limiar, as escuras atravessam-no e as alaranjadas situam-se sempre acima. Oito das doze empresas encontram-se inteiramente de um dos lados, o que significa que para essas a decisão estava tomada antes de o título ser lido.

Os valores confirmam a leitura da figura. Sobre as 36,925 decisões registadas, a amplitude média das pontuações dentro de cada empresa é de 0.072, ao passo que a amplitude entre as medianas das empresas é de 0.392, isto é, 5.4 vezes superior. Duas empresas situavam-se sempre acima do limiar e cinco nunca o alcançavam, pelo que apenas em cinco das doze o limiar chegava a decidir. Em 48% dos títulos distintos o resultado estava determinado pela empresa antes de o título ser lido. Contada por decisão registada, a fração sobe para 60%. Cita-se a primeira: a repontuação dos mesmos títulos a cada ciclo de sessenta segundos inflaciona a segunda, e de forma desigual entre empresas.

Numa janela anterior e mais curta, de 4,366 decisões, os mesmos indicadores valiam 0.064, 0.385 e 6.1 vezes, com três empresas sempre acima e quatro em que o limiar decidia. A conclusão estrutural não depende da janela. Em nenhuma das avaliações registadas a Apple alcançou o limiar, com um máximo observado de 0.472, ao passo que a AMD o ultrapassou em todas.

As duas contagens medem quantidades distintas e ambas são reportadas. O sistema repontua os mesmos títulos a cada ciclo de sessenta segundos, e a duplicação é maior nas empresas com menos notícias, que são as que nunca ultrapassam o limiar, pelo que contar decisões desloca a fração para cima. Contada uma vez por título distinto, sobre uma janela mais ampla de 36,925 decisões, a fração é de 48%. A conclusão estrutural não depende da contagem escolhida: continuam a existir empresas inteiramente de um dos lados do limiar, e a amplitude entre empresas continua a ser várias vezes a amplitude dentro de cada uma.

O defeito tem a mesma forma que o identificado na Secção 5.2, uma camada acima. Naquele caso, um limiar fixo sobre o retorno mede a volatilidade da empresa em vez da raridade do dia; neste, um limiar fixo sobre uma pontuação quase constante por empresa ordena empresas em vez de notícias.

Nenhum dos testes automáticos existentes o cobria. O modelo foi avaliado sobre a distribuição do conjunto de treino e implantado atrás de dois filtros que a alteram, e cada componente cumpria o seu contrato. O teste que o teria apanhado é o que este trabalho passou a executar: comparar a amplitude das pontuações dentro de cada empresa com a amplitude entre empresas. O que estava errado era uma suposição sobre a ligação entre eles, não enunciada em nenhum ponto. Corresponde à classe de custo que D. Sculley et al. (2015) arrumam sob dependências de dados e emaranhamento entre componentes, com o sintoma que os autores descrevem: o sistema continua a produzir valores de aparência normal.

Existe uma segunda causa possível, que os registos não permitem separar da primeira. Uma das nove entradas é o retorno do próprio dia. No treino corresponde ao movimento completo do dia da notícia. Em produção é calculado sobre a última barra diária disponível, que durante a sessão é o fecho da véspera ou uma barra incompleta. As duas causas produzem o mesmo sintoma, e o registo, à data desta medição, guardava a probabilidade e o resultado mas não o valor das entradas no momento da pontuação, pelo que a sua separação exigiria registar essas entradas e voltar a observar durante semanas.

Essa instrumentação foi construída e implantada em 4 de setembro de 2026, e o registo passou a guardar, por decisão, o valor de cada uma das nove entradas, a data da última barra de preço utilizada e a identidade do modelo que pontuou. Duas consequências decorrem dela e nenhuma é um resultado. A primeira é que as decisões anteriores a essa data permanecem não reproduzíveis a partir do registo, uma vez que recalcular hoje as entradas de um dia passado utilizaria uma série de preços que já contém o que veio a seguir; as medições deste capítulo assentam nessas decisões e não são afetadas pela alteração. A segunda é que a janela de observação necessária para separar as duas causas começou nessa data, e o critério de aceitação de qualquer comparação futura foi fixado antes de existir candidato, pela mesma razão que o Capítulo 3 invoca para as restantes experiências. Nada se afirma aqui sobre o seu desfecho.

A correção aparente consistiria em substituir o limiar fixo por um limiar relativo a cada empresa. A simulação dessa alternativa sobre as mesmas decisões resolve metade do problema, uma vez que as doze empresas passam a estar representadas, e introduz outro. Nas empresas cuja pontuação quase não varia, o percentil recai sobre a constante e quase todas as decisões empatam acima dele; uma delas passa a gerar 874 alertas. O limiar relativo deixa de escolher por materialidade e passa a escolher por desempate, que é precisamente o artefacto documentado na Secção 5.4. Dentro de uma empresa a pontuação do modelo quase não varia com o título, e nenhuma regra de decisão aplicada a essa pontuação a pode tornar sensível à notícia, porque a informação que distinguiria um título do seguinte não está presente.

Em consequência, o modelo deixou de constituir porta e passou a constituir critério de ordenação, que é a utilização que a medição sustenta, e o controlo de volume passou para um orçamento diário de cinco alertas. Esta alteração aproxima duas políticas que divergiam: a dissertação avaliava uma política de orçamento e o sistema implantava uma política de limiar, pelo que o valor reportado descrevia um sistema distinto do que estava em funcionamento. As duas políticas passaram a pertencer à mesma família sem se tornarem idênticas.

A alteração foi medida sobre os alertas efetivamente entregues, e o efeito pretendido verifica-se. A quantidade observada é a concentração, definida como a fração dos alertas que pertence às três empresas mais alertadas, e a sua vantagem é ser adimensional, o que permite comparar janelas de dimensões distintas. Nos alertas de notícia, que são os que o orçamento governa, a concentração desce de 0.713 para 0.480, com intervalo de confiança da diferença a excluir zero, e o número de empresas que chegam a produzir um alerta passa de sete para nove das doze monitorizadas.

O que sustenta a leitura não é, porém, a descida isolada, mas a comparação com uma série que a alteração não governa. Os alertas de movimento de preço atravessam o mesmo período, as mesmas empresas e as mesmas condições de mercado, e o orçamento diário não os conta. Nessa série a concentração passa de 0.449 para 0.485, com o intervalo a conter zero. A quantidade governada deslocou-se e a quantidade não governada não, o que afasta a explicação alternativa de que o mercado se teria distribuído de forma mais uniforme no segundo período.

Cinco dias são excluídos da janela posterior, e a razão é registada por ser ela própria um resultado. Entre 25 e 29 de agosto de 2026 o contador diário residia em disco efémero e regressava ao valor inicial a cada arranque do processo, o que produziu séries de exatamente cinco alertas nos instantes dos arranques e um dia com vinte. Nesses dias o orçamento não estava em vigor, e a sua inclusão elevaria a concentração posterior para 0.514. A reamostragem é efetuada por dia, que é a unidade sobre a qual o orçamento atua.

A comparação não constitui um ensaio aleatorizado. A série de controlo partilha o período mas não foi atribuída ao acaso, e as duas janelas diferem em duração e em condições de mercado. O que se estabelece é que a quantidade governada se deslocou e a não governada não, e não que nenhuma outra causa exista.

A métrica avaliada ordena todas as candidatas do bloco de teste e apenas depois preenche cinco lugares em cada dia, o que exige conhecer o dia completo antes de decidir e constitui por isso uma política diferida. O sistema implantado decide em linha, de sessenta em sessenta segundos, e gasta o primeiro lugar na primeira notícia que atravesse as portas, porque no momento da decisão a notícia da tarde ainda não existe. A precisão dentro do orçamento reportada neste capítulo é, por conseguinte, um limite superior do que a política implantada obtém.

5.6.2 O desempenho do modelo sobre as decisões que tomou

O sistema regista todas as decisões de triagem e, decorridos os dias necessários para o preço responder, confronta-as com o resultado efetivo. Duas medições resultam desse registo: a fração de casos que se revelaram materiais de cada lado da porta, e a capacidade de ordenação do modelo, medida pela Receiver Operating Characteristic Area Under the Curve (ROC-AUC). A Figura 5.16 apresenta ambas, sobre 825 decisões já maturadas.

Sobre a população que efetivamente observa, esta amostra não distingue o modelo do acaso em nenhuma das duas medidas. As notícias que deixou passar revelaram-se materiais em 58.9% dos casos, contra 61.7% das que travou, sobre uma taxa-base de 60.2%; os intervalos das duas frações sobrepõem-se, pelo que a ordenação aparente não é estabelecida. O que

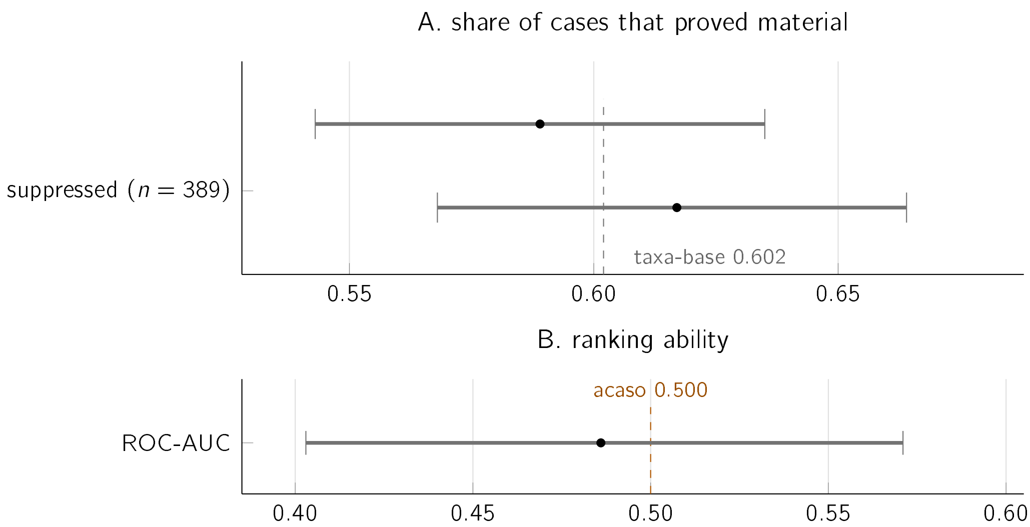

Figure 5.16: Em A, a fração de casos que se revelaram materiais entre as decisões que o modelo deixou passar e entre as que travou, com intervalos de confiança a noventa e cinco por cento e a taxa-base do conjunto. As suprimidas apresentam fração superior às mantidas, mas os dois intervalos sobrepõem-se em quase toda a extensão e não as separam. Em B, a capacidade de ordenação medida sobre as 239 unidades independentes, com o intervalo a conter confortavelmente o valor de acaso.

fica medido é a ausência de benefício detetável com esta dimensão de amostra, e não a ausência de benefício.

Estes valores exigem uma reserva de precisão considerável: as 825 decisões não constituem 825 observações independentes, dado que o rótulo é atribuído por par formado pela empresa e pelo dia, e as linhas repartem-se por apenas 239 pares distintos. Medida sobre essas unidades, a capacidade de ordenação situa-se em 0.486 com intervalo de confiança entre 0.403 e 0.571, num intervalo em que o acaso vale exatamente 0.500. A leitura correta não é a de que o modelo seja pior do que a ausência de modelo, mas a de que esta amostra não o distingue do acaso.

A medição foi repetida quando o registo dispunha de 825 decisões maturadas contra as 530 da primeira leitura, e quase nada se alterou: a diferença entre mantidas e suprimidas estreitou-se, a capacidade de ordenação passou de 0.494 para 0.486 e o intervalo encolheu. Com metade mais evidência, o veredicto mantém-se e torna-se mais firme.

A explicação mais provável não é a de que o modelo esteja avariado, mas a de que esteja em excesso. Quando é invocado, as notícias já atravessaram o filtro de relevância e o filtro de frescura, e esses filtros elementares removeram grande parte daquilo que o modelo foi treinado para remover, restando pouco para separar. Um modelo avaliado isoladamente e implantado atrás de filtros nunca foi avaliado na distribuição que efetivamente observa, e esta é a constatação com maior capacidade de transferência do trabalho.

Três das doze empresas monitorizadas, a AMD, a Netflix e a Meta, não figuram no corpus de treino, e uma quarta, a Microsoft, figura apenas no bloco de teste. O modelo pontua-as na mesma, porque o que recebe são valores numéricos e não a identidade da empresa, mas fá-lo sobre uma combinação de entradas que não observou. Duas das ausentes, a AMD e a Meta, estão entre as que se situam sempre acima do limiar, pelo que a determinação por empresa descrita acima se concentra precisamente em nomes que o modelo nunca observou. Constitui um segundo sentido, mais literal, em que o modelo implantado opera fora da distribuição em que foi avaliado, e as decisões dessas empresas entram nas 825 acima como quaisquer outras.

5.6.3 Deriva das distribuições

Uma segunda razão, independente da anterior, pode fazer com que um modelo implantado deixe de servir: os dados alteram-se. O modelo foi treinado sobre o período de 2018 a 2023 e opera em 2026. A medida utilizada é o Population Stability Index (PSI), que confronta aqui a distribuição de cada entrada no bloco de treino, compreendido entre 2018 e 2022, com a do bloco de teste, de 2023. Não é, portanto, a distância entre o treino e o presente, que a subsecção seguinte discute em separado: é a deriva interna ao próprio protocolo. As bandas são as convencionadas em risco de crédito, segundo as quais valores abaixo de 0.10 indicam estabilidade, valores entre 0.10 e 0.25 deslocamento moderado e valores acima de 0.25 deslocamento significativo. A Figura 5.17 apresenta o resultado.

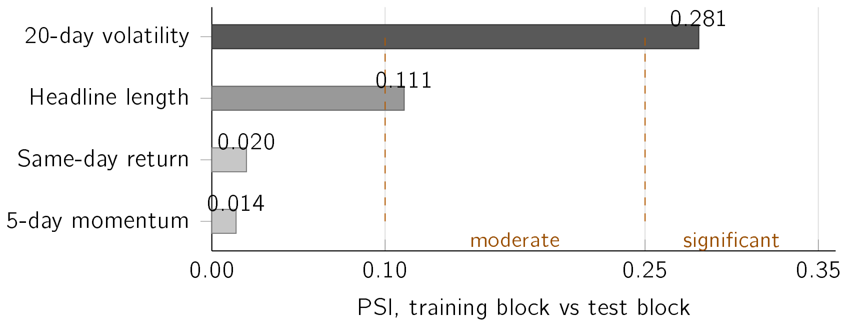

Figure 5.17: Índice de estabilidade populacional de cada entrada entre o bloco de treino, de 2018 a 2022, e o bloco de teste, de 2023, com as bandas convencionadas assinaladas. A entrada de que o modelo mais depende é a que mais se deslocou. A prevalência do rótulo passa de 0.3854 para 0.3781 entre as duas pontas, mas não é estável pelo meio: no bloco de validação, temporalmente entre ambas, chega a 0.4704.

O resultado tem duas metades. A entrada que mais se deslocou é a volatilidade dos vinte dias anteriores, situada na banda significativa, ao passo que as restantes permanecem estáveis ou moderadas. O rótulo desloca-se pouco entre as pontas, com a prevalência de positivos a passar de 0.3854 para 0.3781, mas não é estável: o bloco de validação, temporalmente entre os dois, chega a 0.4704. A prevalência oscila com o regime de volatilidade em vez de derivar numa direção, o que é coerente com a volatilidade ser a entrada que mais se desloca.

A leitura correta desta medição é a de que a deriva não é externa aos valores deste capítulo: é a que o próprio protocolo atravessa, uma vez que treina em 2018–2022 e avalia em 2023. A PR-AUC reportada é já uma medida sob esta deriva, e a combinação é desfavorável porque a entrada de que o modelo mais depende é precisamente a que mais se deslocou.

O que esta medição não estabelece é a distância entre o treino e 2026, que é o período em que o sistema opera. Um instantâneo sobre cotações atuais aponta na mesma direção, com a volatilidade a ser de novo a entrada que mais se desloca. Assenta, porém, em janelas deslizantes consecutivas que partilham a quase totalidade das observações, o que inflaciona o índice e torna a sua magnitude não comparável com a da figura. O valor permanece por medir, e não é estimado por analogia. A deriva é, em qualquer dos casos, medida e não corrigida, e a Secção 6.5 enuncia o que seria necessário para a corrigir.

5.6.4 O sistema completo contra as alternativas existentes

As secções anteriores comparam componente a componente. Falta a comparação ao nível do sistema: dadas as notícias de um dia, a política escolhe as cinco a apresentar melhor do que uma regra disponível sem custo. A Figura 5.18 responde. Todas as políticas escolhem cinco por dia, sobre o mesmo bloco de teste e com o mesmo rótulo, e o desempate é aleatório e explícito em todas.

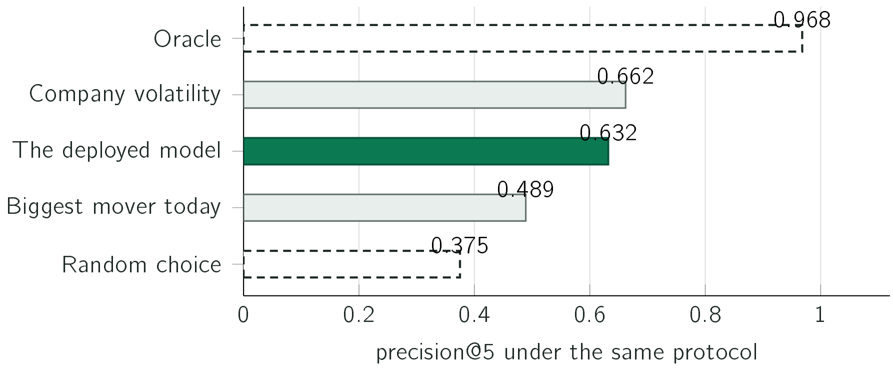

Figure 5.18: Cinco políticas de seleção sobre o mesmo bloco de teste. A escolha aleatória serve de referência; o oráculo é um máximo teórico não utilizável, obtido com conhecimento das respostas. Todas escolhem conhecendo o dia completo, pelo que cada valor é um limite superior do que a mesma regra obteria decidindo ao minuto.

Três leituras resultam da figura. A primeira é que o sistema supera a alternativa realista: apresentar notícias das empresas com maior movimento do dia é o que qualquer aplicação de bolsa já faz sem custo, e obtém 0.489, contra 0.632 do sistema, um ganho de 0.143 sob o mesmo protocolo. Este valor mede a capacidade de ordenar o rótulo aproximado no bloco histórico, e não a utilidade percebida por uma pessoa nem o valor causal dos alertas em funcionamento.

A segunda é que o sistema continua a não superar a volatilidade. Treze constantes calculadas exclusivamente sobre o bloco de treino obtêm 0.662, o que é coerente com a ablação da Secção 5.4.5: o modelo é, na prática, uma tabela por empresa, e a volatilidade é outra tabela por empresa, mais simples, sendo esta a que dispensa treino.

A terceira, e mais útil, é a localização da limitação pelo máximo teórico. O melhor resultado possível, escolhendo com conhecimento das respostas, seria de 0.968, o que significa que em quase todos os dias existem efetivamente cinco notícias materiais no conjunto disponível. O sistema não está limitado por escassez de matéria-prima, mas pela incapacidade de identificar quais são. A margem entre 0.632 e 0.968 é de 0.336 e reside quase integralmente na separação de notícias dentro de cada empresa, que é precisamente o que o modelo implantado não consegue realizar. É a mesma conclusão obtida por um terceiro caminho independente.

Todas as políticas desta comparação escolhem de entre o que a fonte entregou, o que deixa de fora a pergunta anterior a todas, relativa aos dias em que nada foi entregue. Sobre um ano de notícias captadas, existia pelo menos uma notícia relevante em 88.5% dos dias que o detetor considerou invulgares, num total de 417, e em 47 dos 52 dias mais extremos, dimensão que não permite distinguir as duas frações. Nos restantes, nenhuma alteração ao sistema produz efeito, uma vez que uma notícia não associada à empresa pela fonte gratuita não chega a entrar no percurso. A cobertura é ainda muito desigual, com a empresa pior coberta a situar-se nos 50%. Este valor é um limite superior, e a distinção determina o que ele vale: estabelece que existia uma notícia, e não que tenha chegado a notícia correta. A Secção 6.4 retoma-o como limitação.

Uma linha de base não foi medida, e a razão fica registada. A alternativa mais natural consistiria em apresentar as primeiras cinco notícias a chegar. Não é mensurável neste protocolo, uma vez que o conjunto de dados histórico não conserva a hora de publicação e está ordenado por data e por empresa, pelo que utilizar a ordem do ficheiro mediria a ordem alfabética das empresas. Seria o mesmo artefacto documentado na Secção 5.4, desta vez introduzido deliberadamente. Uma linha de base errada é pior do que uma linha de base em falta.

5.6.5 Utilidade percebida dos alertas entregues

O canal passou a acompanhar cada alerta entregue com dois botões, *útil* e *não ajudou*, e a registar o voto de quem carrega. As regras de análise foram fixadas antes de existir um único voto e não foram alteradas depois: mínimo de vinte votos efetivos para reportar qualquer proporção, um voto por pessoa e por alerta com o último a substituir o anterior, salvaguarda que repete o cálculo sem o votante dominante quando este representa mais de quarenta por cento dos votos, sempre sujeita ao mesmo mínimo, e intervalos de confiança de Wilson, apropriados a proporções com amostras pequenas. Contaram apenas votos sobre alertas presentes no registo partilhado.

Foram registados 81 votos válidos entre 2026-09-02 e 2026-09-04, que correspondem a 42 votos efetivos de 3 pessoas distintas, sobre 29 alertas. 10 votos foram posteriormente alterados pela mesma pessoa, e apenas o último de cada par conta. 5 votos foram excluídos por não corresponderem a nenhum alerta do registo partilhado.

Dos 42 votos efetivos, 41 classificaram o alerta como útil, ou seja 98%, com intervalo de confiança de Wilson a 95% entre 88% e 100%. A largura deste intervalo é a medida honesta do que 42 votos permitem afirmar.

Uma única pessoa representa 67% dos votos efetivos, acima do limite de 40% fixado no protocolo. Excluindo-a, restam 14 votos, dos quais 14 classificam o alerta como útil. Este recorte também fica abaixo do mínimo de 20; nenhuma segunda proporção é reportada e a amostra não sustenta uma leitura independente do votante dominante.

Estes votos constituem retorno observacional em contexto real e não substituem o estudo controlado que a Secção 6.4 identifica como a lacuna de maior peso deste trabalho, e que permanece por realizar.

Quatro limitações acompanham este resultado e nenhuma delas se resolve com mais votos. A primeira é a autosseleção: vota quem quer, e quem considera um alerta indiferente tende a não carregar em nada, o que empurra a amostra para os extremos. A segunda é a ausência de contrafactual, uma vez que nenhum grupo recebeu a variação de preço sem a explicação que a acompanha; nada aqui atribui a utilidade à explicação em si. A terceira é a distância entre utilidade percebida e decisão melhor, que é a hipótese fundadora deste trabalho e permanece por testar. A quarta é o desconhecimento de quem vota: o canal é público e não se sabe se quem responde pertence ao público que a dissertação assume.

5.7 Síntese dos resultados

Esta secção reúne as comparações realizadas ao longo do trabalho, incluindo aquelas em que o sistema mantém a opção que a medição não favorece.

Duas das três respostas são afirmativas e uma é negativa. A resposta negativa é a que exigiu maior esforço de verificação, e é também a que estabelece que a avaliação foi construída de modo a poder produzir uma resposta negativa. A Tabela 5.2 apresenta cada escolha técnica ao lado da alternativa medida contra ela.

Três linhas da tabela merecem desenvolvimento. A primeira respeita ao codificador de domínio financeiro, cuja derrota tem explicação no objetivo de treino e não no conhecimento do domínio, conforme a Secção 5.3 estabelece. A segunda respeita à janela de vinte dias, que é a opção mais difícil de justificar, uma vez que a única medição existente sobre este ponto não a sustenta e a justificação de responsividade não é quantitativa. A terceira respeita ao estimador de volatilidade, mantido por explicabilidade contra uma alternativa medida como superior.

O capítulo documenta três ocasiões em que uma técnica mais simples superou uma técnica mais sofisticada em condições equivalentes, e três ocasiões em que sucedeu o inverso e o sistema conservou mesmo assim a opção implantada. Nestas últimas a razão difere em cada caso, e nenhuma delas é o resultado da medição: o estimador de pesos iguais foi mantido por explicabilidade, a janela de vinte dias por responsividade, e o codificador de frases mais pequeno por custo computacional. A distinção entre os dois conjuntos é a distinção entre resultado e escolha, e o Capítulo 6 retoma-a ao estabelecer o que este trabalho demonstra e o que apenas decide.

Table 5.2: As decisões técnicas do trabalho, cada uma com a alternativa medida contra ela. Três linhas registam que a opção implantada não obteve o melhor resultado, e a razão da sua manutenção consta da coluna respetiva. As três últimas linhas são restrições fixadas antes de qualquer medição, e não resultados: apresentá-las como comparação atribuir-lhes-ia uma autoridade que não possuem.

Opção implantada	**Alternativa medida**	**Leitura**
z-score deslizante	Limiar fixo de três por cento: amplitude 0.344 contra 0.015	Um limiar fixo mede a volatilidade da empresa e não a raridade do dia.
z-score deslizante	*Isolation Forest* com F_1 de 0.269 e *Local Outlier Factor* com 0.280, contra 0.530	Os detetores aprendidos assinalam quase tudo e acertam pouco.
Janela de vinte dias	Não obteve o melhor resultado: dez dias produzem F_1 de 0.385, vinte dias 0.516 e sessenta dias 0.678	A medição favorece a janela mais longa. Mantida por responsividade, com a reserva de circularidade do rótulo declarada.
Desvio-padrão de pesos iguais	Não obteve o melhor resultado: o decaimento exponencial produz F_1 de 0.664 contra 0.516	Mantido por ser explicável a um leitor não especializado, e registado como escolha e não como resultado.
Embeddings de frase	Sobreposição de palavras: precisão@5 de 0.346 contra 0.514	Vocabulário comum não implica assunto comum, e assunto comum não exige vocabulário comum.
MiniLM em vez de MPNet	Não obteve o melhor resultado: o MPNet produz 0.538 contra 0.514; um codificador de domínio financeiro produz 0.420	O modelo mais pesado é melhor e foi substituído por custo. O codificador de domínio obteve o pior resultado de todos, por medição e não por suposição.
Regressão logística	Árvores com reforço do gradiente: PR-AUC de 0.469 contra 0.538	O modelo mais expressivo não obteve melhor resultado, e o modelo simples é explicável.
Calibração de Platt	Regressão isotónica, comparada sobre a mesma validação	A calibração de Platt iguala ou supera no índice de Brier em todas as famílias avaliadas.
Semelhança do cosseno	Distância euclidiana	Com vetores normalizados as duas produzem a mesma ordenação, e a primeira dispõe de leitura mais direta.
Dois gatilhos, notícia e preço	Fusão com o volume e a intensidade de notícia, na linha dos painéis que combinam sinais de origens distintas (*World Monitor* 2026): perde em dois dos três orçamentos, com -0.050 e -0.032, e ganha no terceiro por $+0.004$	Um ganho que depende do orçamento escolhido, e uma ordem de grandeza menor do que as perdas, não sustenta a adoção. O volume continua a ser calculado e apresentado como contexto, sem originar alertas.
Apenas fontes gratuitas	—	Restrição fundadora: define o público que o sistema pode servir.
Ausência de previsão de preços	—	Restrição de desenho: é o que torna verificável tudo o restante.
Mercado dos Estados Unidos	—	Restrição de âmbito: é onde as fontes gratuitas apresentam cobertura.

Chapter 6

Conclusions

Chapter 5 measured each component in isolation. This one brings together what those measurements yield. It synthesises the work carried out, presents the answer obtained for each research question, identifies the limitations of the study and sets out the lines of future development.

6.1 Main conclusions

The work produced a system in operation and an evaluation of each of the techniques that compose it. InvestiGator monitors twelve companies in a continuous cycle, delivers explained alerts to a public channel and provides a page on which both the messages sent and the decisions that led to silence are recorded. Over the periods documented in Chapter 5, 367 messages were delivered, between 9 July and 13 August 2026, and 36,925 triage decisions were logged over a wider window, and the precedent base accumulated 38,214 cases with measured impact. These numbers describe a deployed system and not a demonstration built for the purpose.

The component that gives the work the character of a dissertation is not, however, the system, but the evaluation to which it was submitted. Three of the four techniques were compared with simpler alternatives, chosen and declared before the measurement was carried out, which is the condition for the result to be able to be negative. In one of them the result was indeed negative. The fourth technique, the decomposition of the movement, does not admit a comparison of the same nature, since the split of a return has no observable external referent. What was measured of it is what remains falsifiable: the quality of the fit that supports it, and the ability to produce distinct answers for distinct companies.

The engineering contribution does not lie in the size of the learned model. The system trains a single model, with nine inputs, and that model loses to a one-variable baseline. Confidence in that value, and in the negative conclusion it supports, depends on what had to be built around it. Section 4.7 describes the seven pieces. The dataset is labelled from two independent sources. The label is declared as a decision, and nine alternative definitions are measured instead of the one adopted being assumed. The temporal separation is verified by an automatic procedure. The calibration comes with the measurement and the declaration of the bias that affects it. The artefacts are kept under version control. The deployed model is demonstrably the model evaluated. And the system in operation continues to be observed. It is in that set, and not in the model, that the work resides.

A second conclusion runs across the four case studies and belongs to none of them in particular. On three distinct occasions, a simpler technique obtained a better result than a more sophisticated one under equivalent conditions: the rolling z-score against two learned detectors, recent volatility against text representations, and a table of thirteen constants against the trained model. On three other occasions the reverse happened, with the alternative measured obtaining the better result. The system kept the deployed option for distinct reasons in each case: the equally weighted estimator for explainability, the twenty-day window for responsiveness, and the smaller sentence encoder for computational cost.

The distinction between these two cases is the distinction between a result and a choice, and the present chapter keeps it explicit: what the measurement establishes is reported as a result, and what the author decided in spite of the measurement is reported as a decision, accompanied by what that decision costs when the measurement separates the alternatives, and by the statement that it costs nothing in PR-AUC when it does not separate them.

6.2 The answers to the research questions

This section answers the three questions stated in Section 1.3 and delimits, in each case, what the answer does and does not allow to be claimed. Figure 6.1 brings the three together.

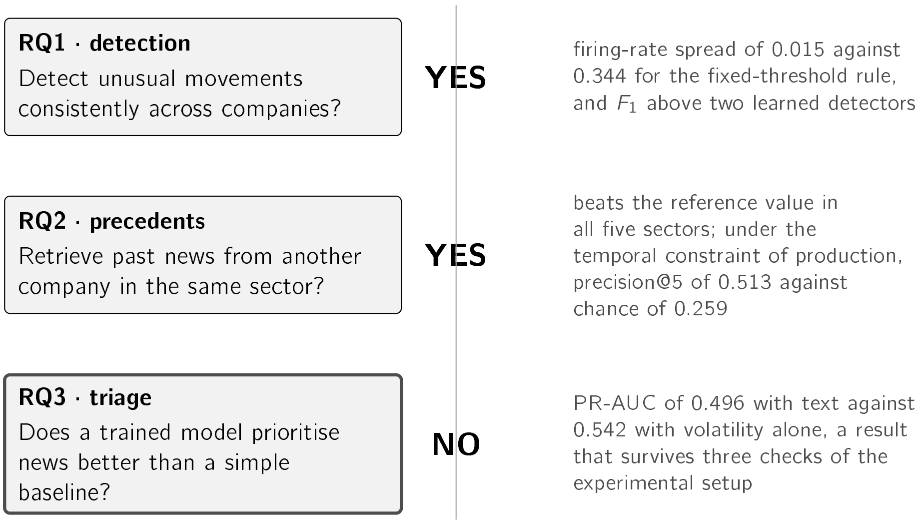

Figure 6.1: The three answers, with the negative answer presented with the same prominence as the other two. The third row is the one that gives credit to the others: a work that reports only the favourable results does not allow one to determine whether the affirmative answers were measured or presumed.

6.2.1 Detection of unusual movements

The answer to RQ1 is affirmative. A z-score over a rolling twenty-day window flags rare days at a practically identical rate in companies of very different volatilities. The spread between the most and the least flagged company is 0.015, against 0.344 for a fixed threshold rule at three per cent. Against an approximate label, the method obtains $F_1 = 0.516$ and the fixed rule 0.218. Subjected to an equivalent causal protocol, the same method further beats two detectors designed for this task, with F_1 of 0.530 against 0.269 for *Isolation Forest* (F. T. Liu, Ting, and Zhou 2008) and 0.280 for the *Local Outlier Factor* (Breunig et al. 2000).

The deployed configuration is not, however, the best measured within the family itself. With the same recall, an exponentially weighted estimator obtains $F_1 = 0.664$ and a sixty-day window obtains 0.678, against the 0.516 of the configuration in production. What RQ1 establishes is that the rolling family beats the fixed threshold and the two learned detectors, and not that the parameterisation chosen is the best of that family.

This result is less surprising than the difference in values suggests, and the literature allows it to be situated. The two learned detectors define rarity from the geometry of the data, one by isolating the point and the other by comparing densities, and were designed for spaces with multiple interacting variables. The problem treated here has a single series of returns per company, in which the relevant structure is temporal and not geometric: agitated periods follow calm ones. It is the property that ARCH and GARCH models formalise (Bollerslev 1986; Engle 1982). The rolling window captures it directly; a detector trained over the complete set would have to learn it. When the assumption built into a simple method coincides with the actual structure of the problem, the margin for beating it is small.

The reach of the answer is delimited by three conditions. The comparison covers fifteen large-capitalisation companies, selected in 2026 and evaluated over the past, which survived by construction; three years of quotes; and a single market. The label used as a secondary argument is itself relative to the volatility of each company, which is why the principal argument is the consistency of the firing rate, which does not depend on labels. The three per cent threshold of the alternative rule was not optimised, so what the comparison establishes is not that no fixed threshold would obtain a better result, but that a fixed threshold rule measures a quantity distinct from the one intended.

6.2.2 Precedent retrieval

The answer to RQ2 is affirmative, and its formulation requires the sector level. Over the recent corpus, retrieval by sentence similarity (Reimers and Gurevych 2019) obtains precision@5 of 0.514, against 0.346 for vocabulary overlap, 0.240 for random choice and 0.126 for presenting the most recent news. Here too the deployed model is not the best measured: a larger variant obtains 0.538 under the same protocol, and was set aside on computational cost.

The aggregate value is not, however, the appropriate way of stating the result. Almost half the news in the corpus belongs to the technology sector, which makes available a strategy that dispenses with any model and does not even look at the query, consisting in invariably returning news from that sector; its precision equals the fraction of technology queries, 0.467. That strategy is useless as a product, since it is right on every query of one sector and on none of the remaining four, but its existence shows that the aggregate measures in part the composition of the corpus.

The comparison that survives the objection is carried out within each sector, where the method beats the respective reference value in all five. The absolute margin is largest in the most abundant sector, at $+0.283$ in technology; in the sparse sectors, where the reference value is around 0.072, the method obtains several times that value with a smaller absolute margin. This is the claim the work supports.

Two further checks delimit the answer. Over the complete historical corpus, with approximately eighty thousand news items from six years, precision rises to 0.595. The reference value rises in the same proportion, and the margin holds. The method does not degrade with scale, nor does it improve with it. Imposing the constraint the system actually faces, under which the precedent must be prior to the query, reduces precision to 0.513 and the reference value to 0.259, with the margin varying by 0.008. The method keeps its advantage when forced to operate as it operates in production.

The second caveat has direct consequences for the product. Retrieval identifies topic and not direction: the agreement between the direction of the movement of the query news item and that of the precedents stands at 0.708 against a chance value of 0.688, a difference of 0.020. Two news items can deal with the same subject and the price move in opposite directions. The result is consistent with what the semi-strong form of the efficient market hypothesis would lead one to expect (Fama 1970), under which public information is already reflected in the price, without the work testing it or relying on it to establish any conclusion. The practical consequence is the decision to present the precedents individually, with date and outcome, rather than summarising them into an average liable to be read as a forecast.

A final delimitation concerns the label. Membership of the same sector is a verifiable approximation of relevance, and not the property one intended to measure. The lexical baseline is the most elementary form of its family, without weighting by the rarity of terms and without normalisation by length. The seventeen percentage points of advantage measure the distance to a starting point, and not to a tuned ranking function (Robertson and Zaragoza 2009).

6.2.3 News triage

The answer to RQ3 is negative. No variant incorporating the text of the headline beats a baseline built only on the volatility of the twenty preceding days, with 0.496 of PR-AUC against 0.542, and the Brier score follows the same ordering. The result survives three checks. The first repeats the comparison granting the text the best conditions available, and raises its best value to 0.533, without reaching the baseline. The second resamples by groups formed by company and day, and keeps the interval excluding zero. The third repeats the measurement for nine alternative combinations of label threshold and horizon, and volatility matches or beats the model with text in all nine.

The answer gained precision over the course of the work and that precision changes what is drawn from it. The earlier comparisons place the representations side by side and therefore answer the question of which of them is better. The question of greater use to whoever builds a system is a different one, and consists in determining whether the text adds to what is already known. Measured against the per-company lookup table, which contains everything the model knows about the company and nothing about the news, the answer is affirmative: the headline adds +0.012 of PR-AUC, with an interval between +0.004 and +0.020, which excludes zero. It is the only occasion on which a text representation shows a detectable increment on this task.

The value falls below the 0.02 limit that Chapter 5 fixed before the measurements, and that limit applies in both directions. What is established is an upper bound on the effect of the text, and not a quantity the system can make use of.

Three measurements prevent this increment from reopening the verdict. The sum does not beat volatility alone, and the interval of the difference contains zero. The increment does not survive the metric that describes the product: precision within the budget of five alerts per day stays at 0.662 with and without text. And the ability to separate news items of the same company remains at the level of chance. What these measurements add to the negative answer is not a caveat but a location: the information the text carries distinguishes companies and periods, when the product needed it to distinguish news items. The formulation that survives is the weakest possible, corresponding to an upper bound on an effect and not to a discovery.

There is an established line of research that uses text to anticipate price movements (Ding et al. 2015; Xing, Cambria, and Welsch 2018), and this result does not contradict it. The difference does not lie in the size of the text, since Ding et al. (2015) likewise operate on headlines, but in the treatment given to it. Those authors extract event structures and train a network to predict the direction of the movement. Here headlines are compared by similarity, and direction is never the object of a prediction. The corpora and the evaluation horizons likewise differ. The claim this work supports is narrower than a refutation: under this data constraint and with this family of methods, the textual signal adds nothing to what recent volatility already indicated. The reach of the result is therefore limited to a regime of resources and to a choice of representation, and does not extend to language.

There is, even so, a theoretical reading that frames it and that is stated with the same caution. The semi-strong form of the efficient market hypothesis (Fama 1970) holds that publicly available information is already reflected in the price. If that is so, a public headline has, at the moment it is read, no information that recent prices do not contain, and the result obtained is what that hypothesis would lead one to expect. The resource condition of this work worsens the expectation rather than easing it: between the publication of a news item and its detection by free sources a median of 353 minutes passes, an interval during which any market reaction has already occurred. The system observes, by construction, information that has ceased to be new.

This reading is a framing and not a conclusion. The work does not test the efficient market hypothesis, and a negative result obtained with one family of methods over a concrete corpus does not confirm it; confirming it would require a design this one does not have. What the reading adds is the reason why the result is less surprising than the initial hypothesis assumed, and why its most likely correction lies not in the representation of the text but in the delay with which it arrives.

6.3 Objectives achieved

Of the four objectives stated in Section 1.3, three are fully met and one is met with the caveat described below.

The first objective, to build the complete system from data acquisition to the delivery of the alert, is met. Both triggers work end to end in production, on free sources, and the path of an event is documented in Chapter 4 with the values of a real case.

The second objective, to evaluate each component against transparent baselines, is met for the three techniques in which an external referent exists. In one of the cases it went beyond what was planned: the detector was compared not only with a fixed threshold rule, but also with two learned methods. The decomposition of the movement is the caveat: there is no ground truth against which to measure the accuracy of a split, so what was measured is what remains falsifiable in it, corresponding to the quality of the fit and to the ability to discriminate. Internal comparisons that would be possible were not carried out, such as that of the one-factor model against the two-factor model, or that of the raw sensitivities against the shrunk ones.

The third objective, to ensure that the explanations presented are faithful to what the system computed, is met by construction and not by subsequent verification. Every value displayed comes from the object that computed it, and the text is composed from those pieces, so it is not possible for the alert to state a value distinct from the one the system obtained. Faithfulness is, however, a property of the system, and not of the relation between the system and whoever reads it; the usefulness of the explanations for a retail investor was not measured, and Section 6.4 deals with that absence.

The fourth objective, to report the results obtained including those that do not confirm the initial expectations, is met. The negative answer to RQ3 appears in Figure 6.1 and in the body of this chapter with the same prominence as the others, accompanied by the three checks to which it was submitted and by the three occasions on which a simpler technique beat a more sophisticated one.

6.4 Limitations

This section lists the limitations of the work in order of importance. Table 6.1 presents them together, with the work that would close each one and the nature of its cost; the text that follows develops the four of greatest consequence and summarises the rest. Most are quantified. Two are not, by their nature: the absence of a controlled study is an absence and not a measure, and the distance between each label and the property it approximates would only be measurable against a human annotation that does not exist.

Table 6.1: The eleven limitations, in order of importance, with the work that would close each one and the nature of the cost of that work. The two that depend on calendar time are the ones that require human judgement, and they are also the two on which the rest most depend: until a sample annotated by people exists, neither the usefulness of the explanations nor the quality of the labels ceases to be an approximation.

Limitation	**What closes it**	**Cost**
No controlled study with people	Run the protocol designed, with six to ten participants	calendar time
The score is almost constant within each company	Inputs that vary from one headline to the next	small
The label may favour the winning baseline	Rebuild the target with estimated sensitivities	small
The labels are verifiable approximations	Annotate a sample with human judgement	calendar time
The alert claims more evidence than it has	Measure impact per news item and not per day	intraday data
The three blocks barely share companies	A split of constant composition	small
Only the headline is read, not the body of the article	Collect and represent the full text	new data
The split is not always well estimated	Display the fit alongside each split	small
Drift is measured and not corrected	Retrain with the production log	small
Source coverage is incomplete and uneven	Add sources, or declare the limit	outside our control
The three causes of the delay are not separated	Log when each item appears in the feed	instrumentation

No controlled study with users was carried out. This is the gap of greatest weight. The system rests on the hypothesis that an explanation accompanied by its evidence leads a non-specialist to a better decision, and that hypothesis was not submitted to an experimental design. The protocol exists, with participants, tasks and criteria defined, and was not executed. What does exist, and is reported in Section 5.6.5, is feedback collected in a real context: readers of the public channel rate the alerts they receive, under analysis rules fixed before the first vote, and the collection continues. The literature is explicit about the weight of this absence: claims of interpretability demand evaluation and are not established by declaration (Doshi-Velez and Kim 2018). The reason is not a formal one. The effect of an explanation on a person is not deducible from its construction: it is known that trust in automated systems fails in both directions, with the user ignoring a correct warning or accepting an incorrect one (J. D. Lee and See 2004), and that the mere presence of an explanation increases the acceptance of wrong answers (Bansal et al. 2021). The system was built to produce appropriate trust, presenting evidence rather than issuing a verdict, but that it succeeds remains a hypothesis. In the absence of the study, the two failure modes are indistinguishable in the logs: an ignored alert and an unduly believed alert produce the same absence of data.

The distinction between the two determines what each one allows to be claimed. The channel feedback is observational: whoever wants to votes, which shifts the sample towards the extremes; no group received the price change without the explanation that accompanies it, so nothing attributes the usefulness to the explanation itself; and the channel is public, so it is not known whether those who answer belong to the audience this work assumes. What those votes measure is perceived usefulness among those who chose to answer. The founding hypothesis concerns a better decision, which is another quantity, and it is that one which remains untested.

The deployed model could not perform the function assigned to it. The variant put into production as a decision gate receives seven company-level inputs, one day-level input and a single one that varies from one headline to the next, the length of the headline. Its score is, by arithmetic and not by accident, almost constant within each company: the mean spread within each company is 0.072 and the spread between the medians of the companies is 0.392. In 48% of the distinct headlines, the outcome was determined before the headline was read.

Two claims must be separated, since only one of them is an error. The result of RQ3 stands, because the variant with text had three hundred and eighty-four numbers per headline and still lost. The product decision was wrong, and the reason is methodological: the choice was made by a metric that interrogates the ordering of the complete set, when the product needed the distinction between two headlines of the same company.

The system was changed and the model ceased to be a gate and became a ranking criterion. That change is not validated. Over the 825 decisions matured in production, grouped into the 239 independent units the label admits, the ranking ability is 0.486, with an interval between 0.403 and 0.571, which contains chance. Ranking is the least indefensible use of the model, and not a demonstrated use.

This failure belongs to a class described in the literature on machine learning systems in production. D. Sculley et al. (2015) identify a cost that appears in no quality metric, and that comes from data dependencies and from entanglement between components: the validity of one piece depends on assumptions about another that were never stated. The model was evaluated in isolation, over the distribution of the training set, and deployed behind two filters that alter it. No test failed and no log flagged an anomaly. A learned component is evaluated where it is trained and is useful where it is placed, and when those two places differ the difference does not announce itself.

The label may favour the winning baseline. The label discounts the movement of the market assuming unit sensitivity, so its error is not random: it is larger in the stocks most sensitive to the market, which are typically the most volatile. The winning baseline is precisely volatility. With what is measured it is not possible to separate the hypothesis that volatility better predicts materiality from the hypothesis that volatility better predicts the error of the label. Both the level and the ordering of the RQ3 metrics are therefore conditional on the label used.

The alert claims more evidence than it has. When it displays three precedents that agree on the direction of the movement, the alert presents that agreement as resulting from three observations. The impact is, however, measured per pair formed by company and day, so three headlines of the same company on the same day share the value by construction. Over the 247 alerts with precedents delivered, in 36.8% the cases displayed rest on fewer distinct days than cases displayed, and in 11.3% they all correspond to the same day; of the 120 alerts that claim unanimity, 23.3% rest on a single day. The wording of the text was corrected to count days and not headlines, which makes the claim true. The cause remains: while impact is measured per day, two news items of the same day are not two independent observations.

The remaining seven, in summary. The **labels are approximations**: relevance is approximated by membership of the same sector, materiality by the occurrence of a movement above a threshold, and rarity by a percentile of the company's own returns. All three are objective and verifiable, and none corresponds exactly to the property one intended to measure. Dependence on approximations is not peculiar to this work: in neighbouring financial domains, such as fraud detection, unsupervised methods occupy a central position because labelled anomalies are scarce (Ahmed, Mahmood, and Islam 2016).

The **three data blocks barely share companies**. The training block contains thirteen companies and the test block nine, five companies from training do not appear once in the test, and one company accounting for 17.1% of the test rows never appears in training. The finding acts in both directions: it reinforces the negative result, because a model whose signal is essentially the identity of the company estimates it over a set almost absent from the block where it is evaluated; and it limits it, because it does not allow the hypothesis that the model is worse to be separated from the hypothesis that identity does not transfer between these two periods.

The **news is read only at headline level**. The system represents the headline and not the body of the article, which limits what the text representation can capture and is a plausible, though unconfirmed, explanation for the negative result of RQ3. Reviews of textual analysis in finance describe works that operate over complete documents (Kearney and S. Liu 2014), and a headline is the part of the document where the most information has been discarded.

The **split of a movement is not always well estimated**. The sum of the three parts holds by construction and validates nothing. The informative measure is the coefficient of determination in the estimation window, whose median was 0.460 and which in one of the seventeen companies took a negative value; in those cases the split continues to be displayed and should be read as indicative. Precision-weighted shrinkage (Vasicek 1973) limits the damage of a bad estimate without turning it into a good one.

Drift is measured and not corrected. The parameters were estimated between 2018 and 2023 and were not readjusted. Between the training block and the test block, the population stability index places twenty-day volatility in the band of significant displacement, at 0.281, and that is the input on which the model most depends. The concept drift literature deals with this problem, with a taxonomy of the change and strategies for detection and adaptation (Gama et al. 2014); here detection exists and adaptation does not. The architecture that would supply it is defined in Section 4.7, with a conditional trigger, a rolling window and a promotion gate, and was not executed: the limitation is one of execution and not of design.

Source coverage is incomplete and uneven. There was at least one relevant news item on 88.5% of the 417 days the detector considered unusual, and the worst covered company stands at 50%. On the remaining days no change to the system has any effect, since a news item not associated with the company by the source never enters the path. The value is an upper bound: it establishes that a news item existed, and not that the appropriate news item arrived.

The **delay is in discovery, and its three causes are not separated**. Over the 278 alerts with a publication time and a send time, the median between publication and detection is 353 minutes, and the median between detection and delivery is 5 seconds. The move from a best-effort scheduler to a sixty-second cycle did not reduce the delay, which rules out the polling cadence as the explanation. Three causes remain that the history does not separate, and only one of them, the source listing late, would be mitigated by a paid service; the other two persist with any source.

6.5 Future work

The lines stated in this section derive from the limitations of the previous section, according to the correspondence presented in Table 6.1, and are ordered by the relation between expected effect and cost. A list of future work that does not derive from measured limitations is a justification in another form. Figure 6.2 groups them by what each one requires before it can be carried out.

1. **Run the study with users.** Six to ten participants, with sessions of approximately fifteen minutes, comparing the reading of an isolated fact with the reading of the complete alert. It closes the central claim of the work, which is that the evidence presented alongside the alert leads to a better decision. It shares with the construction of an annotated target, the third item of this list, the property that no computational means accelerates it, since both depend on human judgement. Other claims remain to be verified independently of it, among them the quality of the three labels, the usefulness of the model over the population it actually observes and the stability of the result across periods.

2. **Give the decision model inputs that vary with the news.** It is the direct correction of the limitation of greatest technical consequence, and its cost is small, since the two most obvious quantities are already computed by the system: the similarity of the best precedent retrieved and the direction agreement among the precedents. Unlike volatility and sector, both vary from one headline to the next.

3. **Build a target with human judgement.** Annotating a sample of days, identifying which of the news items suppressed today should pass and which of those that pass add nothing, would supply for the first time a real target to optimise, in place of tuning constants against an approximation. The same sample would allow the quality of the three labels used to be estimated.

4. **Rebuild the label with estimated sensitivities.** The triage label discounts the movement of the market assuming unit sensitivity, which introduces an error correlated with the volatility of the company, which is precisely the winning baseline. Redoing the target with the shrunk sensitivities described in Section 3.4 and repeating the six families of models would determine whether the ordering obtained survives an alternative definition of materiality.

5. **Read the body of the articles and not only the headline.** It is the most direct hypothesis for explaining the negative result of RQ3 and is testable with the protocol already built, requiring only the collection of the full text.

6. **Detect repeated stories by meaning.** Suppression of repetitions rests today on vocabulary overlap, and doing it by similarity would use the technique the system already has. The point matters more than its technical size suggests: if the retail investor buys the assets that capture their attention (Barber and Odean 2008), a system that communicates the same story three times is not informing, but amplifying the same stimulus.

7. **Remove the asymmetry of reference in the same-day return.** The input corresponds, in training, to the complete movement of the day of the news, and in production to the last bar available at the moment of scoring. Replacing it with the return accumulated up to publication, or, more conservatively, with the previous day's return, removes the difference and allows performance to be reported under a single reference. The direction of the effect is known and favours the negative result, so the correction would reinforce it rather than undo it; what the experiment adds is the measurement of that reinforcement instead of its deduction.

8. **Group the precedents by company-day pair before selection.** The correction applied to the text of the alert made the claim true without removing the cause. Selecting the candidates after grouping by day, displaying from each day only the closest headline, eliminates at source the possibility of one day observed three times being presented as three agreeing cases.

9. **Expose the quality of the fit in the message itself.** The split of a movement is displayed with the same confidence when the coefficient of determination of the window is high and when it is negative. Presenting it alongside the split, and adding an explicit note below a minimum threshold, turns a limitation declared today only in the document into a property observable by whoever reads the alert.

10. **Retrain the model with the production log.** Drift is measured and the new labels derive from prices, so execution is possible. The impediment is not technical, since a retraining would change the values reported in this document and would require evaluation time of its own.

One path was considered and not followed, and the reasons are recorded because they are needed by whoever takes up the work. Conformal prediction allows sets of answers to be returned with a distribution-free guarantee (Angelopoulos and Bates 2023; Vovk,

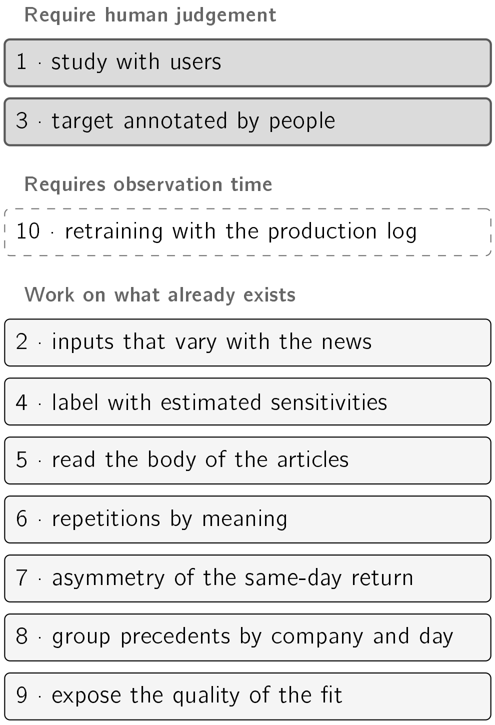

Figure 6.2: The ten lines of future work, grouped by what each one requires before it can be carried out. Seven operate on components the system already has and depend only on technical work. One awaits observation time, which began to run in September 2026. Two require human judgement and are the only ones that no computational means accelerates, which places them on the critical path despite not being the ones of greatest technical cost. The numbering is that of the list in this section, which is ordered by the relation between expected effect and cost.

Gammerman, and Shafer 2005), which suits a tool that refuses to claim more than it knows. Two reasons ruled it out. The first is that the guarantee is marginal, holding on average over cases and not case by case, which is precisely the reading a non-specialist user would make.

The second is more decisive: the guarantee requires exchangeability between calibration and test, and these data are a time series, in which the headlines arrive in order. It is the same assumption this work refuses when it splits the set by date and with an embargo. Applying the method without addressing that point would produce intervals with a guarantee the data do not support, replacing an honest value with a value that has the appearance of rigour.

6.6 Final considerations

The work started from three questions a retail investor asks when faced with a price movement, and which free tools do not answer. Whether the movement is unusual. Whether the origin is the company or the market. And whether precedents with a known outcome exist. A system was built that answers all three with verifiable evidence and without issuing forecasts, and each component was evaluated against simpler alternatives declared before the measurement.

The result with the greatest informative value is negative. The hypothesis that the text of the news would carry information absent from the prices was not confirmed over this corpus. Investigating it produced a diagnosis of wider reach than the question that gave rise to it. The learned component was evaluated over one distribution and deployed over another, and the metric by which it was selected posed a question distinct from the one the product needed. No automatic test flagged the discrepancy, because each component fulfilled its contract and what was wrong was an assumption about the link between them.

The founding constraint of the work, under which the system produces no price forecasts, proved to be the decision of widest reach. It determines what can be claimed: every claim of the system is confirmable against the record at the moment it is read. It determines the form of the explanation, which presents individual cases instead of averages. And it determines the evaluation criterion, which bears on the ability to identify what is unusual and not on the ability to anticipate. A system that presents evidence instead of verdicts delivers less than a system that announces directions, and delivers only what it can support.

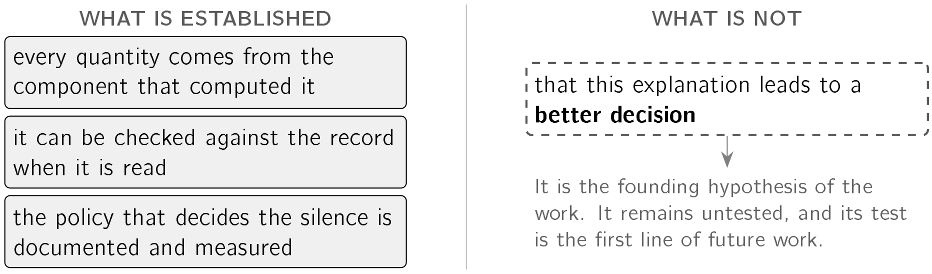

Figure 6.3: The exact limit of the claim. On the left, the three properties the system demonstrates and that are verified by inspecting what it produces. On the right, the proposition the work did not test: that presenting the evidence alongside the alert leads to a better decision. The separation is not a courtesy caveat, since the second is the hypothesis that motivated the work, and distinguishing it from what was demonstrated is what prevents the first from being read as though the second were proved.

The contribution is therefore stated at its exact level. What this work establishes is that the system is technically prepared to support explanations resting on verifiable historical evidence: every quantity presented comes from the component that computed it, can be checked against the record at the moment it is read, and the policy that decides the silence is documented and measured. What this work does not establish is that this form of explanation leads the retail investor to a better decision. That is the founding hypothesis, it remains untested, and its test is the first item of the previous section. Figure 6.3 presents the separation.

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Appendix A

Reprodutibilidade

Este apêndice documenta a proveniência dos resultados apresentados no corpo da dissertação. Enumera os artefactos conservados, estabelece a correspondência entre cada resultado reportado e o procedimento que o produz, apresenta o estado em que ficou cada afirmação substantiva depois de medida, e regista as condições de licenciamento dos dados utilizados. Encerra com o desenho do estudo com utilizadores que a Secção 6.4 identifica como a lacuna de maior peso do trabalho.

A.1 Artefactos conservados

Esta secção enumera o que é conservado no repositório de código e o que não é, com a razão de cada caso.

O ambiente de execução está descrito na Secção 3.7, com as versões exatas das bibliotecas que produziram os valores deste documento. As versões estão fixadas, e não expressas em intervalos, porque uma biblioteca que se atualize sozinha altera resultados sem emitir aviso. Nenhum dos procedimentos exige unidade de processamento gráfico.

São conservadas três classes de artefacto. A primeira é o componente aprendido, que compreende o modelo treinado, o calibrador ajustado sobre o bloco de validação e um ficheiro descritivo da execução que os produziu, com a dimensão de cada bloco, a prevalência de casos positivos, a semente utilizada e as métricas obtidas. A segunda é o conjunto de ficheiros de resultados, um por procedimento de avaliação, que o texto cita. A terceira é a base de precedentes que a aplicação implantada consulta, com cinquenta e seis megabytes de vetores e oito de metadados. Esta última não constitui uma amostra: é o artefacto de que o sistema necessita para arrancar, e sem ele não existiriam precedentes para apresentar.

Os corpora de origem não são conservados no repositório. O conjunto histórico ocupa várias centenas de megabytes e é reconstruído pelos procedimentos de preparação descritos na Secção 3.2. Acompanham o código amostras reduzidas, cuja finalidade é permitir que a verificação automática decorra sem acesso à rede, incluindo o teste que impõe a separação temporal descrita na Secção 3.6.

A.2 Origem de cada resultado

Esta secção estabelece a correspondência entre cada resultado reportado no corpo do documento e o procedimento que o gera.

Nenhum resultado desta dissertação foi introduzido manualmente no texto. Cada um é produzido por um procedimento automático de semente fixa, que escreve o seu próprio ficheiro de resultados, e é esse ficheiro que o texto cita. Uma segunda execução sobre os mesmos dados devolve os mesmos valores. A Tabela A.1 reúne os resultados reportados e identifica a secção em que cada um aparece; o repositório de código nomeia, para cada resultado, o ficheiro que o contém e o procedimento que o produz.

A garantia de determinismo aplica-se ao bloco superior da tabela. As linhas do bloco inferior são de outra natureza, e por isso estão separadas: constituem medições sobre a operação do sistema, cujo valor depende do momento em que o registo foi lido e não de uma semente. A secção em que cada uma é apresentada declara o período e o conjunto sobre que foi realizada.

Uma dessas medições não é regenerável, e a distinção fica registada em vez de omitida. A repartição do funil por porta, apresentada na Secção 4.4, foi lida do registo de decisões num dia concreto, e esse registo conserva apenas três dias, pela razão indicada na Secção 3.2: é republicado a cada ciclo, pelo que o custo é de publicação e não de armazenamento. A execução do mesmo procedimento numa data posterior produz outro dia, e não o mesmo. O valor consta de um ficheiro de resultados, com a data da leitura e o procedimento que a produziu, de modo que qualquer leitura futura tenha origem declarada ainda que aquele dia não se repita.

Os restantes valores numéricos do documento pertencem a três categorias. Os valores extraídos da literatura estão citados no local em que aparecem. As constantes de configuração constam da Tabela 4.2, com a distinção entre as que foram derivadas de uma medição e as que foram escolhidas. A terceira categoria reúne medições que servem uma passagem e não um resultado, como as linhas descartadas pelo embargo, o desvio-padrão do caso trabalhado da Secção 3.3 ou as contagens por fonte de notícias; cada uma declara, no local em que aparece, o conjunto sobre que foi realizada.

A.3 Matriz de evidência

Esta secção submete cada afirmação substantiva do trabalho à evidência que a sustenta e regista o estado em que ficou.

A tabela da secção anterior enumera resultados. A Tabela A.2 enumera afirmações, exame mais exigente, uma vez que uma afirmação pode não sobreviver à evidência que a suportava. As afirmações retiradas permanecem na tabela, dado que uma matriz que enumere apenas as sobreviventes não constitui uma auditoria.

Duas observações incidem sobre a tabela no seu conjunto e não sobre qualquer linha isolada.

A primeira é que as afirmações retiradas o foram por medições do próprio trabalho, e nenhuma por revisão externa. Constitui o comportamento pretendido de um plano de avaliação fixado antes das experiências, cuja utilidade reside precisamente na capacidade de produzir uma resposta negativa.

A segunda é que as duas linhas assinaladas como não afirmadas têm o mesmo peso das restantes. Uma delas é a utilidade das explicações para o utilizador, que não foi medida por ausência de estudo com pessoas. Onde não existe evidência, não é formulada afirmação.

Table A.1: Os resultados reportados no corpo do documento e a secção em que aparecem. O bloco superior é determinístico e reproduz-se exatamente sobre os mesmos dados; o bloco inferior reúne medições sobre a operação, cujo valor depende da data de leitura do registo.

Resultado	Valor	Onde aparece
Amplitude da taxa de disparo, desvio relativo contra limiar fixo	0.015 contra 0.344	§5.2
F_1 contra o critério aproximado	0.516 contra 0.218	§5.2
Comparação com dois detetores aprendidos	0.530 / 0.269 / 0.280	§5.2
Ablação à janela e ao peso do desvio	0.678 e 0.664 contra 0.516	§5.2
Precisão@5 da recuperação, quatro métodos	0.514 / 0.346 / 0.240 / 0.126	§5.3
Valor de referência do setor mais representado	0.467	§5.3
Precisão@5 no protocolo simétrico à escala	0.595 com acaso 0.333	§5.3
Precisão@5 apenas com precedentes anteriores	0.513 com acaso 0.259	§5.3
Concordância de direção contra o valor de acaso	0.708 contra 0.688	§5.3
Triagem, PR-AUC das seis famílias	de 0.378 a 0.542	§5.4
Precisão dentro do orçamento, quatro políticas	0.163 / 0.379 / 0.662 / 0.632	§5.4
Grelha de nove definições de critério	9 de 9	§5.4
Ablação da identidade, tabela de consulta contra modelo	0.534 contra 0.538	§5.4.5
Acréscimo do texto sobre a tabela de consulta	0.547 contra 0.534	§5.4.5
Qualidade do ajuste da decomposição, mediana e casos negativos	0.460, 1 em 17	§5.5
Seletividade da porta, dentro e entre empresas	0.072 contra 0.392	§5.6
Deriva de distribuição, PSI da volatilidade	0.281	§5.6
Sistema completo contra as alternativas	0.489 / 0.632 / 0.662 / 0.968	§5.6
Verificação da geração ancorada, ataques e controlos	23 de 23 e 8 de 8	§4.7
Medições sobre a operação: o valor depende da data de leitura do registo		
Decisões maturadas, mantidas contra suprimidas	58.9% contra 61.7%	§5.6
Capacidade de ordenação sobre as decisões tomadas	0.486	§5.6
Cobertura de notícias em dias invulgares	88.5%	§5.6
Latência, descoberta e entrega	353 min e 5 s	§4.6
Funil de seletividade, títulos distintos até entregues	743 → 15	§4.4
Funil de um dia, repartido por porta	5,060 avaliações	§4.4

Table A.2: Cada afirmação substantiva, a evidência que a sustenta e o estado em que ficou. O estado *retirada* indica que o trabalho a abandonou depois de a medir; o estado *estreitada* indica que subsiste numa formulação mais restrita do que a inicial.

Afirmação	**Evidência**	**Estado**
Deteção de movimentos invulgares		
Dispara a ritmo comparável entre empresas de volatilidade muito distinta	Amplitude de 0.015 contra 0.344, sem recurso a rótulos	Mantida
Supera um limiar fixo no critério aproximado	F_1 de 0.516 contra 0.218	Mantida, com o critério declarado como aproximação
Supera dois detetores aprendidos com a mesma informação	F_1 de 0.530 contra 0.269 e 0.280	Mantida
Não utiliza informação posterior ao dia avaliado	A janela exclui esse dia, e um teste automático impõe a exclusão	Mantida
A janela de vinte dias e o desvio de pesos iguais são as melhores opções	Variantes alternativas obtêm 0.678 e 0.664 contra 0.516	**Retirada**: mantidas por explicabilidade, não por desempenho
Recuperação de precedentes		
Supera as alternativas lexical e triviais	Precisão@5 de 0.514 contra 0.346, 0.240 e 0.126	Mantida
Supera por larga margem o melhor valor de referência trivial	A escolha do setor mais representado obtém 0.467, o que reduz a margem de +0.274 para +0.047	**Estreitada**: a afirmação passa a ser por setor, onde se mantém nos cinco
Resiste ao aumento de escala do corpus	Precisão@5 de 0.595 com acaso de 0.333	Mantida como teste de escala
Resiste à restrição temporal da produção	Precisão@5 de 0.513 com acaso de 0.259: a margem varia 0.008	**Estreitada**: o valor simétrico descreve tarefa menos exigente
Capta tema e não direção	Concordância de 0.708 contra acaso de 0.688	Mantida, e reportada como limitação
A pertença ao mesmo setor equivale a relação entre notícias	—	**Não afirmado**: é uma aproximação declarada
Triagem de notícias		
Nenhuma variante com texto supera a volatilidade	PR-AUC de 0.496 contra 0.542, com nove definições de critério e reamostragem por grupos	Mantida
O texto acrescenta informação sobre a tabela de consulta	PR-AUC de 0.547 contra 0.534, com intervalo entre +0.004 e +0.020	**Nova**, positiva, e sem efeito na precisão dentro do orçamento
As entradas não incorporam informação futura	Alterar o futuro de um exemplo deixa as entradas iguais e altera o critério	Mantida
A triagem quadruplica a precisão dentro do orçamento	O valor de referência de 0.163 resulta de ordenação alfabética; a escolha aleatória obtém 0.379	**Retirada**: o ganho é de 1.67 vezes
O ganho dentro do orçamento provém da aprendizagem	Treze constantes por empresa obtêm 0.662 contra 0.632 do modelo	**Retirada**
O modelo implantado seleciona melhor do que o acaso	58.9% de casos materiais entre as mantidas contra 61.7% entre as suprimidas	**Retirada**
Sistema		
O texto do alerta não pode divergir do cálculo	É composto a partir dos objetos calculados, sem geração de linguagem	Mantida
A restrição de gratuitidade limita a cobertura	Pelo menos uma notícia relevante em 88.5% dos dias invulgares	Mantida como limite superior
As explicações são úteis a uma pessoa	—	**Não afirmado**

A.4 Dados, licenças e atribuição

Esta secção regista a origem dos dados utilizados e as obrigações que o seu licenciamento impõe.

O conjunto histórico que sustenta a base de precedentes é o FNSPID (Dong, Fan, and Peng 2024), utilizado ao abrigo da licença Creative Commons BY-SA 4.0, cuja atribuição é aqui cumprida. Do corpus constam no documento apenas excertos mínimos de títulos, a título ilustrativo.

Essa licença comporta uma segunda obrigação, que é acionada neste caso. Uma vez que o repositório distribui ficheiros derivados do FNSPID, a cláusula de partilha nos mesmos termos aplica-se a esses ficheiros. A licença do código constitui uma decisão que permanece em aberto neste documento, e é esta a restrição que a condiciona.

Os preços provêm de serviços gratuitos de dados de mercado, através da cadeia de alternativas descrita na Secção 4.2; as notícias em tempo real provêm de três interfaces gratuitas, dentro dos respetivos limites de utilização, comparadas na Secção 4.2.2.

Quanto a dados pessoais, a formulação exata consta da Secção 3.8.1: o sistema não conhece as posições dos utilizadores e não as solicita, e o canal público não identifica destinatários. A exceção é a subscrição opcional de alertas personalizados, em que ficam associados o identificador de conversa atribuído pela plataforma e a lista de empresas escolhida, e esses são os únicos elementos armazenados.

A.5 Infraestrutura: o que entra no sistema, e a que custo

A Figura A.1 reúne as peças externas de que o sistema depende, pelo ponto do percurso em que cada uma entra, e o texto que se lhe segue apresenta o custo efetivo da operação. O codificador de frases é executado no formato Open Neural Network Exchange (ONNX), que permite corrê-lo sem a biblioteca de treino. Nenhuma peça foi escolhida por preferência: todas decorrem da restrição fundadora de utilizar apenas fontes gratuitas, e as duas exceções a essa restrição, que são o alojamento e

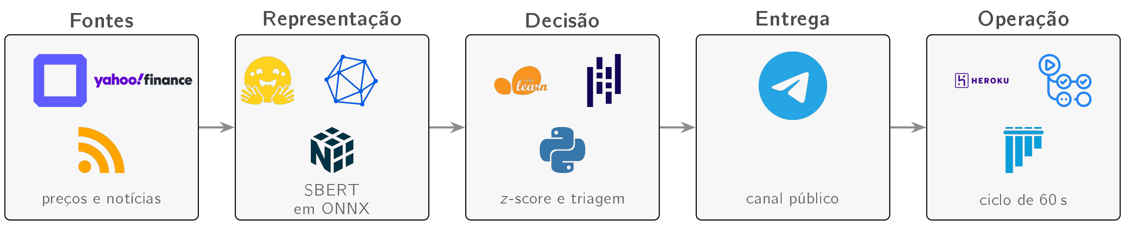

Figure A.1: As peças externas de que o sistema depende, pela ordem em que entram. As marcas pertencem aos respetivos titulares e a sua utilização aqui é nominativa, destinando-se a identificar a ferramenta empregue. A proveniência de cada ficheiro, com o resumo criptográfico correspondente, consta do manifesto que acompanha o repositório.

Custo da operação, e a razão do escalão escolhido

O sistema correu durante a maior parte do período documentado em contentores do escalão mais baixo, ao custo de sete dólares mensais por processo. Em 4 de setembro de 2026 o processo de recolha passou a exceder de forma persistente a quota de memória de 512 MB, com avisos de excesso a uma frequência de aproximadamente dois por minuto e leituras entre 518 e 970 MB.

Quatro causas foram medidas e corrigidas: a base de casos viva era carregada em listas de *float* de Python, ao custo de 136.7 MB para 10,968 registos, contra 21.7 MB da mesma informação numa matriz de precisão simples; as bases e o codificador eram reconstruídos a cada ciclo de sessenta segundos; o ficheiro de casos pendentes era lido uma vez por empresa; e a publicação de um ficheiro de 40 MB produzia cinco cópias dele em memória. O conjunto reduziu o pico de 970 para 663 MB.

As correções não bastaram, e é esse o resultado honesto. O conjunto de trabalho da aplicação excede genuinamente um contentor de 512 MB: as bibliotecas numéricas e o motor de inferência ocupam, por si, 158 MB, e a matriz de precedentes torna-se residente assim que a primeira consulta lhe percorre as páginas. O processo assentava em 536 MB mesmo entre picos.

A alternativa técnica seria retirar do processo a base de 38,214 precedentes, o que degradaria a qualidade da recuperação, isto é, trocaria uma limitação de infraestrutura por uma limitação de resultado. A opção tomada foi a inversa, e o custo consta da Tabela A.3.

Table A.3: Escalões de contentor antes e depois de 4 de setembro de 2026. O fornecedor não permite misturar escalões na mesma aplicação, pelo que o processo web, que nunca excedeu a quota, subiu por arrasto. É a razão de o acréscimo ser de sessenta e um dólares e não dos quarenta e três que a alteração de um só processo custaria.

Processo	Escalão	Memória	Custo	Razão
Até 4 de setembro de 2026				
web	Basic	0.5 GB	\$7	nunca excedeu a quota
recolha	Basic	0.5 GB	\$7	excedia de forma persistente
A partir de 4 de setembro de 2026				
web	Standard-1X	0.5 GB	\$25	subiu por não ser possível misturar escalões
recolha	Standard-2X	1.0 GB	\$50	é o primeiro escalão acima de 0.5 GB

Após a alteração, e sobre a mesma carga, não se registou um único aviso de excesso de memória. O saldo de crédito disponível cobre aproximadamente quatro meses de operação neste escalão, o que excede a duração prevista do trabalho. A restrição de utilizar apenas fontes gratuitas mantém-se para os *dados*, que são a matéria da dissertação; o alojamento e o canal de mensagens são custos de operação e estão declarados como tal.

A.6 Desenho do estudo com utilizadores

Esta secção apresenta o desenho do estudo referido na Secção 6.5.

O estudo compara duas condições sobre seis alertas reais extraídos do canal. A condição A apresenta o facto sem explicação e a condição B apresenta o alerta completo. Dois dos seis casos são situações em que o tema da notícia e a direção do movimento discordam, configuração em que a Secção 5.3 mostra o sistema ser mais suscetível de interpretação incorreta.

O contrabalanço cruza dois fatores, que são a ordem das condições e a metade dos estímulos que serve de condição A. O segundo fator é necessário porque as duas condições recorrem a alertas distintos, sob pena de a segunda medir memória em vez de compreensão; sem o cruzamento, qualquer diferença entre condições ficaria confundida com a maior acessibilidade de uma das metades.

A pergunta central é a que a verificação automática não alcança. A correspondência entre cada frase do alerta e o facto que a sustenta está garantida por construção, mas a afirmação do produto descreve um percurso realizado por um leitor: dada uma frase com remissão, aceder ao facto correspondente e julgar autonomamente se este a sustenta. A questão é qualitativa e não exige amostra de grande dimensão.

Duas salvaguardas metodológicas estão fixadas no código antes de existirem dados: o limiar de oito participantes para o teste estatístico, cuja alteração ficaria registada no histórico de versões, e o procedimento de análise, que sobre a folha por preencher não produz resultado algum. Um terceiro bloco do desenho original, relativo a texto gerado, não é executável, uma vez que as vias que o serviam foram retiradas do sistema.

Appendix B

Correspondência com o plano curricular

Este apêndice estabelece a correspondência entre as unidades curriculares do mestrado e os pontos do trabalho em que a respetiva matéria foi aplicada. A Tabela B.1 apresenta essa correspondência e identifica igualmente as unidades cuja matéria não teve aplicação, uma vez que uma correspondência que encontre aplicação para todas as unidades em todos os pontos não transmite informação.

Duas entradas da tabela declaram ausência de aplicação: a matéria de Ambientes Inteligentes e a componente de programação lógica e por restrições. O seu registo é mais informativo do que a construção de uma ligação artificial.

Table B.1: Aplicação da matéria de cada unidade curricular do plano de estudos do mestrado no trabalho descrito nesta dissertação. A última coluna indica a secção em que essa aplicação é visível.

Unidade	**Onde é aplicada**	**Secção**
Fundamentos de sistemas inteligentes		
Aprendizagem Automática 1	Construção de um conjunto de treino rotulado a partir de duas fontes desalinhadas no tempo; regressão logística com normalização e reponderação de classes; conjuntos de árvores por reforço do gradiente como referência superior; calibração *a posteriori*	§3.6, §5.4
Engenharia do Conhecimento	Raciocínio baseado em casos: recuperação e reutilização de precedentes em substituição de uma regra geral, com a interrupção deliberada do ciclo antes da adaptação	§2.6, §3.5
Paradigmas de Programação em AI	Separação entre lógica pura e efeitos externos, que é a propriedade que permite verificar o núcleo do sistema sem acesso à rede e sem carregar modelos. A matéria de programação lógica e por restrições não teve aplicação	§3.1
Planeamento e Tomada de Decisão	Política de decisão sob orçamento: portas em cascata, limiar derivado de uma análise de custos de erro, e distinção entre uma política que dispõe do dia completo e uma política que decide a cada ciclo	§4.4, §5.4
Inteligência artificial avançada e aplicações especializadas		
Aprendizagem Automática 2	Representações contextuais de frase produzidas por uma arquitetura *Transformer*; exportação do codificador para um formato leve, com medição da fidelidade da conversão, imposta pela restrição de memória do ambiente de execução	§2.4, §3.5
Linguagem Natural e Sistemas Conversacionais	Semelhança semântica entre frases e avaliação de recuperação sensível à ordem. O sistema constitui uma arquitetura de recuperação sem geração: recupera evidência e recusa produzir texto sobre ela, com a justificação apresentada no estado da arte	§2.4, §5.3
Ambientes Inteligentes	Sem aplicação. O trabalho não comporta componente física, robótica nem de Internet das Coisas	—
Profissionalismo, segurança e inovação		
Aspetos Sociais e Éticos em AI	Confiança apropriada em sistemas automáticos e risco de a explicação aumentar a aceitação de respostas incorretas; recusa de aconselhamento; fadiga de alertas como requisito de desenho	§2.7, §3.8
Privacidade e Segurança em Sistemas Inteligentes	Minimização de dados por desenho, com a exceção declarada; gestão de credenciais e mascaramento de segredos em registos; verificação de integridade do modelo transferido	§3.8.1, §3.8.2
Investigação e Inovação em AI	Enquadramento em investigação por desenho; critérios de sucesso fixados antes das medições; linhas de comparação declaradas com o respetivo valor de referência; matriz de evidência com as afirmações retiradas	§3.1, Apêndice A